1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

| REV | ECN | DESCRIPTION OF REVISION | CK APPD DATE |
|---|---|---|---|
| <REV> | <ECN> | <ECO_DESCRIPTION> | <ECODATE> |

# J50I MLB-I

Fri Sep 9 19:19:37 2011


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| 56 | 58 | AUDIO:HEADPHONE AMP | J40I | 05/26/2011 |
| 57 | 59 | AUDIO:SPEAKER AMP | J40I | 05/16/2011 |
| 58 | 60 | AUDIO:JACKS | J40I | 05/16/2011 |
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ALIASES RESOLVED

DOCUMENTS / BOARDS / ASSEMBLIES

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 051-9341 | 1 | SCH,MLB-I,J50I | SCH | | |
| 820-3227 | 1 | PCBF,MLB-I,J50I | MLB | | |
| 639-3210 | | PCBA,MLB-I,J50I | | | |
| 085-4179 | 1 | DEV LIST,MLB-I,J50I | DEV1 | | DEVELOPMENT_LIST |

DRAWING TITLE=SHORTSTACK
ABBREV=DRAWING

DRAWING TITLE: SCH,MLB_I,J50I

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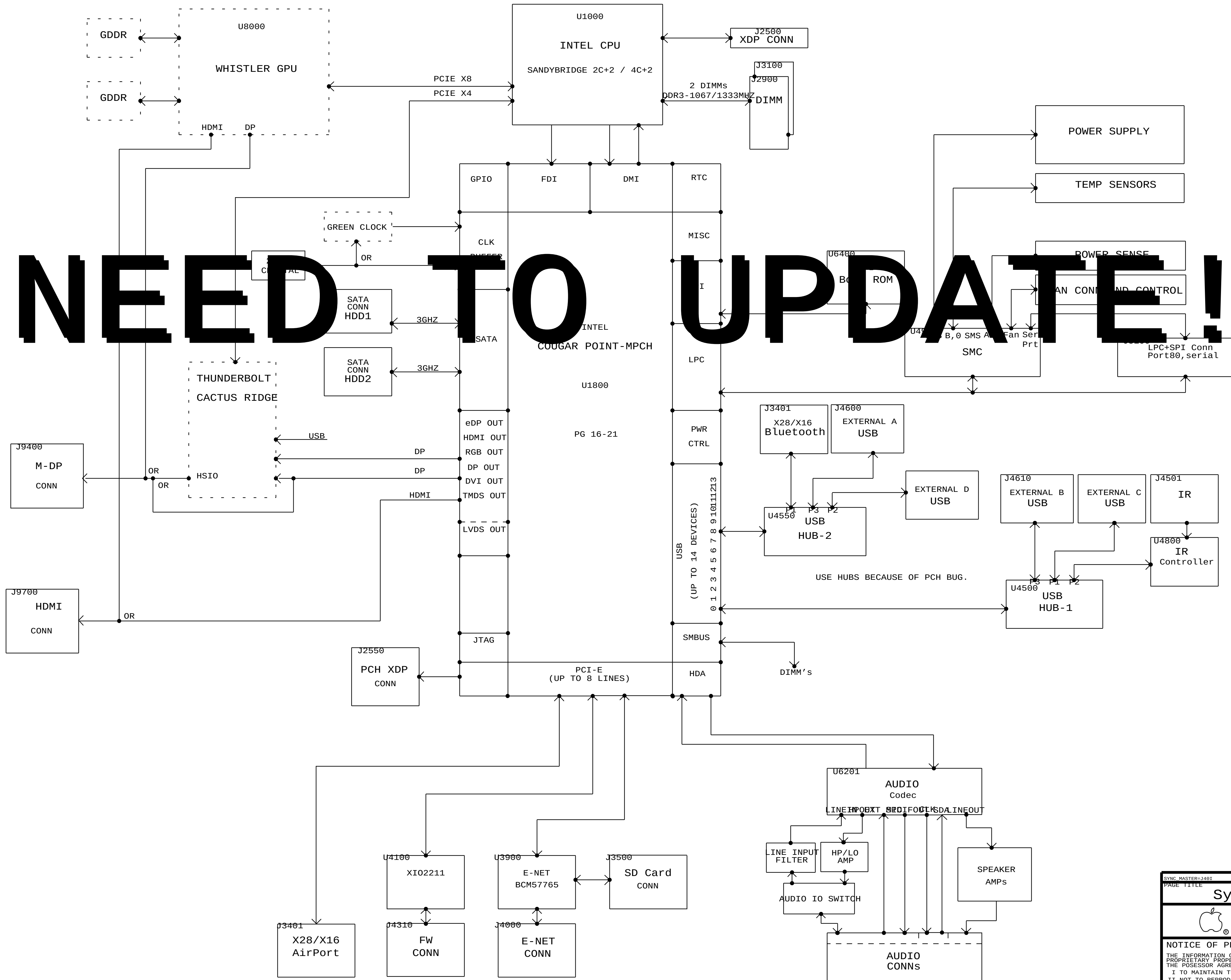


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# System Block Diagram

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# Power Block Diagram

Apple Inc.



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## PLATFORM CONTROL HUB (PCH)

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 337S4143 | 1 | IC,PCH,PANTHER_POINT,BGA,SSU,ES1 | U1800 | CRITICAL | PANTHER_POINT:ES1 |

## PROGRAMMED PARTS

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION | |
|---|---|---|---|---|---|---|
| 341T0999 | 1 | IC,EFI BOOTROM,J50 | U6400 | CRITICAL | | BASE: 335S0770,335S0769 |
| 341T0999 | 1 | IC,SMC,J50G | U4900 | CRITICAL | SMC:J50G | BASE: 338S0968 |
| 341T0999 | 1 | IC,SMC,J50I | U4900 | CRITICAL | SMC:J50I | BASE: 338S0968 |
| 341T0999 | 1 | IC,SMC,J50S | U4900 | CRITICAL | SMC:J50S | BASE: 338S0968 |
| 341T0340 | 1 | IC,IR CONTROLLER,J40 | U4700 | CRITICAL | | BASE: 338S0375 |
| 341T0341 | 1 | IC,ENET ROM J40 | U3990 | CRITICAL | | BASE: 335S0539, 335S0663, OR 335S0800 |

## NON-SCHEMATIC PARTS

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION | |
|---|---|---|---|---|---|---|
| 825-6838 | 1 | LBL,SERIAL NUMBER | LBL | | | |
| 946-3189 | 1 | MLB UV EB,J40G | UVEB | CRITICAL | J40X:J40G | 0.78 GRAM |
| 946-3379 | 1 | MLB UV EB,J40I | UVEB | CRITICAL | J40X:J40I | 0.61 GRAM |
| 946-3379 | 1 | MLB UV EB,J40S | UVEB | CRITICAL | J40X:J40S | 0.61 GRAM |
| 946-3190 | 1 | MLB ADHESIVE,LOCTITE | ADHS | CRITICAL | | 0.048GRAM |

## Alternate Parts

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 128S0286 | 128S0248 | ? | ? | 39UF CAP FROM SANYO AND CHEMI-CON |
| 132S0099 | 132S0203 | ? | ? | 0.1UF CAP,X7R |
| 138S0676 | 138S0691 | | ? | 22UF CAP FROM MURATA |
| 138S0681 | 138S0638 | | ? | 10UF CAP FROM TAIYO YUDEN |
| 152S0138 | 152S0694 | ? | ? | INDUCTOR FROM VISHAY |
| 152S1447 | 152S1269 | ? | ? | INDUCTOR FROM CYNTEC |
| 152S1282 | 152S1155 | ? | ? | INDUCTOR FROM CYNTEC |
| 152S1366 | 152S1289 | ? | ? | INDUCTOR FROM TOKO |
| 155S0641 | 155S0397 | ? | ? | FERRITE FROM LAIRD |
| 155S0457 | 155S0329 | | ALL | MAG LAYERS ALT TO MURATA |
| 157S0058 | 157S0055 | | ALL | DELTA ALT TO TDK MAGNETICS |
| 353S3196 | 353S3326 | ? | ? | DIFFERENT TEMP RATING |
| 371S0679 | 371S0652 | ? | ? | PIN DIODE FROM NXP |
| 376S0914 | 376S0881 | | ? | SI7121DN AS ALTERNATE |
| 376S0957 | 376S0999 | | ? | DUAL FET FROM FAIRCHILD |
| 376S0958 | 376S0953 | | ? | DUAL FET FROM FAIRCHILD |
| 376S1000 | 376S0960 | | ? | DUAL FET FROM FAIRCHILD |
| 376S0972 | 376S0612 | | ? | FET FROM ROHM |
| 376S0613 | 376S0859 | | ? | OLDER DUAL FET FROM TOSHIBA |
| 376S0855 | 376S0859 | | ? | DUAL FET FROM DDS |
| 376S0977 | 376S0859 | | ? | DUAL FET FROM DDS |
| 138S0703 | 138S0648 | | ? | CAP FROM MURATA |
| 197S0339 | 197S0179 | | ? | OSC. FROM SITIME |
| 376S0926 | 376S0610 | | ? | DUAL FET FROM FAIRCHILD |
| 376S0572 | 376S0659 | | ? | FET FROM ALPHA OMEGA |
| 377S0124 | 377S0057 | | ? | VARISTOR FROM AMOTECH |
| 152S1493 | 152S1300 | ? | ? | INDUCTOR FROM COILCRAFT |

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# COMMON PARTS TABLE

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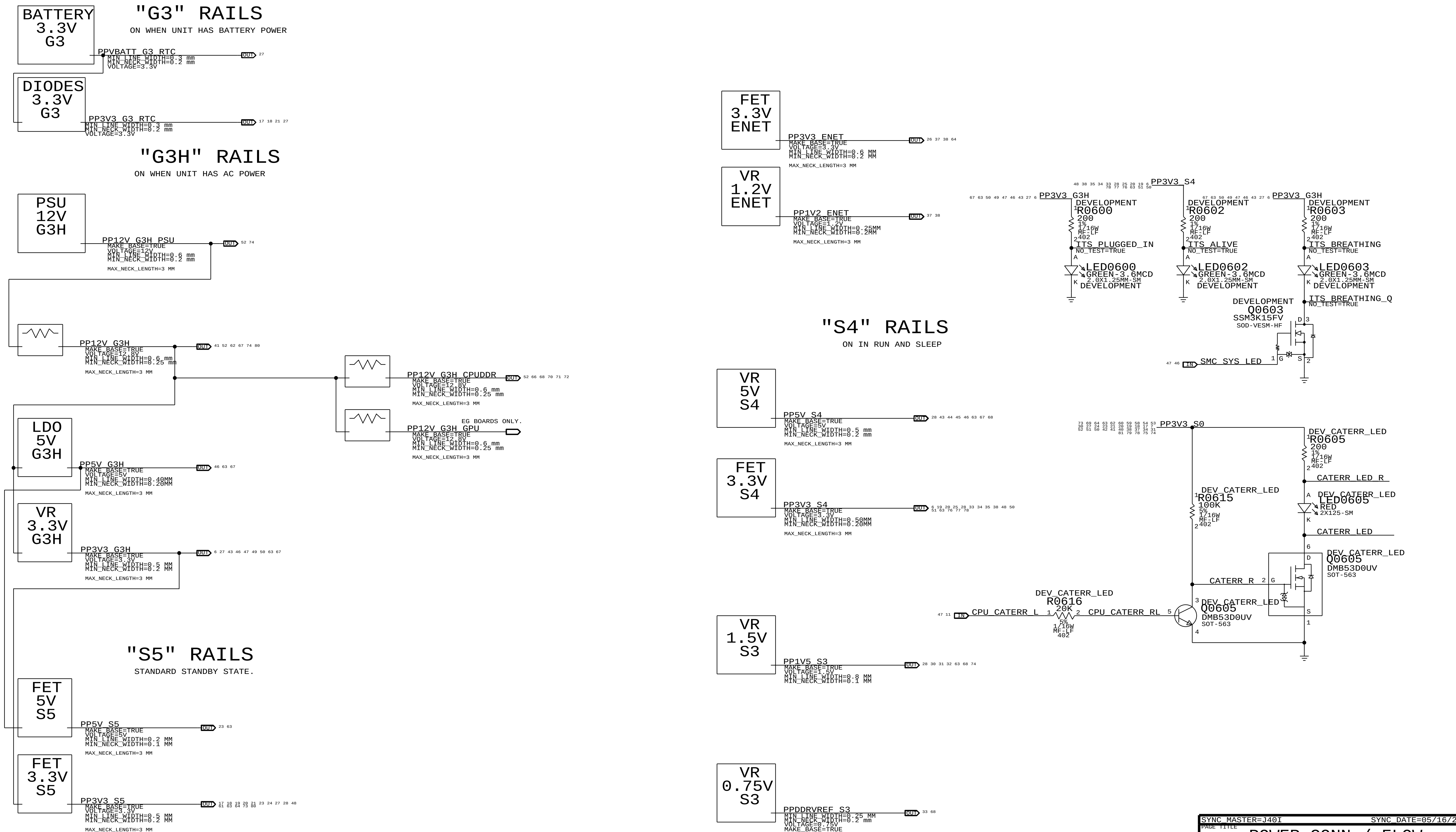


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POWER CONN / FLOW

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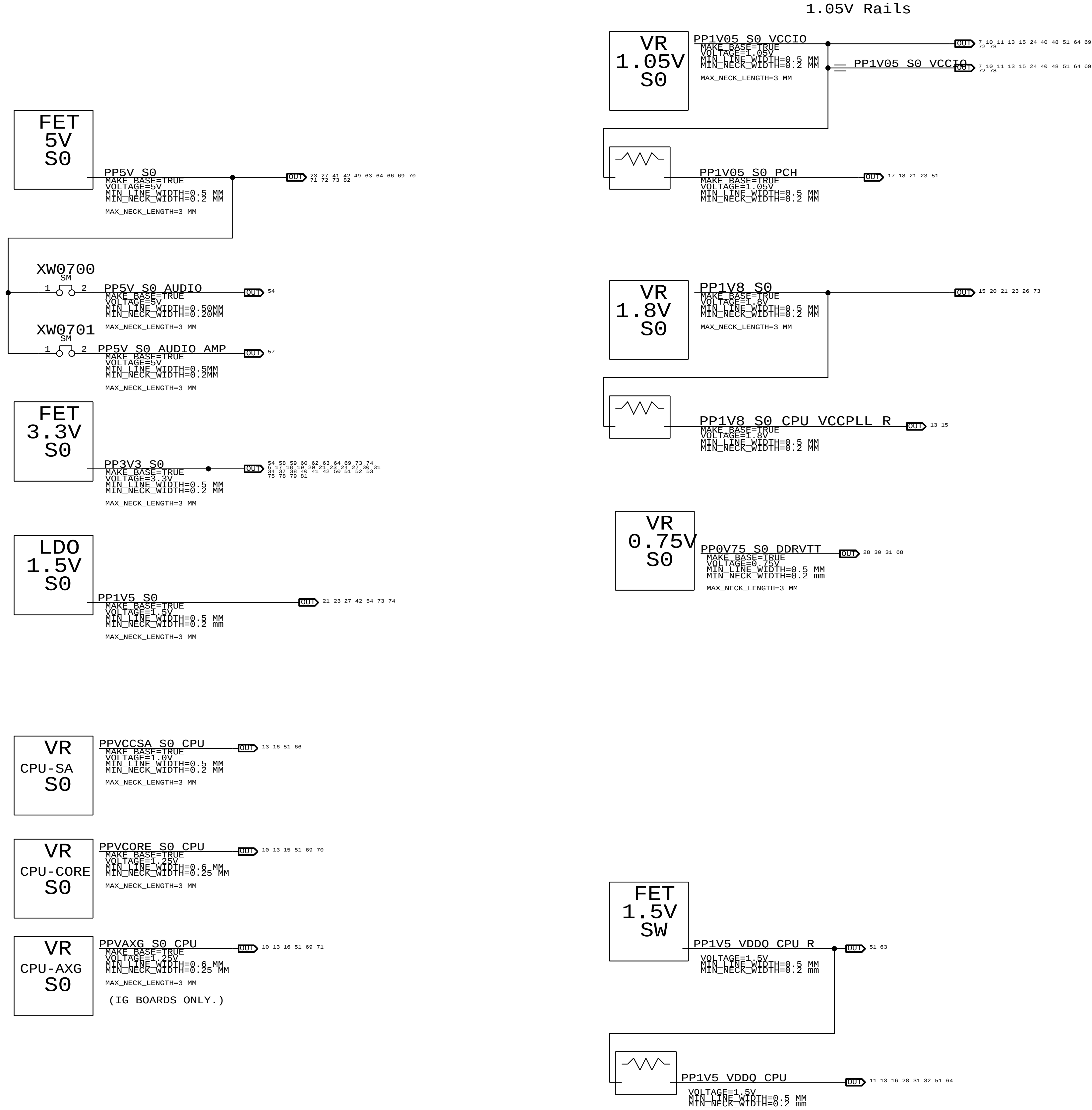


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# INTEL INTEGRATED GRAPHICS

FDI BUS.

| | | |
|---|---|---|
| IN =FDI_DATA_P<7..0> | MAKE_BASE=TRUE | FDI_DATA_P<7..0> OUT |
| IN =FDI_DATA_N<7..0> | MAKE_BASE=TRUE | FDI_DATA_N<7..0> OUT |
| OUT FDI_INT | MAKE_BASE=TRUE | FDI_INT IN |
| OUT =FDI_FSYNC<1..0> | MAKE_BASE=TRUE | FDI_FSYNC<1..0> IN |
| OUT =FDI_LSYNC<1..0> | MAKE_BASE=TRUE | FDI_LSYNC<1..0> IN |

NO PCIE TO EXTERNAL GRAPHICS

| | | |
|---|---|---|
| OUT =PEG_D2R_N<7..0> | MAKE_BASE=TRUE | NC_PEG_D2RN<7..0> |
| OUT =PEG_D2R_P<7..0> | MAKE_BASE=TRUE | NC_PEG_D2RP<7..0> |
| IN =PEG_R2D_C_N<7..0> | MAKE_BASE=TRUE | NC_PEG_R2D_CN<7..0> |
| IN =PEG_R2D_C_P<7..0> | MAKE_BASE=TRUE | NC_PEG_R2D_CP<7..0> |
| IN NC_PEG_CLK100MN | MAKE_BASE=TRUE | NC_PEG_CLK100MN |
| IN NC_PEG_CLK100MP | MAKE_BASE=TRUE | NC_PEG_CLK100MP |

HDMI

| | | |
|---|---|---|
| BI HDMI_LS_SCL | MAKE_BASE=TRUE | HDMI_LS_SCL BI |
| BI HDMI_LS_SDA | MAKE_BASE=TRUE | HDMI_LS_SDA BI |
| OUT HDMI_LS_HPD | MAKE_BASE=TRUE | HDMI_LS_HPD IN |
| IN =IG_HDMI_DATA_C_N<2..0> | MAKE_BASE=TRUE | HDMI_DATA_C_N<2..0> OUT |
| IN =IG_HDMI_DATA_C_P<2..0> | MAKE_BASE=TRUE | HDMI_DATA_C_P<2..0> OUT |
| IN HDMI_CLK_C_N | MAKE_BASE=TRUE | HDMI_CLK_C_N OUT |
| IN HDMI_CLK_C_P | MAKE_BASE=TRUE | HDMI_CLK_C_P OUT |

USED FOR T29 IC

ZT0840
STDOFF-4.00D1.85H-SM

ZT0841
STDOFF-4.00D1.85H-SM

# THUNDERBOLT (TBT)

PCIE

| | | |
|---|---|---|
| OUT =PCIE_TBT_D2R_N<3..0> | MAKE_BASE=TRUE | PCIE_TBT_D2R_N<3..0> IN |
| OUT =PCIE_TBT_D2R_P<3..0> | MAKE_BASE=TRUE | PCIE_TBT_D2R_P<3..0> IN |
| IN =PCIE_TBT_R2D_C_N<3..0> | MAKE_BASE=TRUE | PCIE_TBT_R2D_C_N<3..0> OUT |
| IN =PCIE_TBT_R2D_C_P<3..0> | MAKE_BASE=TRUE | PCIE_TBT_R2D_C_P<3..0> OUT |

TBT DISPLAY PORT INPUT

| | | |
|---|---|---|
| BI DP_TBTSNK0_CTRL_CLK | MAKE_BASE=TRUE | DP_TBTSNK0_CTRL_CLK BI |
| BI DP_TBTSNK0_CTRL_DATA | MAKE_BASE=TRUE | DP_TBTSNK0_CTRL_DATA BI |
| BI DP_TBTSNK0_AUXCH_C_N | MAKE_BASE=TRUE | DP_TBTSNK0_AUXCH_C_N BI |
| BI DP_TBTSNK0_AUXCH_C_P | MAKE_BASE=TRUE | DP_TBTSNK0_AUXCH_C_P BI |
| OUT DP_TBTSNK0_HPD | MAKE_BASE=TRUE | DP_TBTSNK0_HPD IN |
| IN =IG_DP_TBTSNK0_ML_C_N<3..0> | MAKE_BASE=TRUE | DP_TBTSNK0_ML_C_N<3..0> OUT |
| IN =IG_DP_TBTSNK0_ML_C_P<3..0> | MAKE_BASE=TRUE | DP_TBTSNK0_ML_C_P<3..0> OUT |

TBT DISPLAY PORT INPUT

| | | |
|---|---|---|
| BI DP_TBTSNK1_CTRL_CLK | MAKE_BASE=TRUE | DP_TBTSNK1_CTRL_CLK BI |
| BI DP_TBTSNK1_CTRL_DATA | MAKE_BASE=TRUE | DP_TBTSNK1_CTRL_DATA BI |
| BI DP_TBTSNK1_AUXCH_C_N | MAKE_BASE=TRUE | DP_TBTSNK1_AUXCH_C_N BI |
| BI DP_TBTSNK1_AUXCH_C_P | MAKE_BASE=TRUE | DP_TBTSNK1_AUXCH_C_P BI |
| OUT DP_TBTSNK1_HPD | MAKE_BASE=TRUE | DP_TBTSNK1_HPD IN |
| IN =IG_DP_TBTSNK1_ML_C_N<3..0> | MAKE_BASE=TRUE | DP_TBTSNK1_ML_C_N<3..0> OUT |
| IN =IG_DP_TBTSNK1_ML_C_P<3..0> | MAKE_BASE=TRUE | DP_TBTSNK1_ML_C_P<3..0> OUT |

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 338S1047 | 1 | IC,TBT,CR-4C,ES1,228 FCBGA,12X12MM | U9000 | CRITICAL | |
| 335S0865 | 1 | IC,EEPROM,SERIAL,SPI,32KX8,1.8V,2X3QFN | U9090 | CRITICAL | TBTROM:BLANK |
| 341T0342 | 1 | IC,CONF ROM,TBT,J50I | U9090 | CRITICAL | TBTROM:PROG_I |
| 341T0394 | 1 | IC,CONF ROM,TBT,J50S | U9090 | CRITICAL | TBTROM:PROG_S |
| 341T0380 | 1 | IC,CONF ROM,TBT,J50G | U9090 | CRITICAL | TBTROM:PROG_G |

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SPECIFIC ALIASES, TABLES

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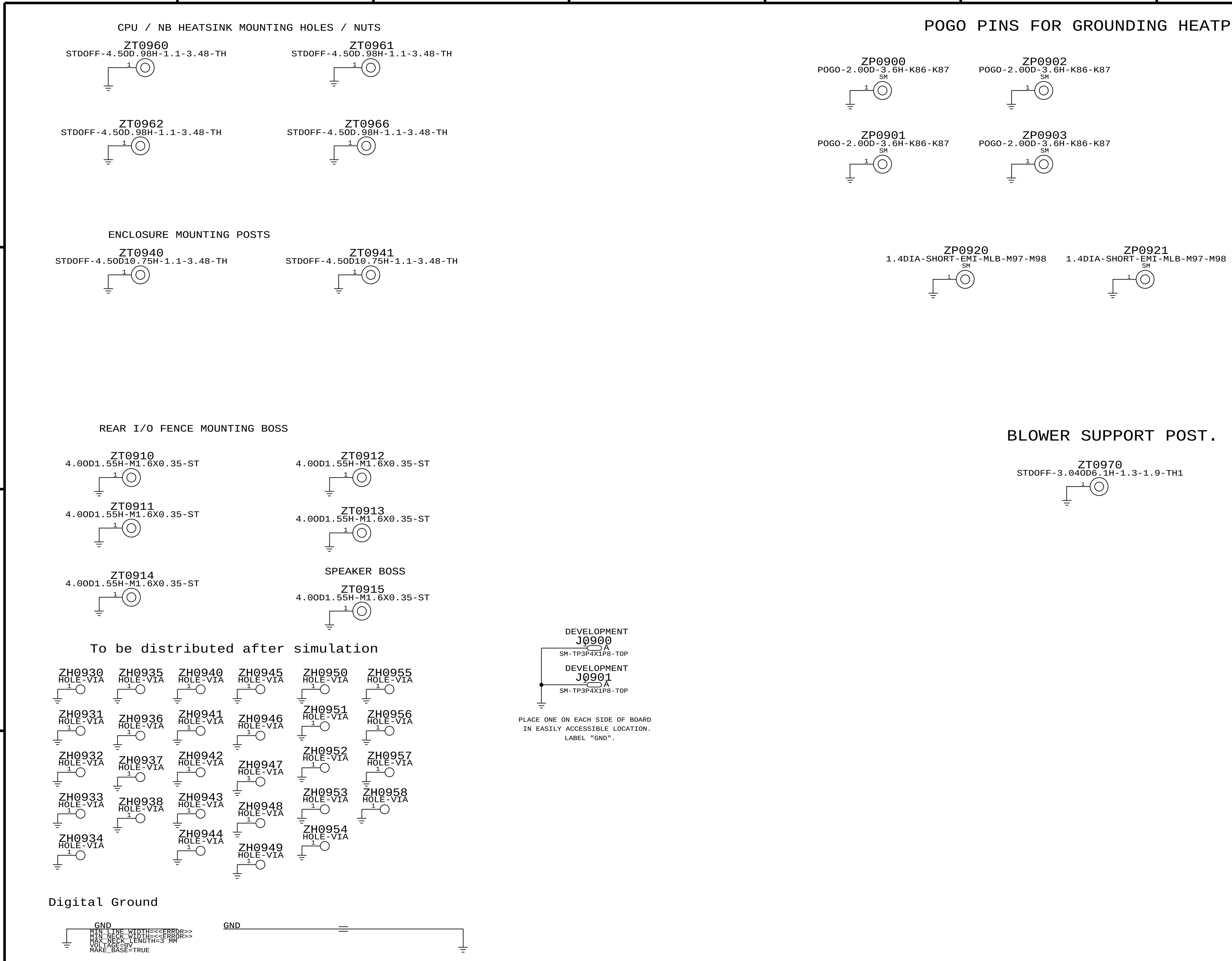


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MECHANICAL, GROUND VIAS

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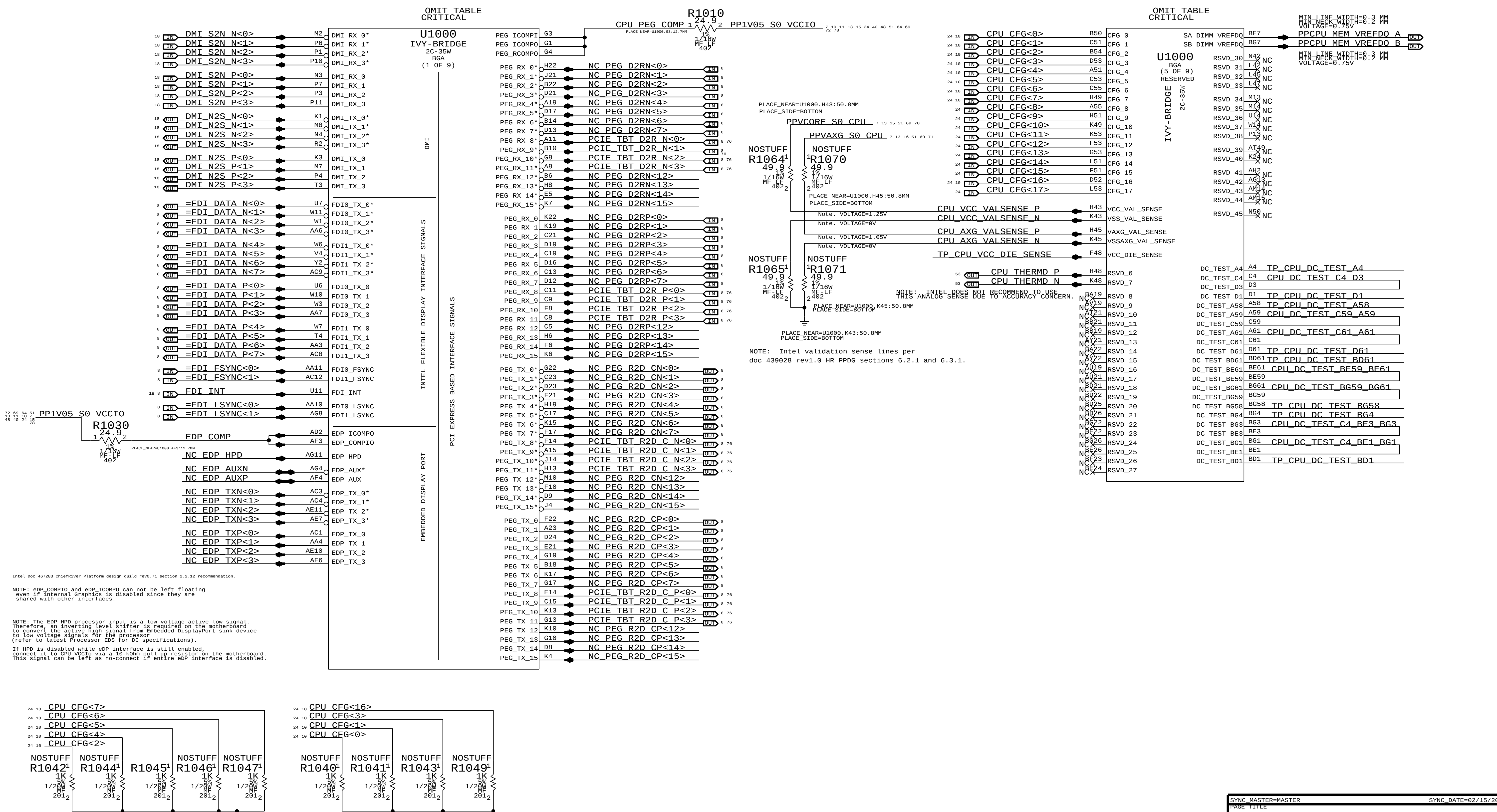
NOTE: Intel provides an internal pull-up OF 5-15k to VCCIO on all CFG signals.




Intel Doc 467283 ChiefRiver Platform design guild rev0.71 section 2.2.12 recommendation.

NOTE: eDP_COMPIO and eDP_ICOMPO can not be left floating even if internal Graphics is disabled since they are shared with other interfaces.

NOTE: The EDP_HPD processor input is a low voltage active low signal. Therefore, an inverting level shifter is required on the motherboard to convert the active high signal from Embedded DisplayPort sink device to low voltage signals for the processor (refer to latest Processor EDS for DC specifications).

If HPD is disabled while eDP interface is still enabled, connect it to CPU VCCIO via a 10-kOhm pull-up resistor on the motherboard. This signal can be left as no-connect if entire eDP interface is disabled.

NOTE: Intel validation sense lines per doc 439028 rev1.0 HR_PPDG sections 6.2.1 and 6.3.1.

NOTE: INTEL DOES NOT RECOMMEND TO USE THIS ANALOG SENSE DUE TO ACCURACY CONCERN.

These can be Placed close to J2500 and Only for debug access

FOR IVYBRIDGE PROCESSOR

| | | |
|---|---|---|
| CFG [7] | :PEG DEFER TRAINING | 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB  0 = WAIT FOR BIOS |
| CFG [6:5] | :PCIE BIFURCATION | 11 = 1 X16 (DEFAULT)  10 = 2 X8  01 = RSVD  00 = X8, X4, X4 |
| CFG [4] | :eDP ENABLE/DISABLE | 1 = DISABLED  0 = ENABLED |
| CFG [3] | :PCIE x4 LANE REVERSAL | 1 = NORMAL OPERATION  0 = LANES REVERSED |
| CFG [2] | :PCIE x16 LANE REVERSAL | 1 = NORMAL OPERATION  0 = LANES REVERSED |

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 337S4113 | 1 | IC,IVB,35W,2+2,BGA | U1000 | CRITICAL | CPU:IVB-2C |

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## CPU DMI/PEG/FDI/RSVD

Apple Inc.

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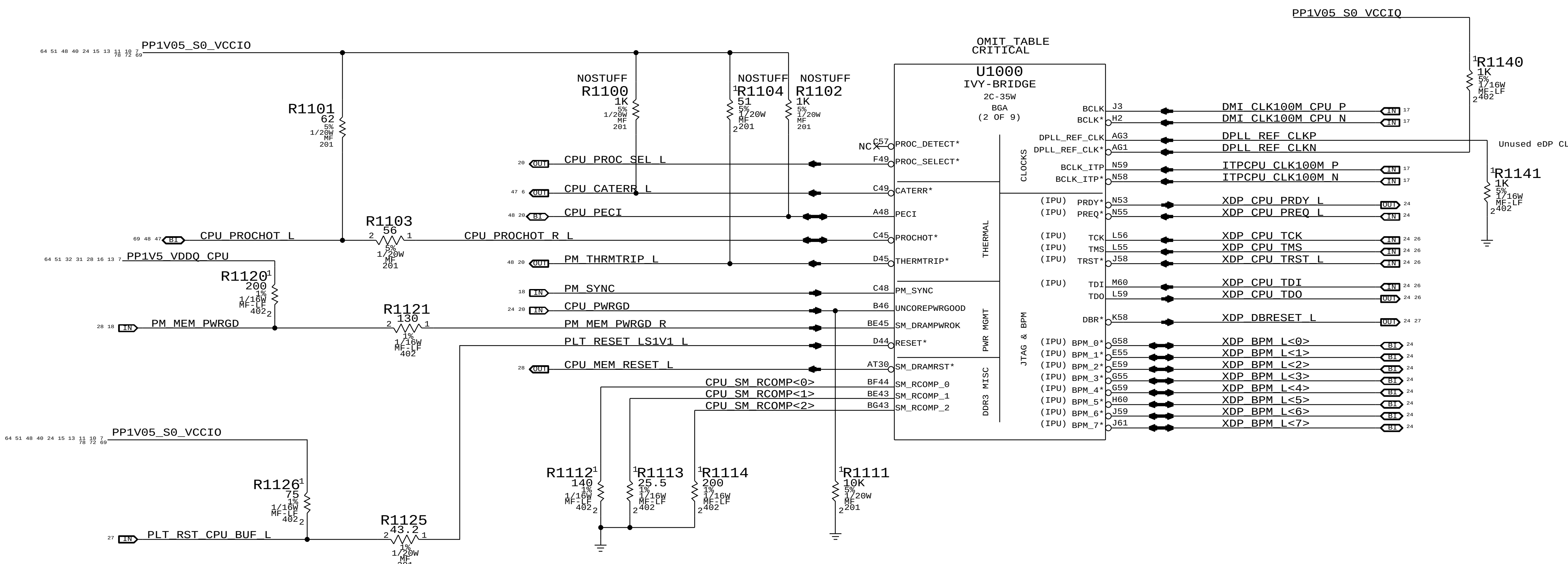


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CPU CLOCK/MISC/JTAG

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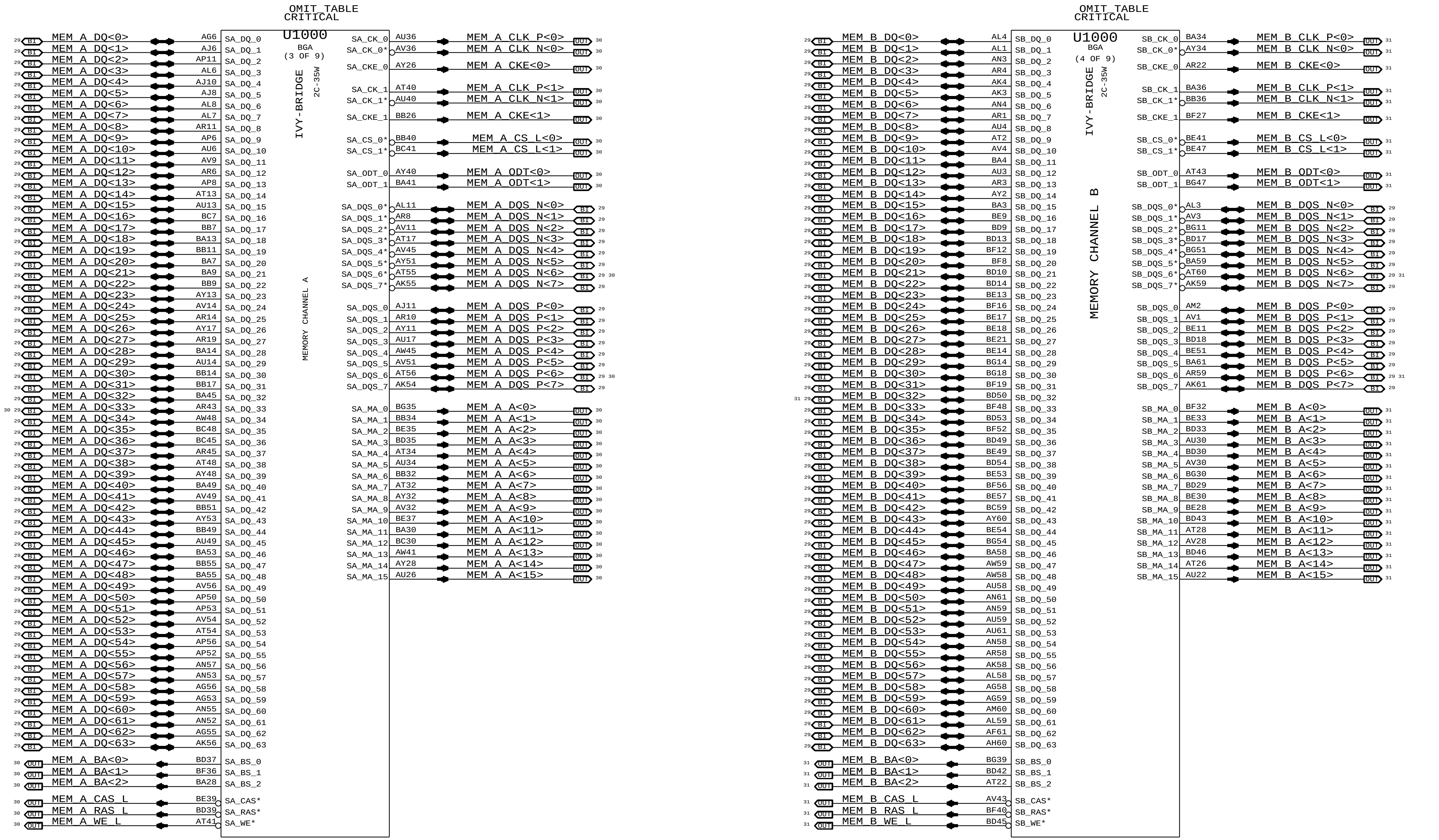


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CPU DDR3 INTERFACES

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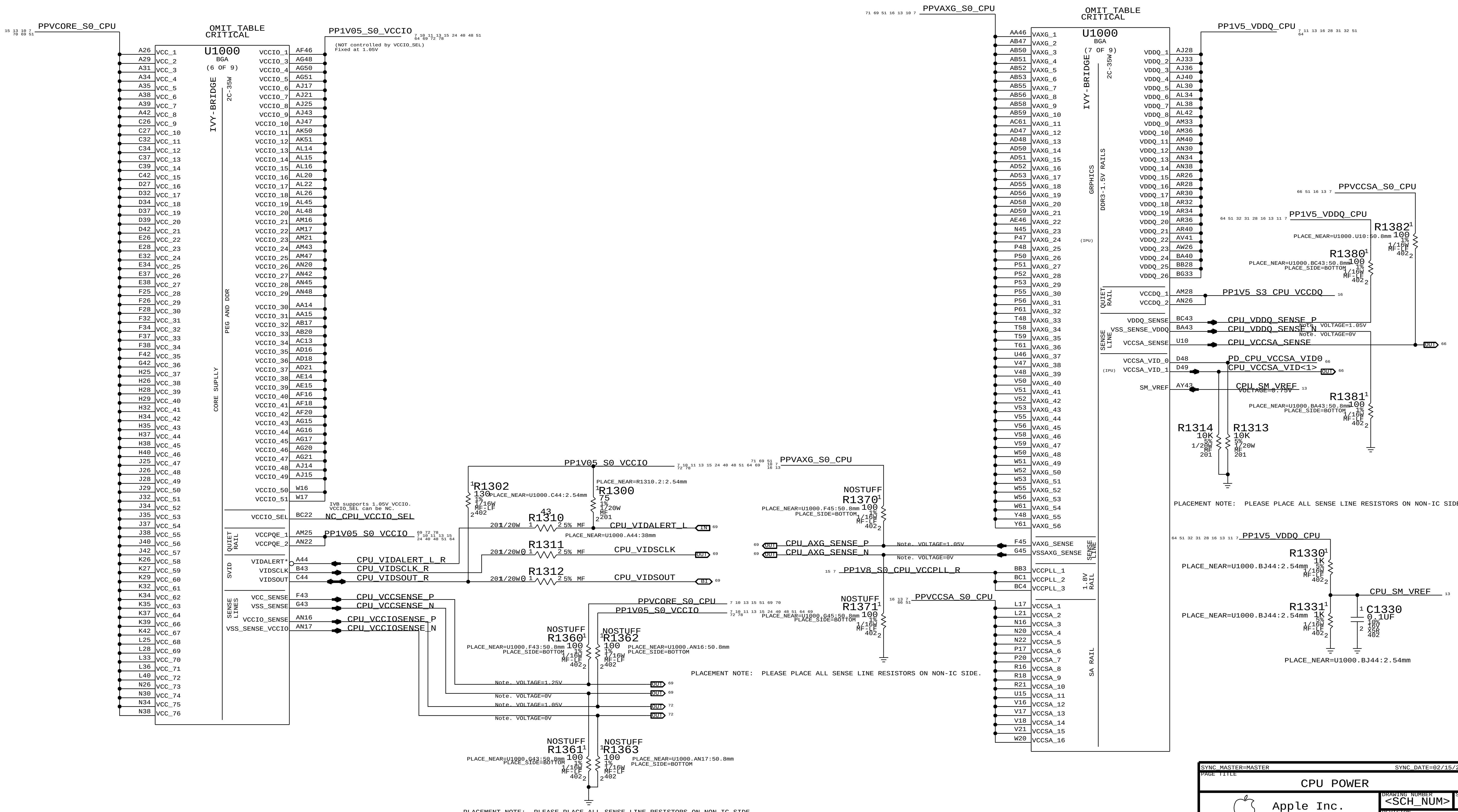


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# CPU POWER

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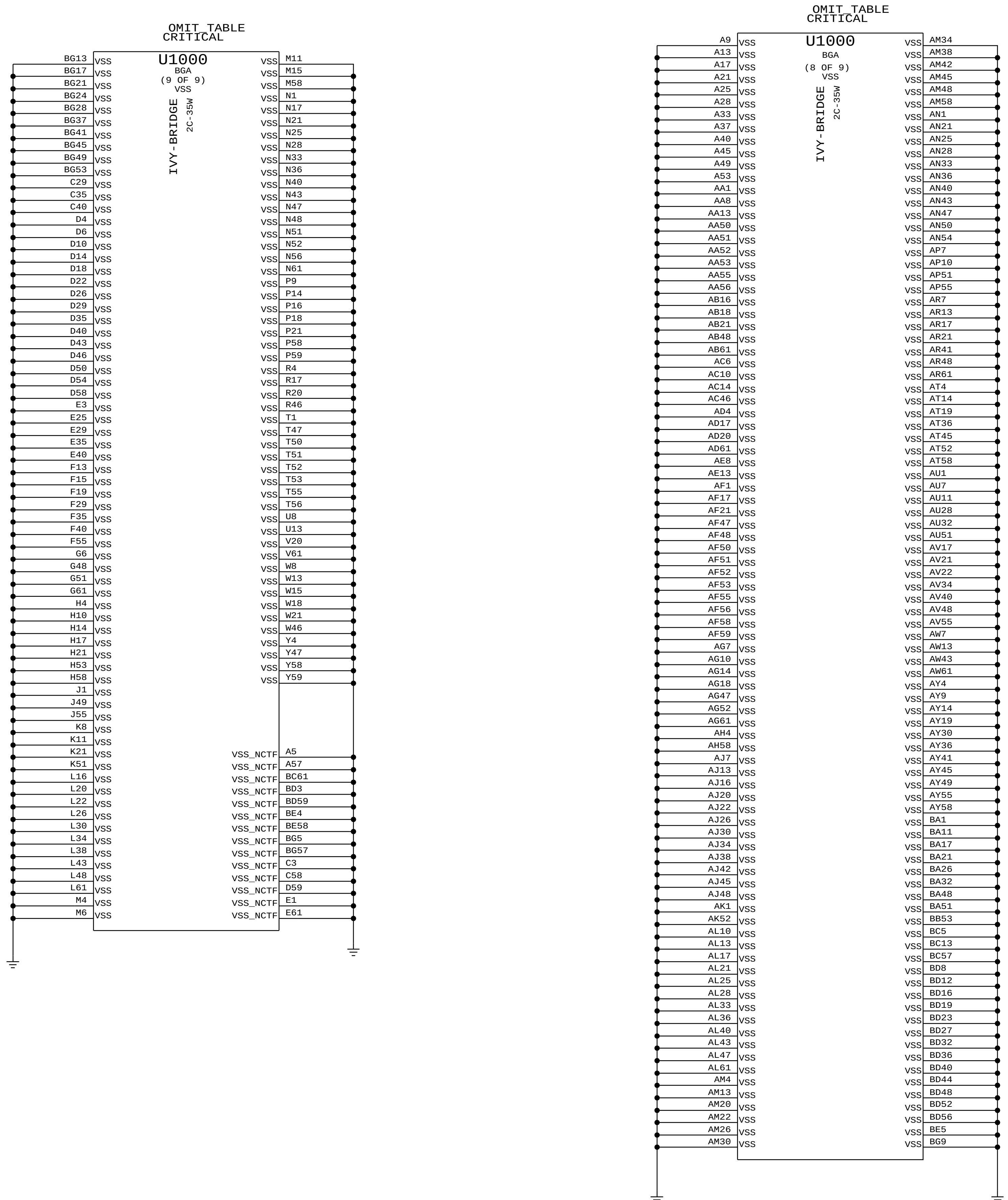


SYNC_MASTER=MASTER SYNC_DATE=02/15/2011

PAGE TITLE

CPU GROUNDS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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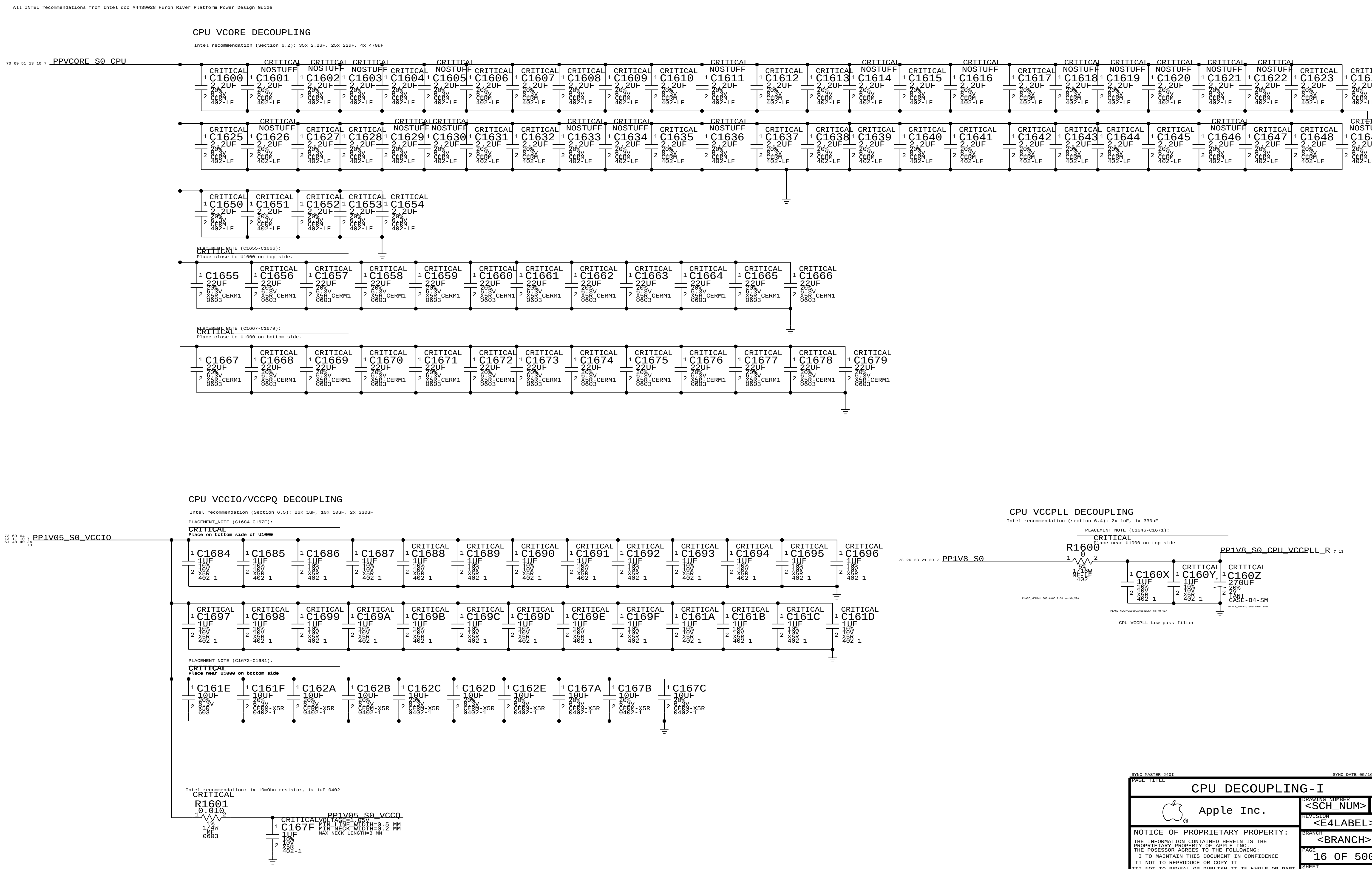


SYNC_MASTER=J40I
SYNC_DATE=05/16/2011

PAGE TITLE

CPU DECOUPLING-I

Apple Inc.

DRAWING NUMBER <SCH_NUM>
SIZE D
REVISION <E4LABEL>
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PAGE 16 OF 500
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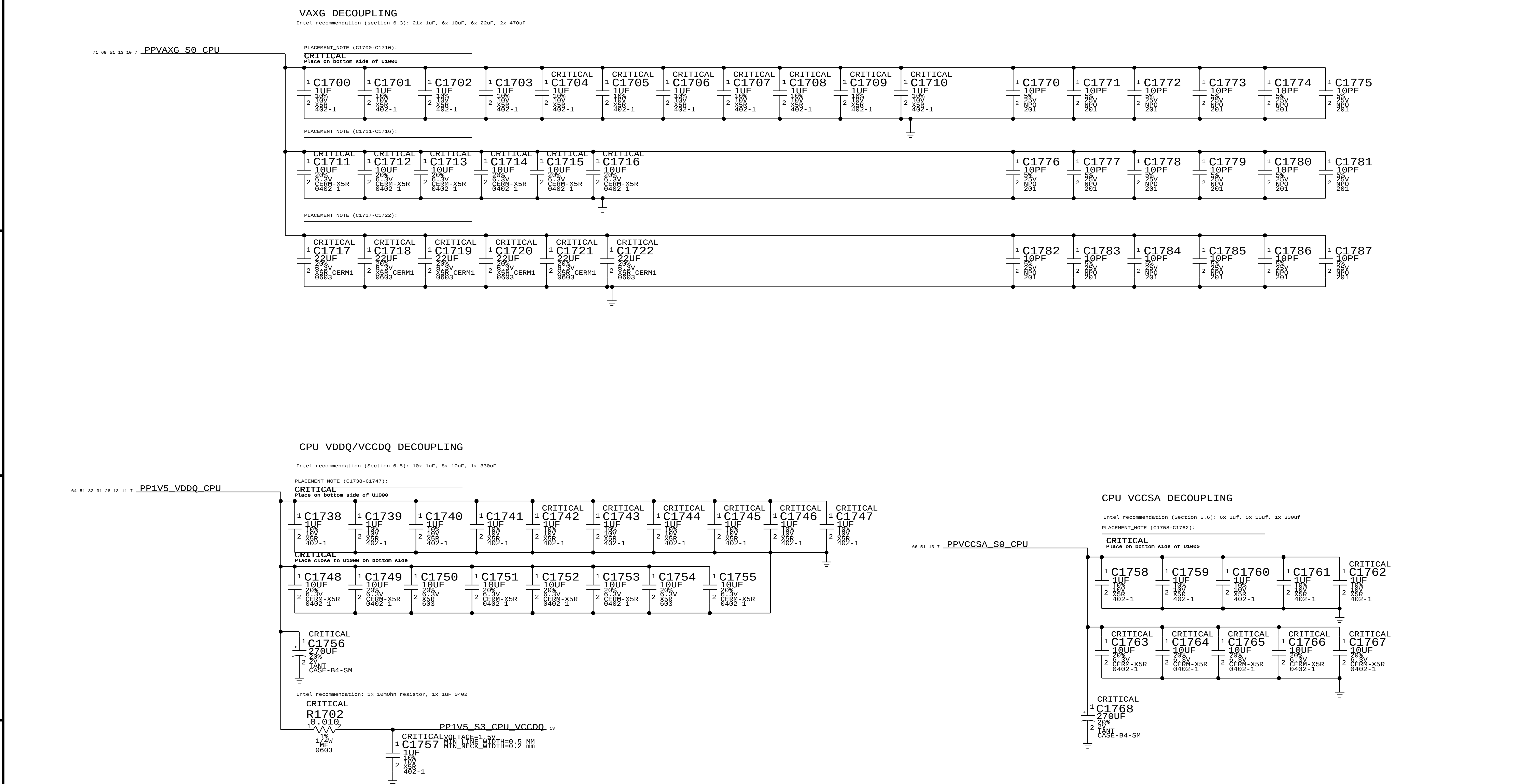


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SYNC_DATE=05/16/2011

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# CPU DECOUPLING-II

Apple Inc.

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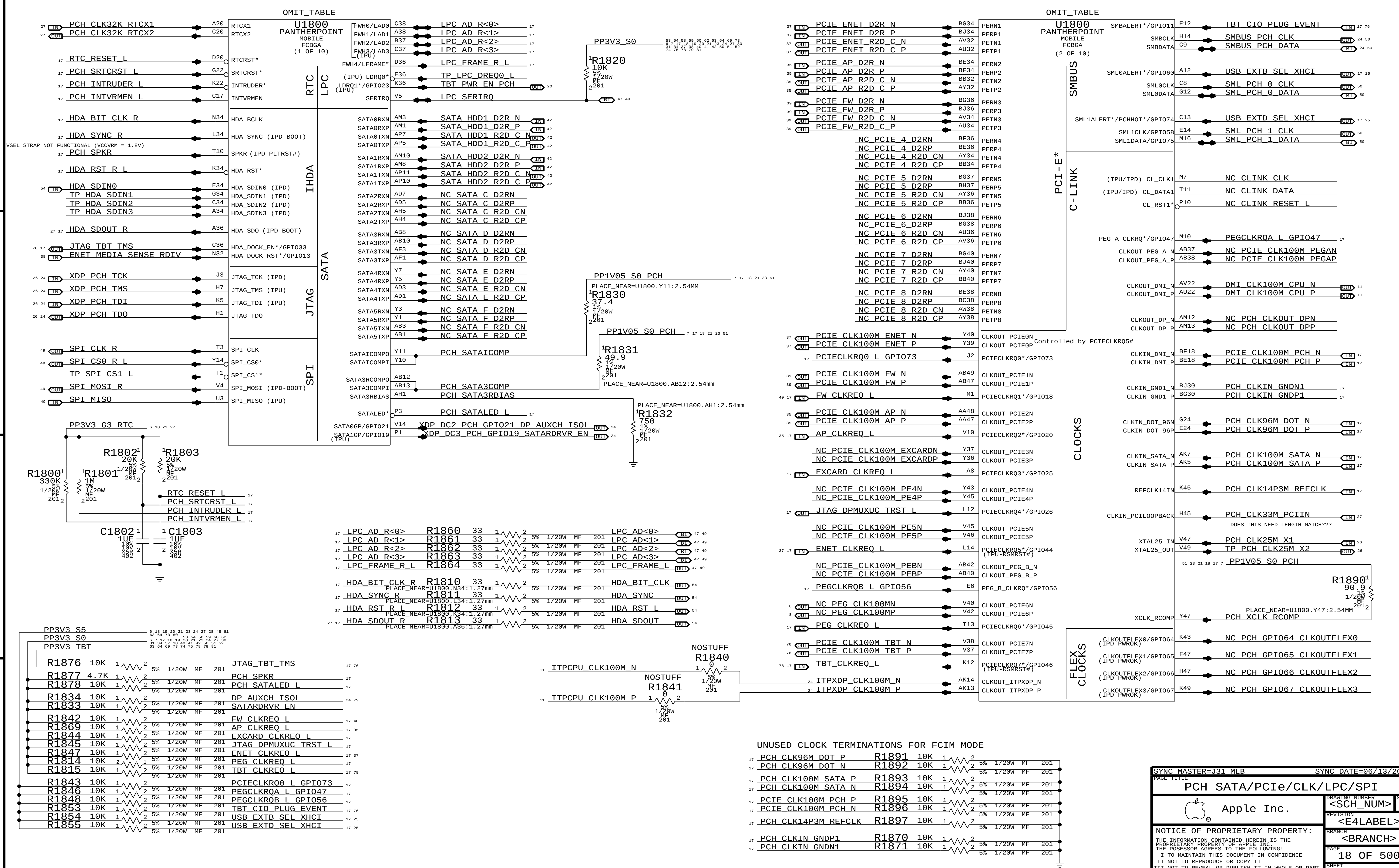


SYNC_MASTER=J31_MLB SYNC_DATE=06/13/2011

PAGE TITLE

PCH SATA/PCIe/CLK/LPC/SPI

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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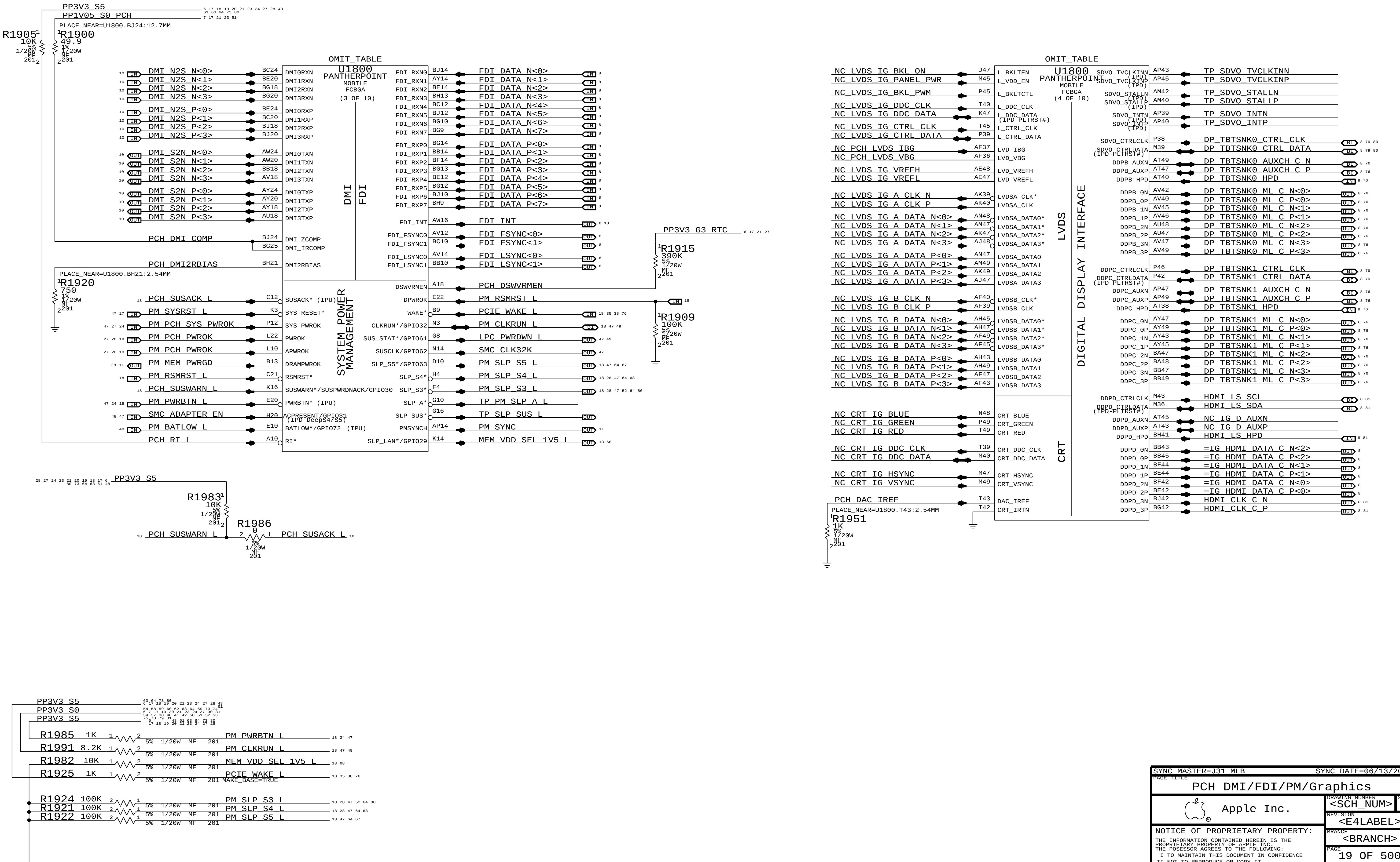


SYNC_MASTER=J31_MLB SYNC_DATE=06/13/2011

PAGE TITLE

# PCH DMI/FDI/PM/Graphics

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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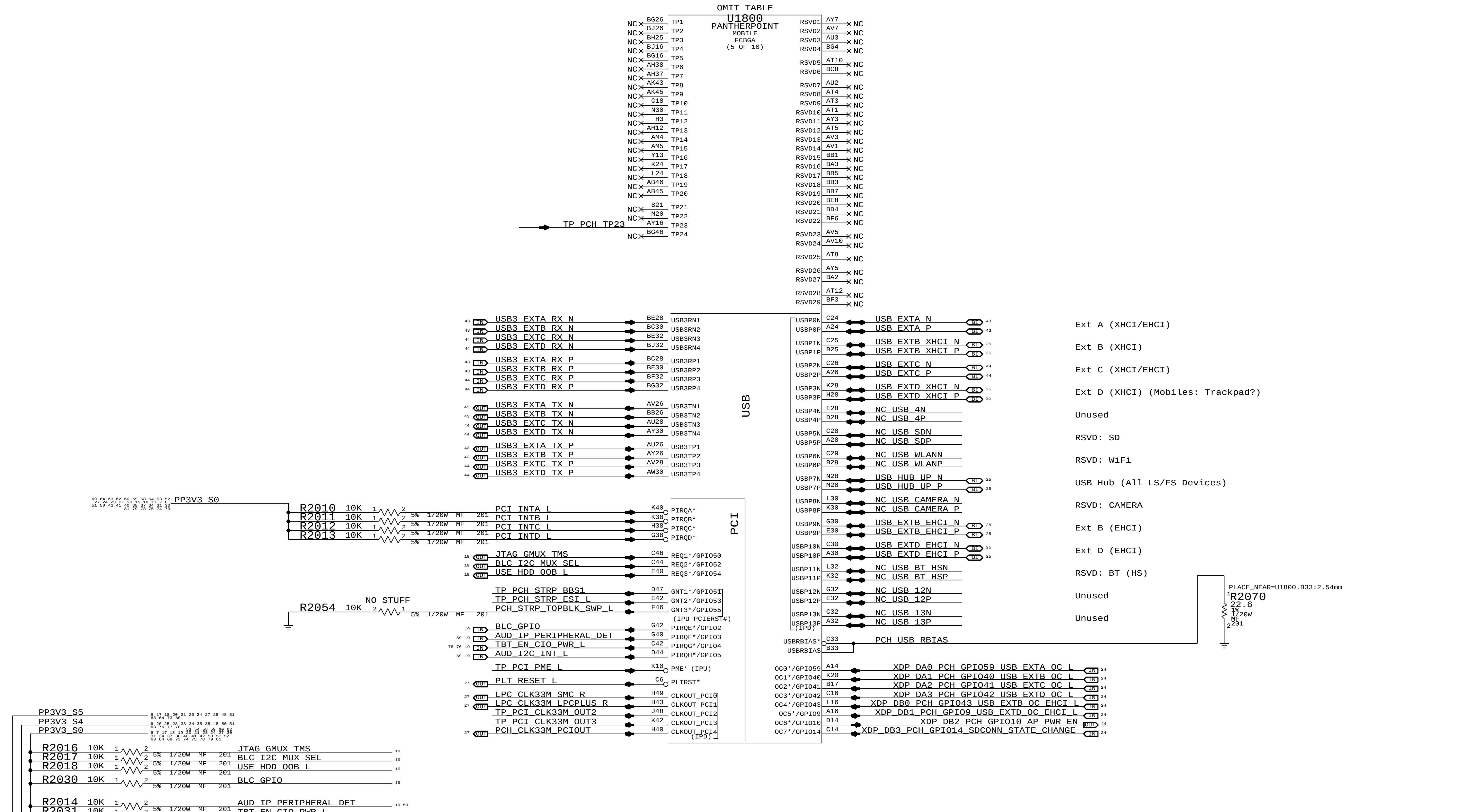


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PAGE TITLE

PCH PCI/USB/TP/RSVD

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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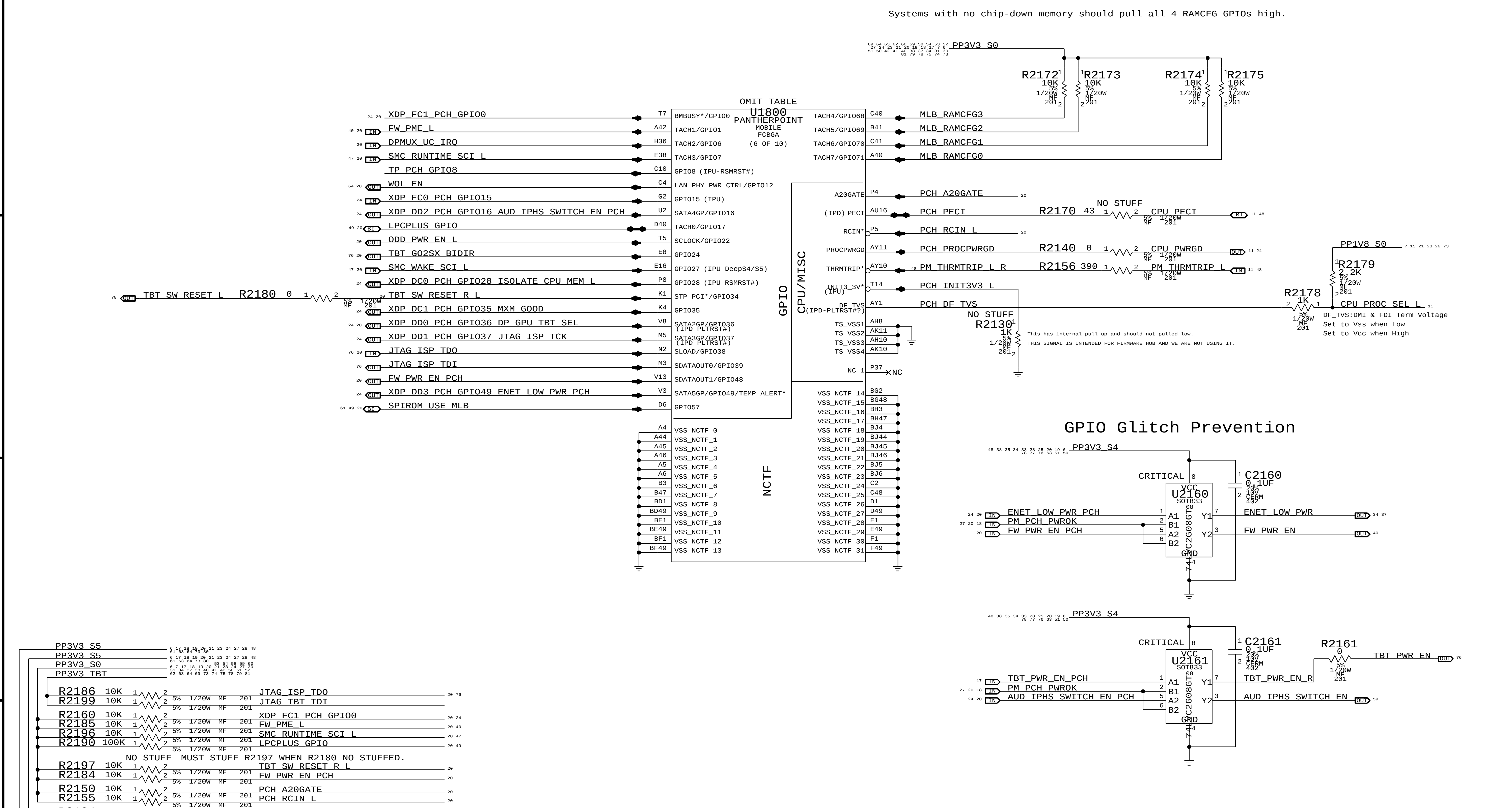


SYNC_MASTER=J31_MLB SYNC_DATE=06/13/2011

PAGE TITLE

PCH GPIO/MISC/NCTF

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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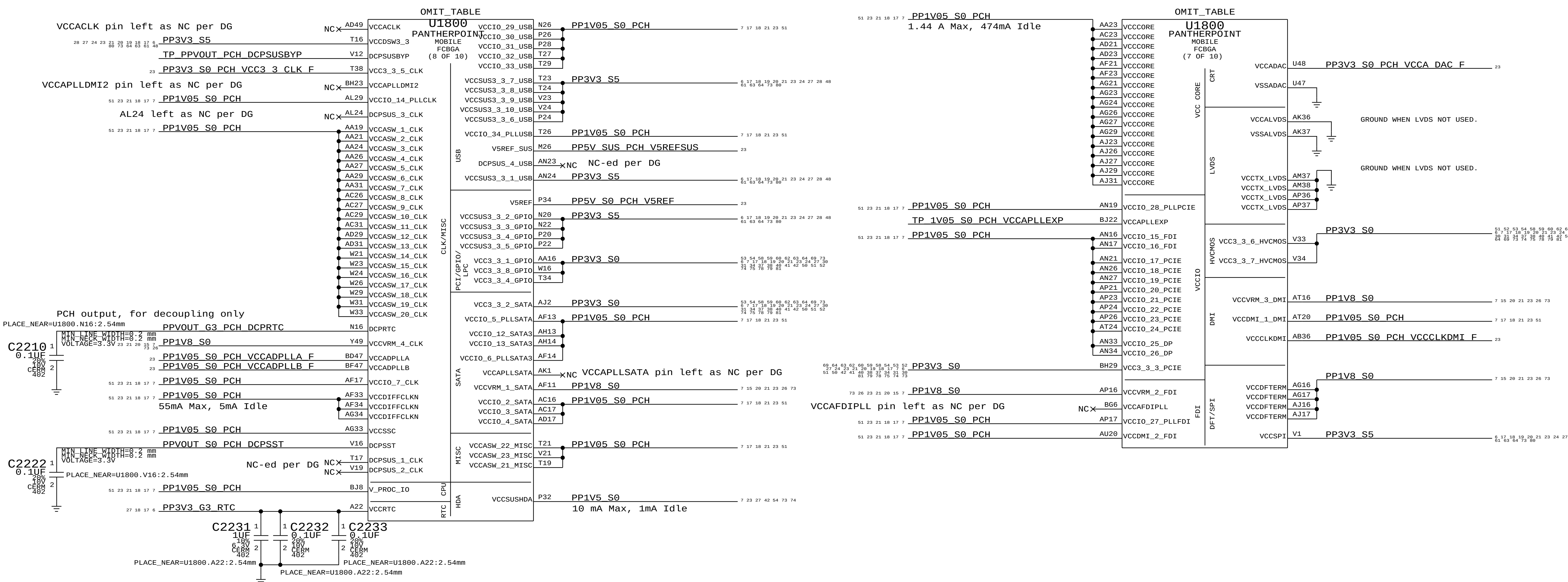


| | | |
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| PAGE TITLE: PCH POWER | | |
| Apple Inc. | DRAWING NUMBER: <SCH_NUM> | SIZE: D |
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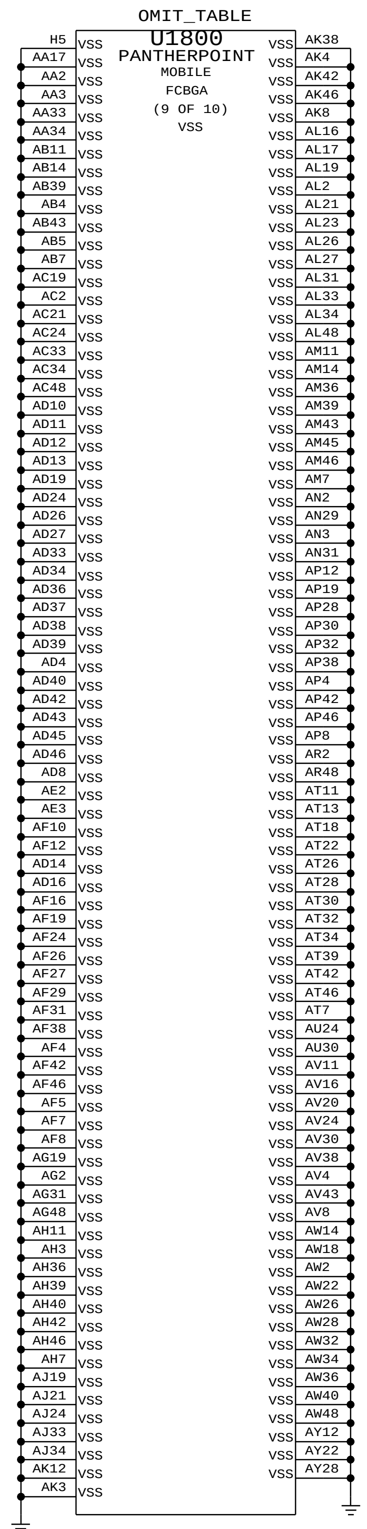
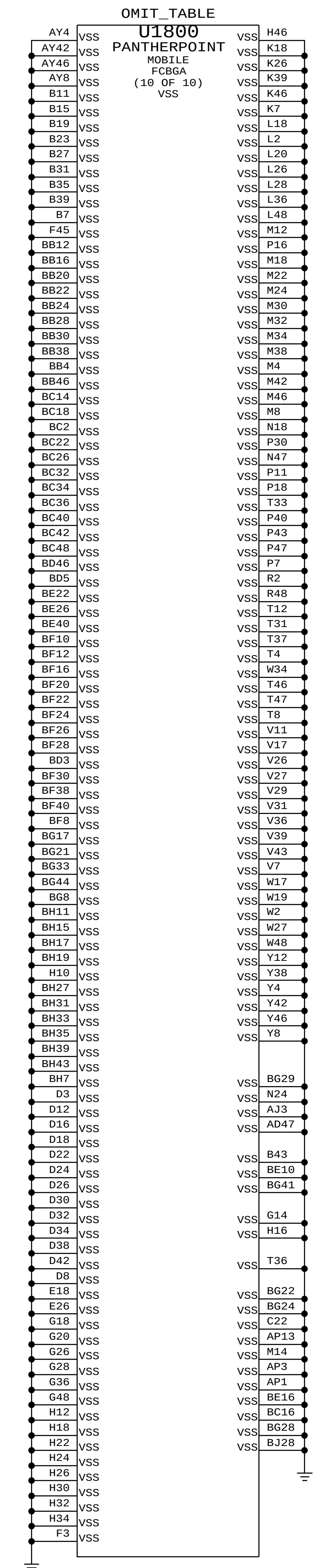
| | PAGE: 22 OF 500 | |
| | SHEET: 21 OF 83 | |




SYNC_MASTER=J31_MLB SYNC_DATE=06/13/2011

PAGE TITLE

# PCH GROUNDS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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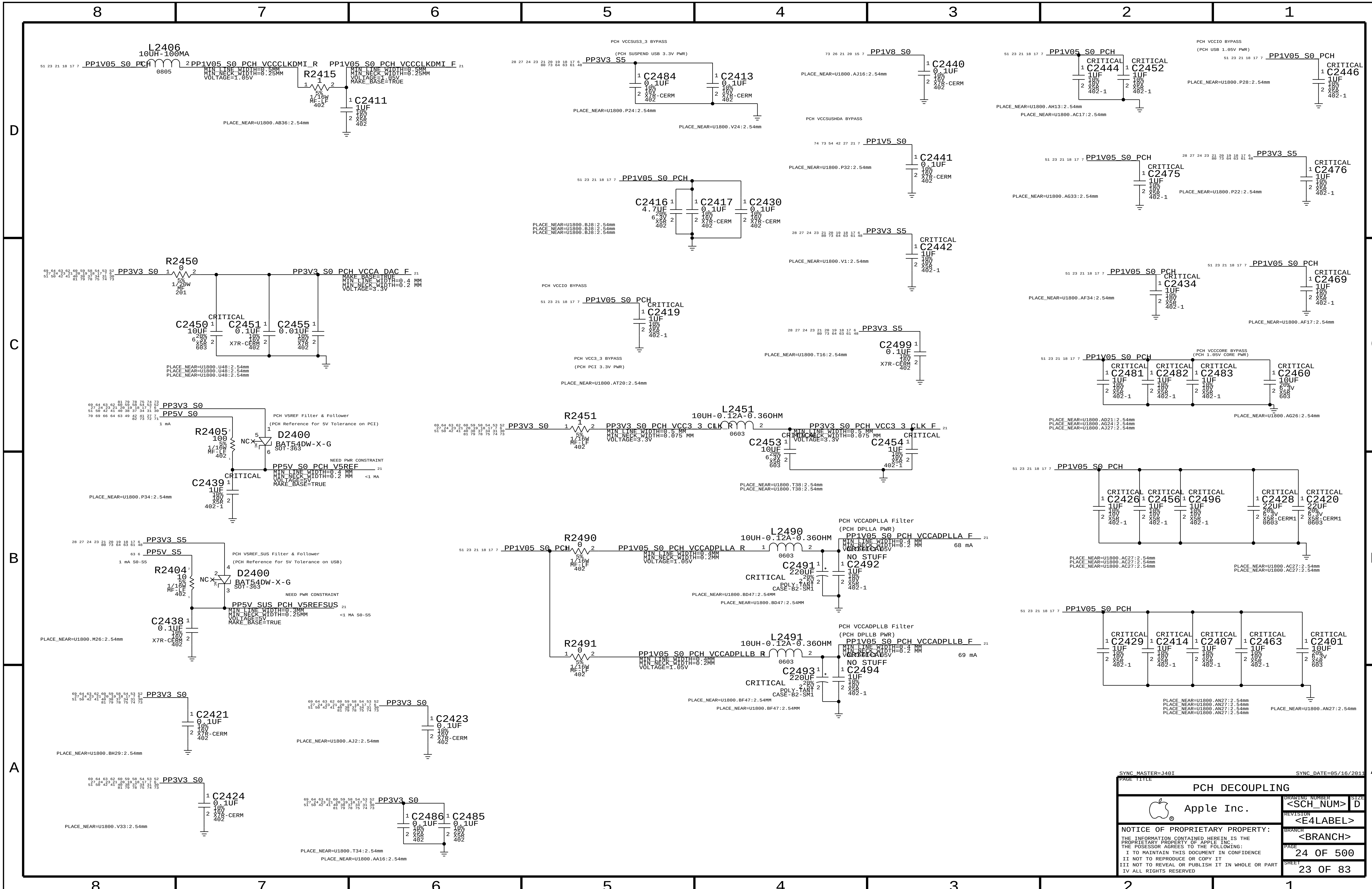

L2406
10UH-100MA
0805
PP1V05_S0_PCH
PP1V05_S0_PCH_VCCCLKDMI_R
PP1V05_S0_PCH_VCCCLKDMI_F
MIN_LINE_WIDTH=0.5MM
MIN_NECK_WIDTH=0.25MM
VOLTAGE=1.05V
MAKE_BASE=TRUE
R2415
1
5%
1/16W
MF-LF
402
C2411
1UF
10%
10V
X5R
402
PLACE_NEAR=U1800.AB36:2.54mm
PCH VCCSUS3_3 BYPASS
(PCH SUSPEND USB 3.3V PWR)
PP3V3_S5
C2484
0.1UF
10%
16V
X7R-CERM
402
C2413
0.1UF
10%
16V
X7R-CERM
402
PLACE_NEAR=U1800.P24:2.54mm
PLACE_NEAR=U1800.V24:2.54mm
PP1V8_S0
C2440
0.1UF
10%
16V
X7R-CERM
402
PLACE_NEAR=U1800.AJ16:2.54mm
PCH VCCSUSHDA BYPASS
PP1V5_S0
C2441
0.1UF
10%
16V
X7R-CERM
402
PLACE_NEAR=U1800.P32:2.54mm
PP1V05_S0_PCH
CRITICAL
C2444
C2452
1UF
10%
10V
X5R
402-1
PLACE_NEAR=U1800.AH13:2.54mm
PLACE_NEAR=U1800.AC17:2.54mm
PCH VCCIO BYPASS
(PCH USB 1.05V PWR)
C2446
PLACE_NEAR=U1800.P28:2.54mm
C2475
PLACE_NEAR=U1800.AG33:2.54mm
PP3V3_S5
C2476
PLACE_NEAR=U1800.P22:2.54mm
C2416
4.7UF
20%
6.3V
X5R
402
C2417
0.1UF
10%
16V
X7R-CERM
402
C2430
0.1UF
PLACE_NEAR=U1800.BJ8:2.54mm
C2442
PLACE_NEAR=U1800.V1:2.54mm
R2450
0
5%
1/20W
MF
201
PP3V3_S0
PP3V3_S0_PCH_VCCA_DAC_F
MAKE_BASE=TRUE
MIN_LINE_WIDTH=0.4 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=3.3V
C2450
10UF
20%
6.3V
X5R
603
C2451
0.1UF
10%
16V
X7R-CERM
402
C2455
0.01UF
10%
50V
X7R
402
PLACE_NEAR=U1800.U48:2.54mm
PCH VCCIO BYPASS
C2419
PLACE_NEAR=U1800.AT20:2.54mm
PCH VCC3_3 BYPASS
(PCH PCI 3.3V PWR)
C2499
0.1UF
PLACE_NEAR=U1800.T16:2.54mm
C2434
PLACE_NEAR=U1800.AF34:2.54mm
C2469
PLACE_NEAR=U1800.AF17:2.54mm
PCH VCCCORE BYPASS
(PCH 1.05V CORE PWR)
C2481
C2482
C2483
C2460
10UF
20%
6.3V
X5R
603
PLACE_NEAR=U1800.AD21:2.54mm
PLACE_NEAR=U1800.AG24:2.54mm
PLACE_NEAR=U1800.AJ27:2.54mm
PLACE_NEAR=U1800.AG26:2.54mm
PP5V_S0
R2405
100
5%
1/16W
MF-LF
402
1 mA
PCH V5REF Filter & Follower
(PCH Reference for 5V Tolerance on PCI)
D2400
BAT54DW-X-G
SOT-363
NC
NEED PWR CONSTRAINT
PP5V_S0_PCH_V5REF
MIN_LINE_WIDTH=0.4 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=5V
MAKE_BASE=TRUE
<1 MA
C2439
1UF
PLACE_NEAR=U1800.P34:2.54mm
R2451
1
5%
1/16W
MF-LF
402
L2451
10UH-0.12A-0.360HM
0603
PP3V3_S0_PCH_VCC3_3_CLK_R
PP3V3_S0_PCH_VCC3_3_CLK_F
MIN_LINE_WIDTH=0.5 MM
MIN_NECK_WIDTH=0.075 MM
VOLTAGE=3.3V
C2453
10UF
C2454
1UF
PLACE_NEAR=U1800.T38:2.54mm
PP1V05_S0_PCH
C2426
C2456
C2496
C2428
C2420
22UF
20%
6.3V
X5R-CERM1
0603
PLACE_NEAR=U1800.AC27:2.54mm
PP3V3_S5
PP5V_S5
1 mA S0-S5
PCH V5REF_SUS Filter & Follower
(PCH Reference for 5V Tolerance on USB)
R2404
10
5%
1/16W
MF-LF
402
PP5V_SUS_PCH_V5REFSUS
MIN_LINE_WIDTH=0.3MM
MIN_NECK_WIDTH=0.25MM
VOLTAGE=5V
MAKE_BASE=TRUE
<1 MA S0-S5
C2438
0.1UF
10%
16V
X7R-CERM
402
PLACE_NEAR=U1800.M26:2.54mm
R2490
0
PP1V05_S0_PCH_VCCADPLLA_R
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.2MM
VOLTAGE=1.05V
L2490
10UH-0.12A-0.360HM
0603
PCH VCCADPLLA Filter
(PCH DPLLA PWR)
PP1V05_S0_PCH_VCCADPLLA_F
68 mA
C2491
220UF
20%
2.5V
POLY-TANT
CASE-B2-SM1
NO STUFF
C2492
1UF
PLACE_NEAR=U1800.BD47:2.54mm
R2491
0
PP1V05_S0_PCH_VCCADPLLB_R
L2491
PCH VCCADPLLB Filter
(PCH DPLLB PWR)
PP1V05_S0_PCH_VCCADPLLB_F
69 mA
C2493
220UF
C2494
PLACE_NEAR=U1800.BF47:2.54mm
C2429
C2414
C2407
C2463
C2401
PLACE_NEAR=U1800.AN27:2.54mm
C2421
0.1UF
PLACE_NEAR=U1800.BH29:2.54mm
C2423
0.1UF
PLACE_NEAR=U1800.AJ2:2.54mm
C2424
0.1UF
PLACE_NEAR=U1800.V33:2.54mm
C2486
C2485
PLACE_NEAR=U1800.T34:2.54mm
PLACE_NEAR=U1800.AA16:2.54mm
SYNC_MASTER=J40I
SYNC_DATE=05/16/2011
PAGE TITLE
PCH DECOUPLING
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
PAGE
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SHEET
23 OF 83
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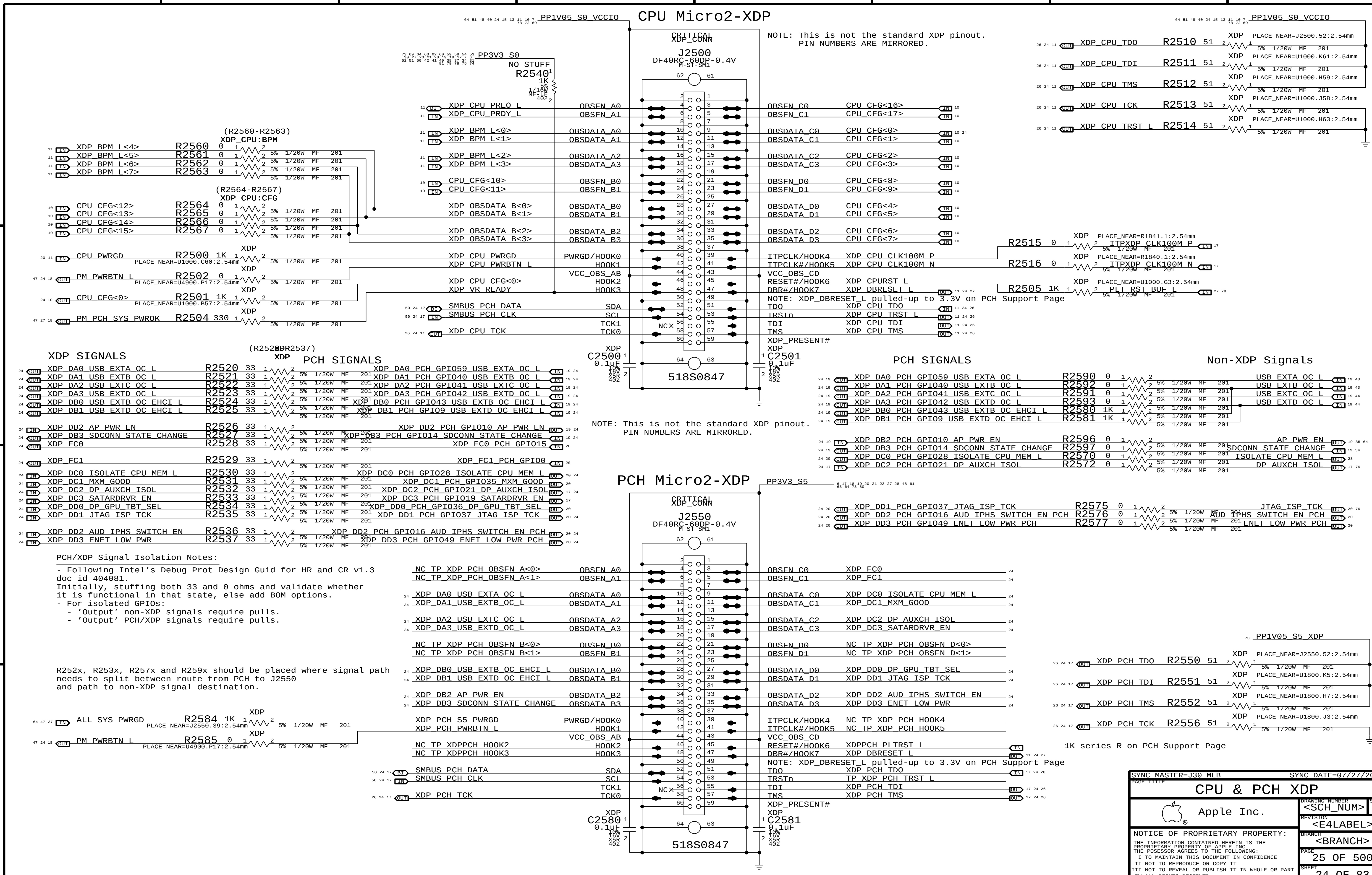


SYNC_MASTER=J30_MLB SYNC_DATE=07/27/2011

PAGE TITLE

CPU & PCH XDP

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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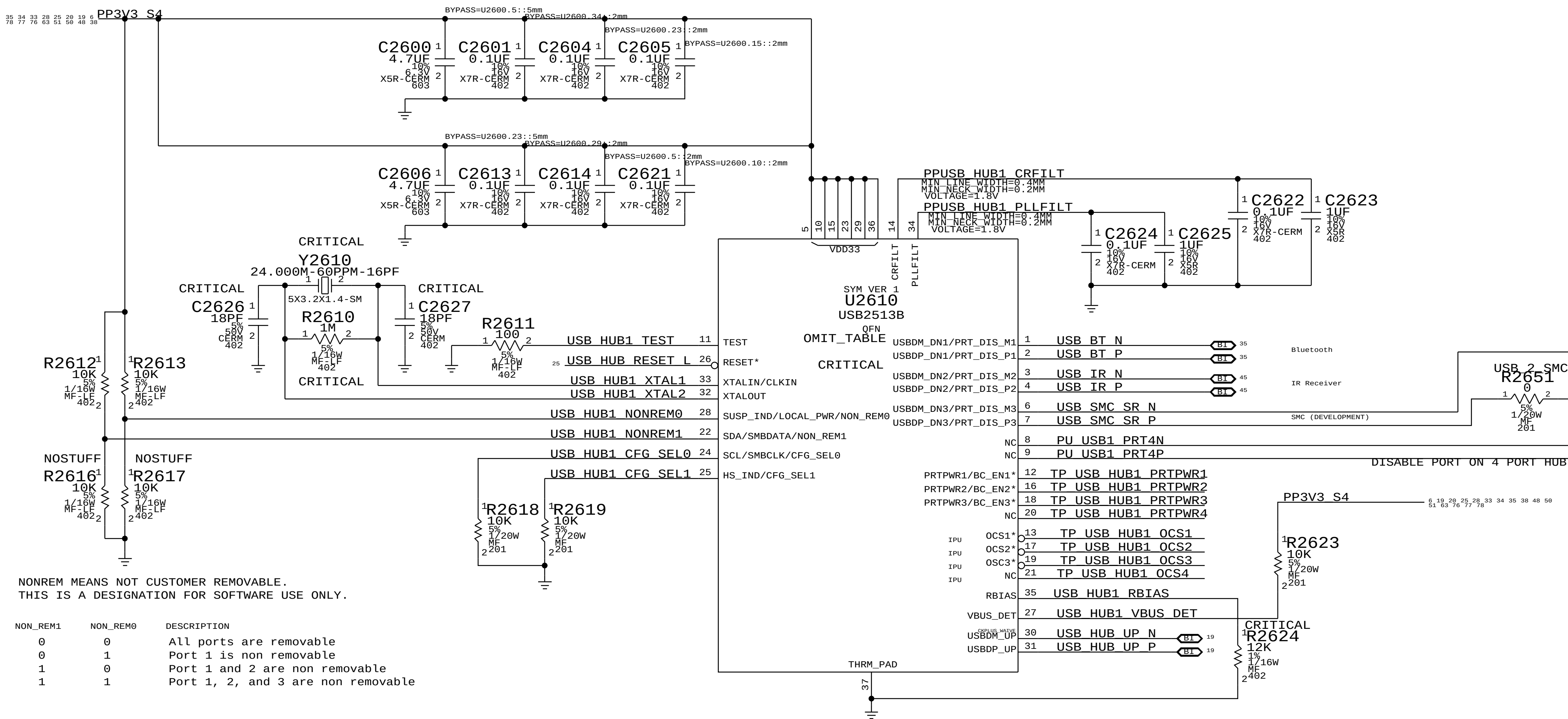
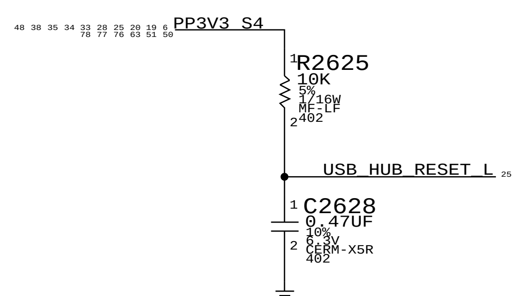



## BOM TABLE

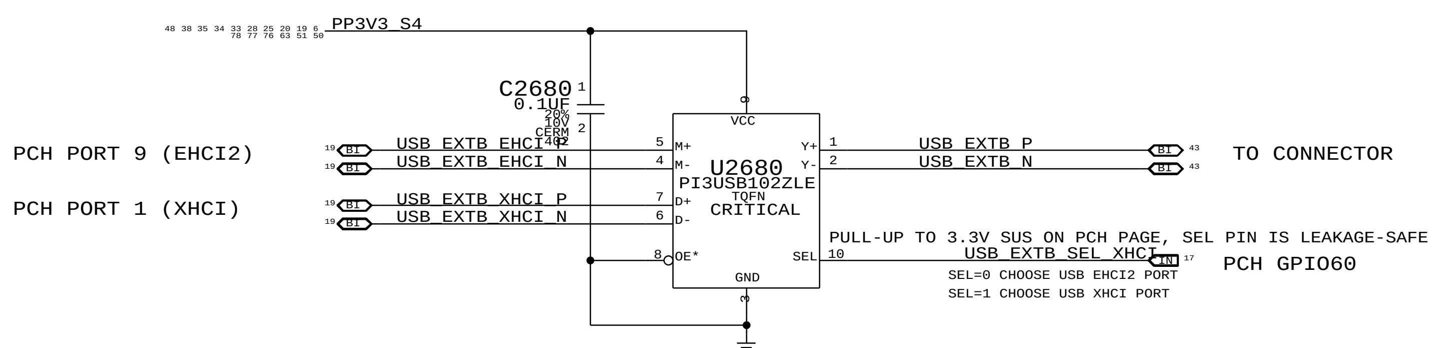
| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 338S0824 | | SMSC USB2514B | | | |
| 338S0923 | 1 | SMSC USB2513B | U2600 | CRITICAL | USB_SMC_LINK:YES |
| 338S0983 | 1 | SMSC USB2512B | U2600 | CRITICAL | USB_SMC_LINK:NO |

NONREM MEANS NOT CUSTOMER REMOVABLE.
THIS IS A DESIGNATION FOR SOFTWARE USE ONLY.

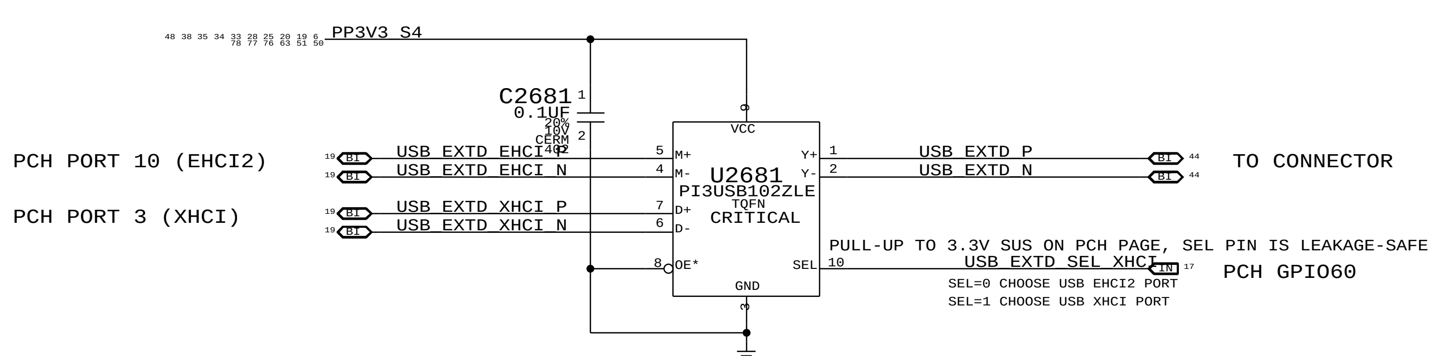
| NON_REM1 | NON_REM0 | DESCRIPTION |
|---|---|---|
| 0 | 0 | All ports are removable |
| 0 | 1 | Port 1 is non removable |
| 1 | 0 | Port 1 and 2 are non removable |
| 1 | 1 | Port 1, 2, and 3 are non removable |




## USB XHCI/EHCI2 PORT MUX FOR EXT B




## USB XHCI/EHCI2 PORT MUX FOR EXT B




SYNC_MASTER=J40I SYNC_DATE=05/16/2011
PAGE TITLE
USB HUB, MUXES
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
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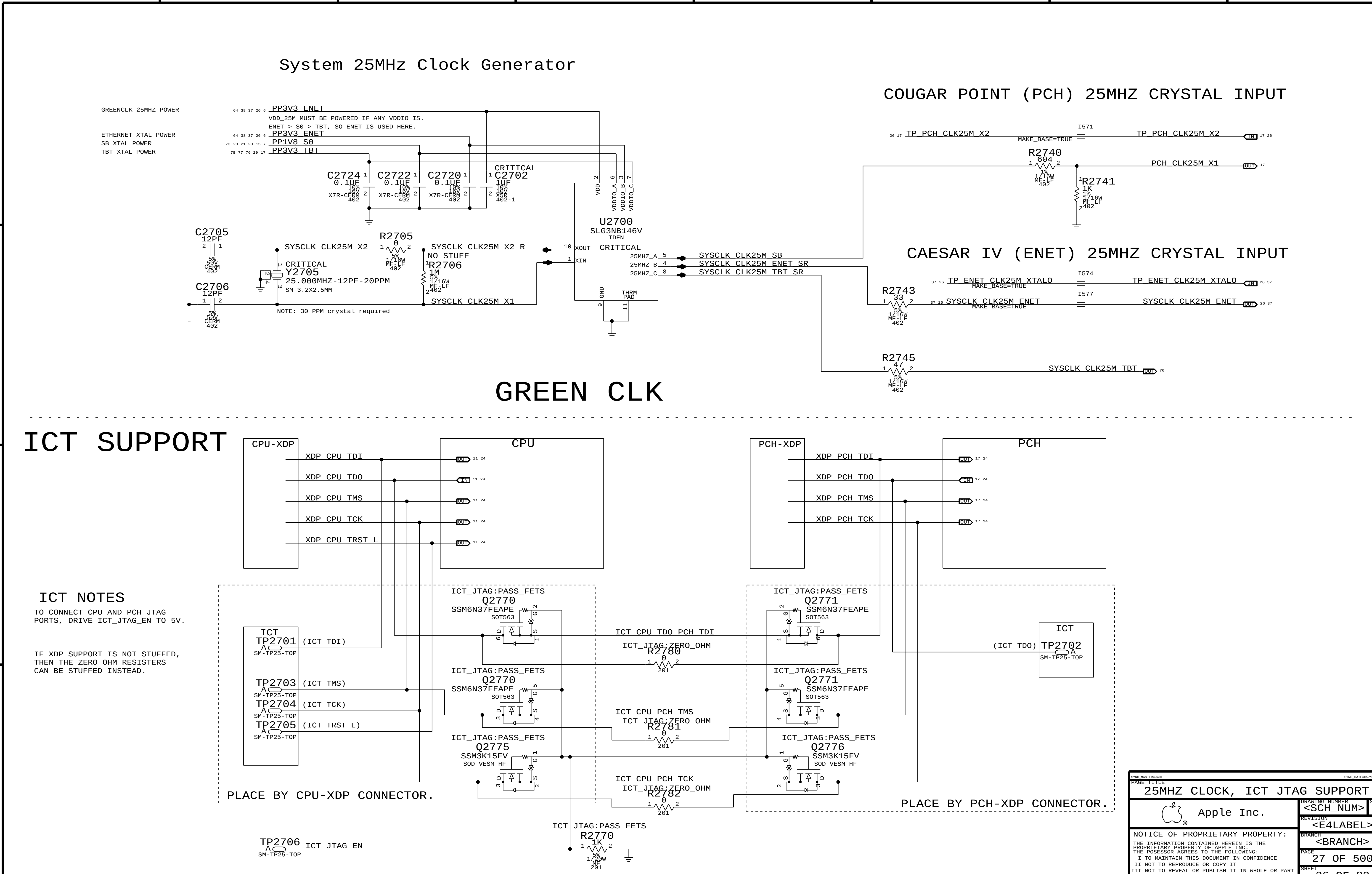



25MHZ CLOCK, ICT JTAG SUPPORT

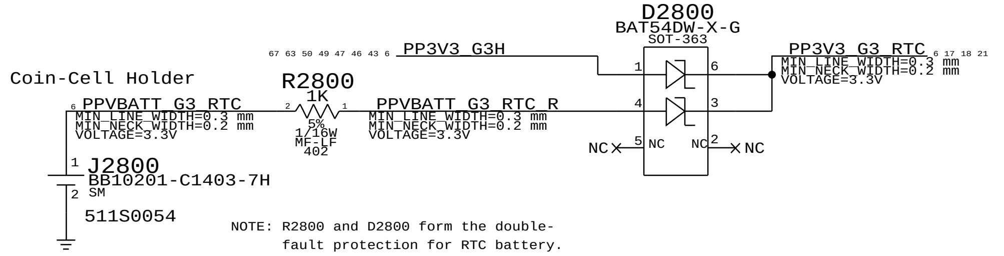
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

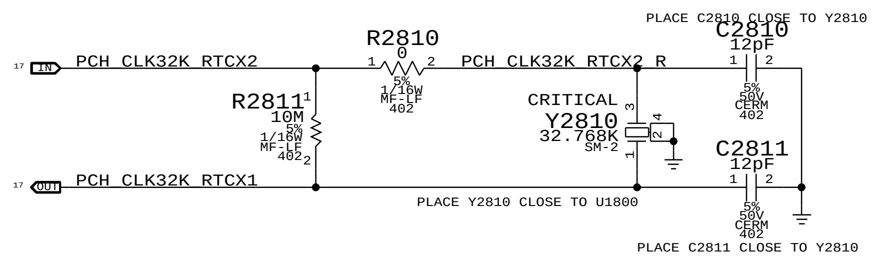
## RTC Power Sources




| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 742-0062 | 1 | BAT,COIN,LI/MAG,BR2032,3V,190MAH | BAT | CRITICAL | |

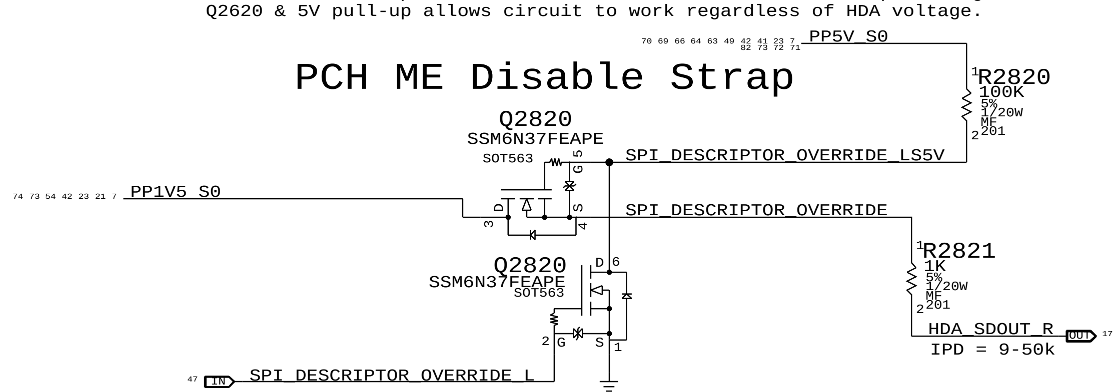
| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 511S0074 | 511S0054 | ? | ? | BATTERY HOLDER WITHOUT ALIGNMENT PEGS |

## PCH RTC Crystal

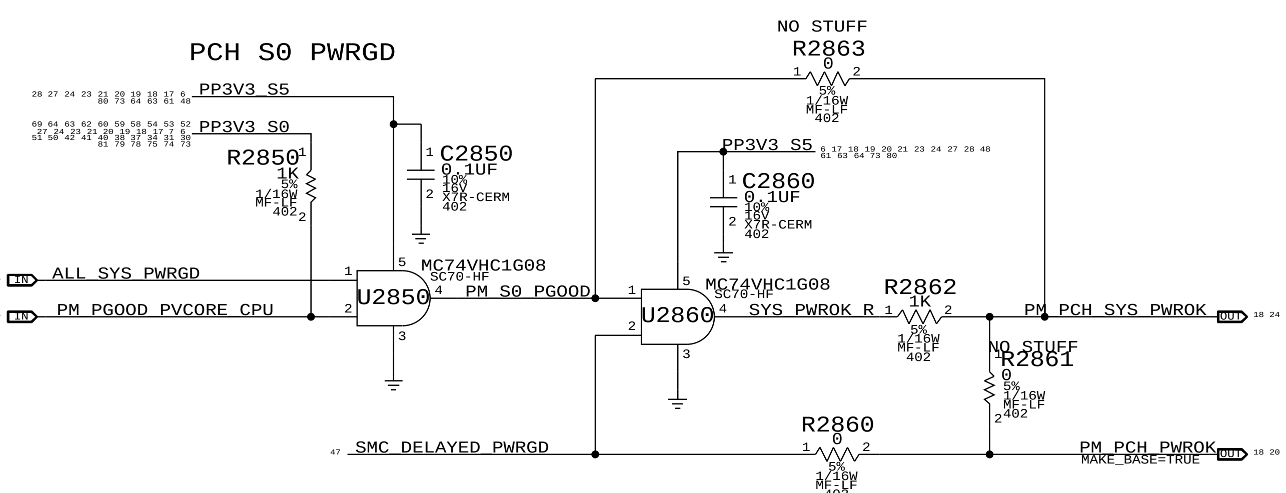



PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.
Q2620 & 5V pull-up allows circuit to work regardless of HDA voltage.

## PCH ME Disable Strap

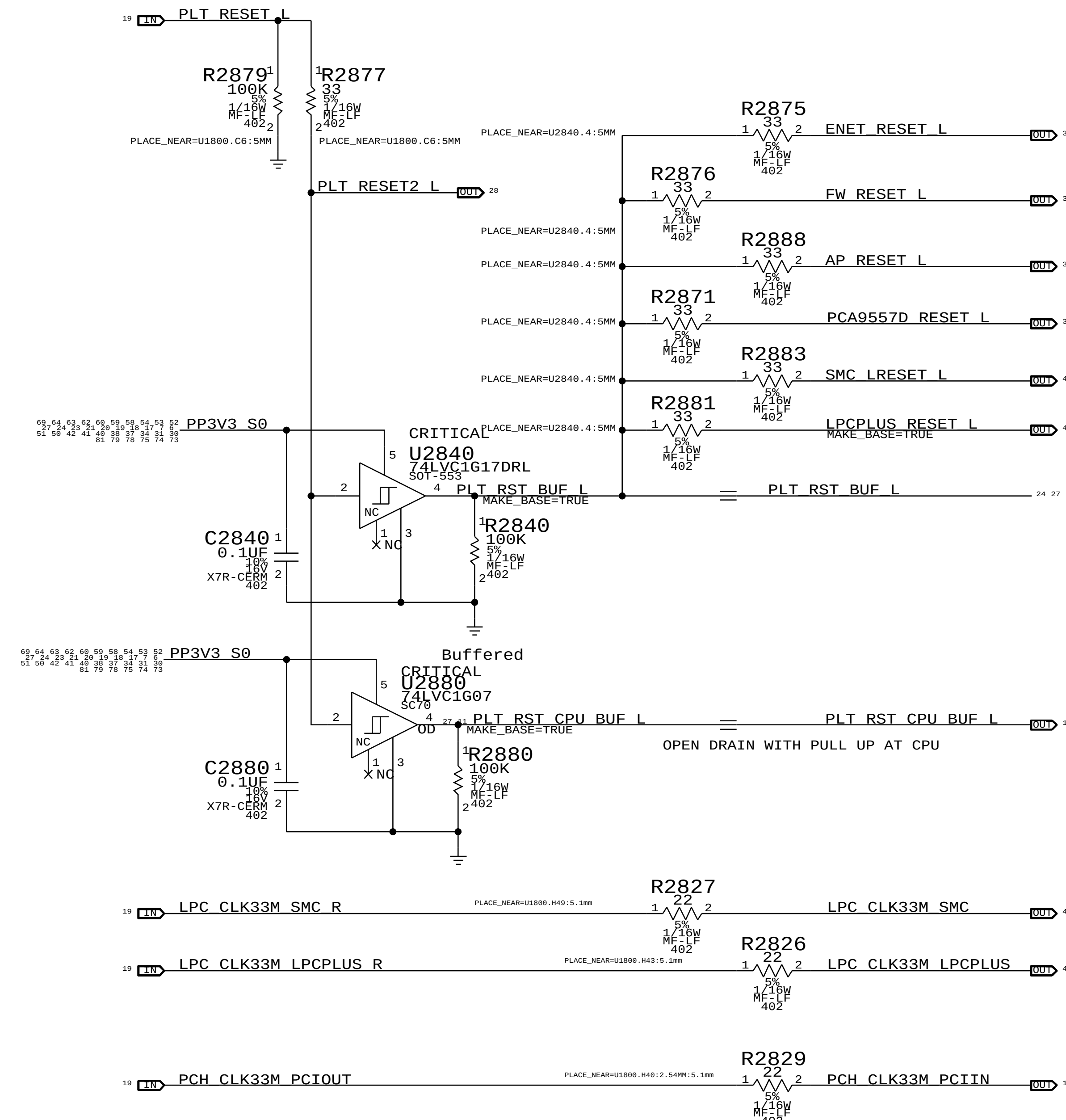



## PCH S0 PWRGD

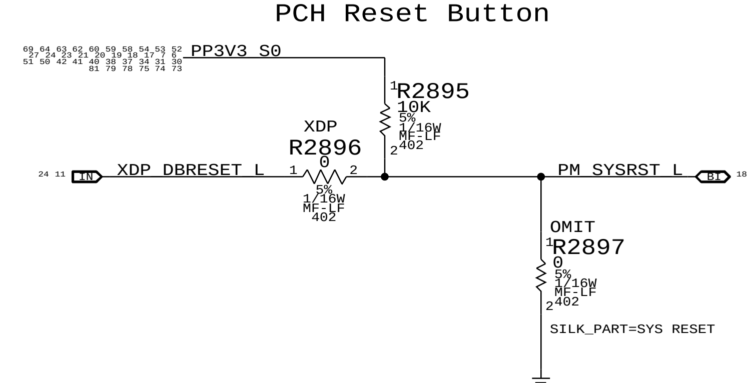



## Platform Reset Connections

Unbuffered




## PCH Reset Button




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

RTC SUPPORT, RESETS

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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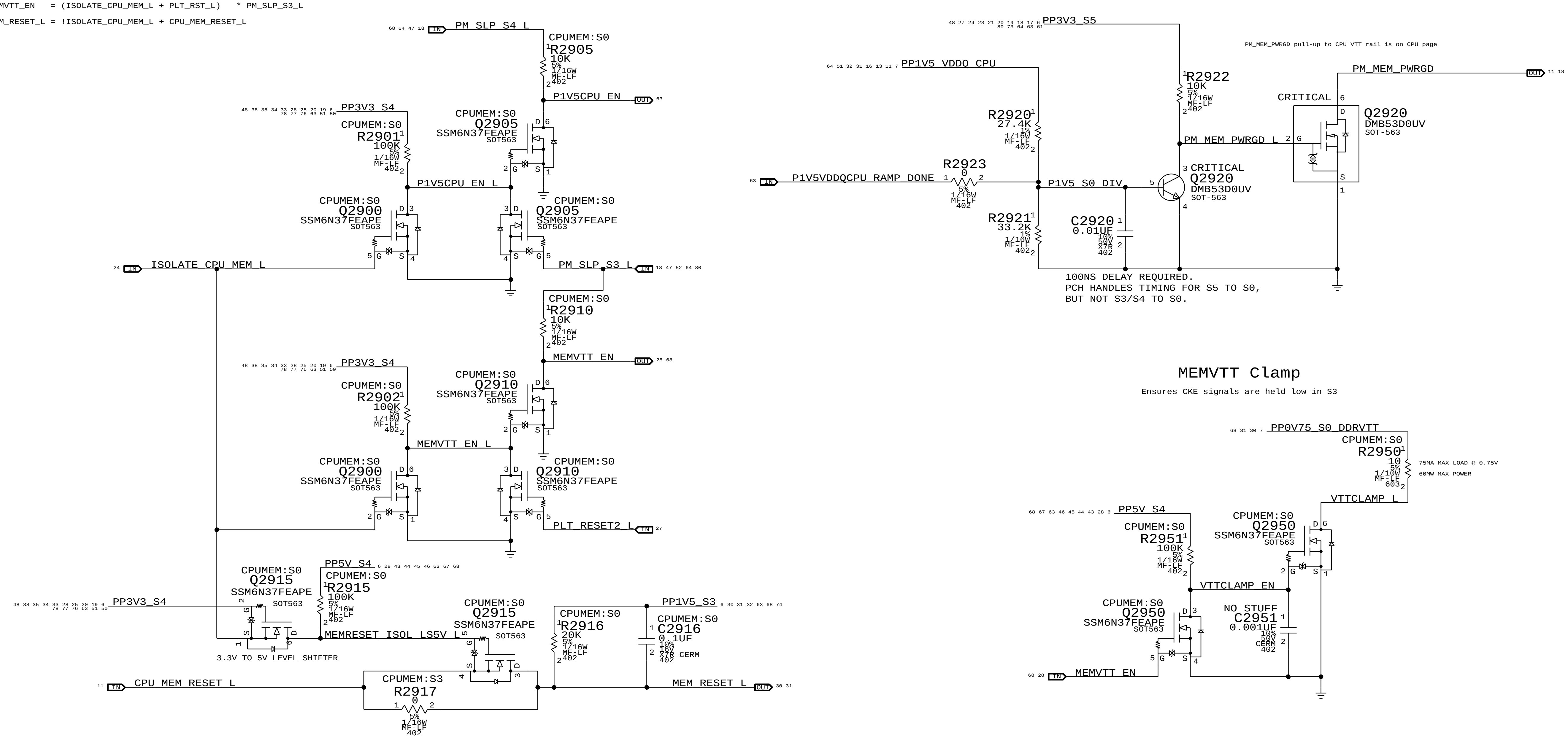
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

## 1V5 S0 "PGOOD" for CPU




| | Step | ISOLATE_CPU_MEM_L | PLT_RESET_L | PM_SLP_S3_L | PM_SLP_S4_L | CPU_MEM_RESET_L | MEM_RESET_L | MEMVTT_EN | P1V5CPU_EN |
|---|---|---|---|---|---|---|---|---|---|
| S0 | 0 | 1 | 1 | 1 | 1 | 1 | CPU_MEM_RESET_L | 1 | 1 |
| to | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| | 2 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 |
| S3 | 3 | 0 | 0 | 0 | 1 | X | 1 | 0 | 0 |
| | 4 | 0 | 0 | 1 | 1 | X | 1 | 0 | 1 |
| to | 5 | 0 | 1 | 1 | 1 | 0 (*) | 1 | 1 | 1 |
| | 6 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| S0 | 7 | 1 | 1 | 1 | 1 | 1 | CPU_MEM_RESET_L | 1 | 1 |

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
CPU Memory S3 Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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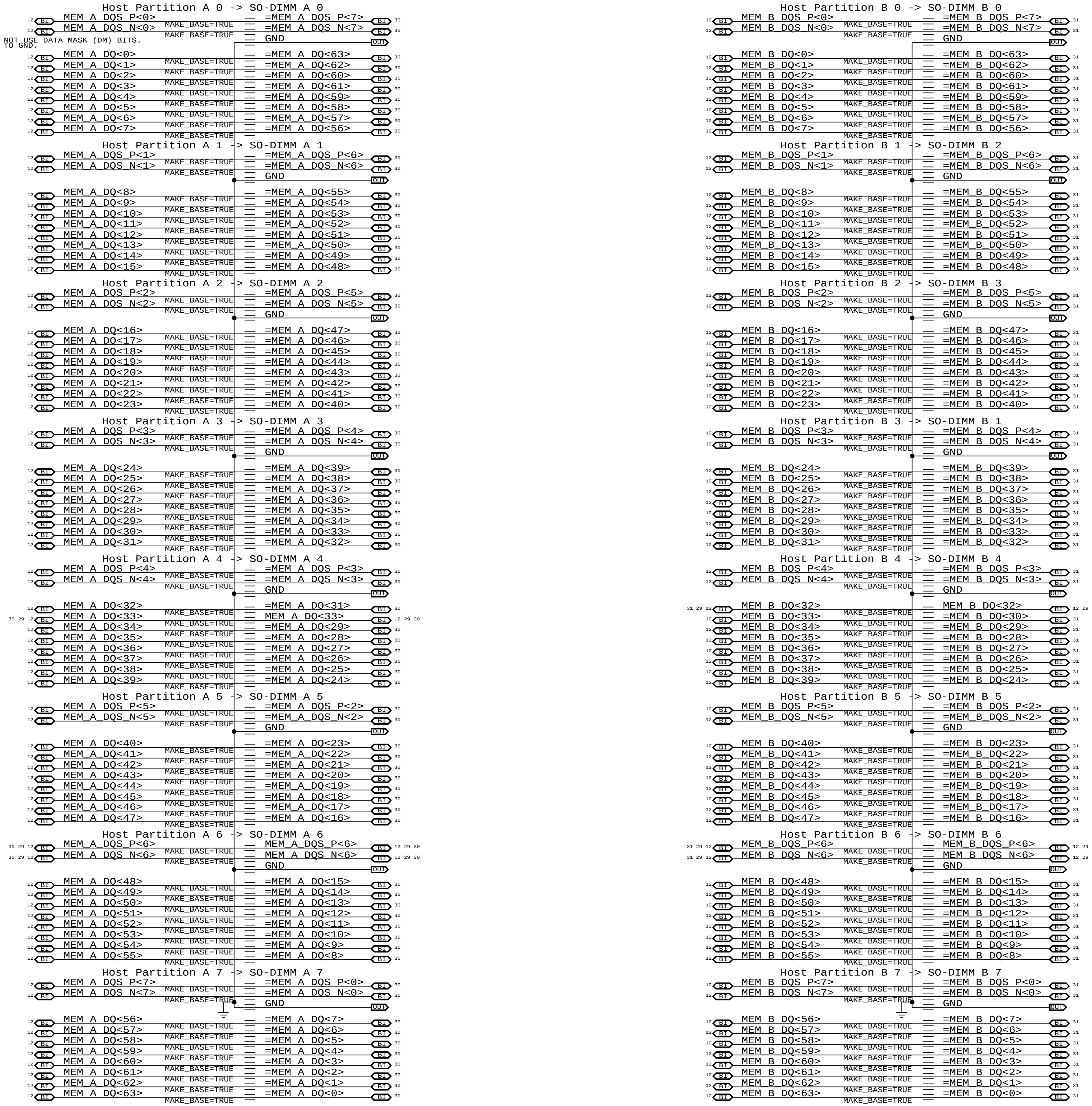


SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# SO-DIMM Pinswaps

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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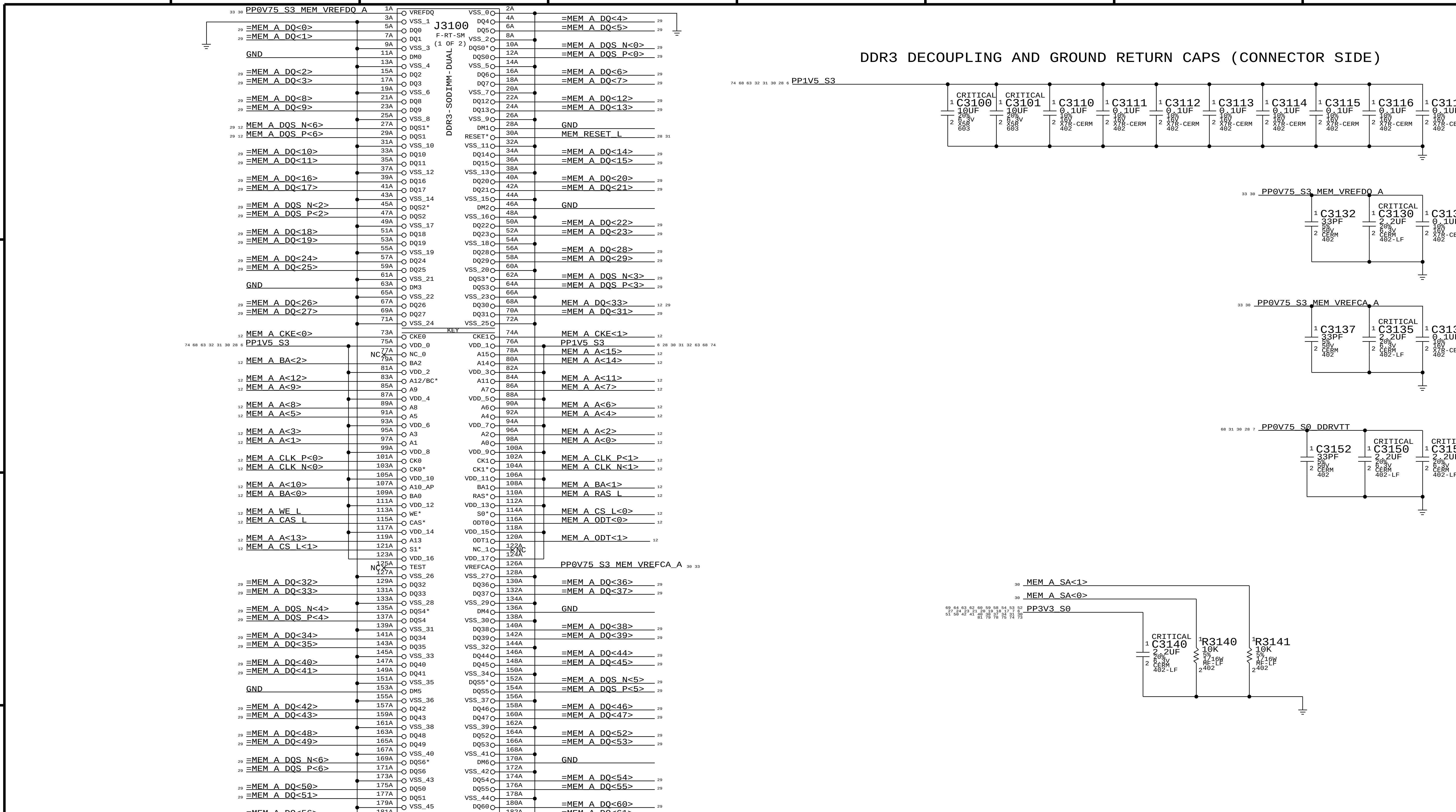


SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# DDR3 SO-DIMM Connector A-1

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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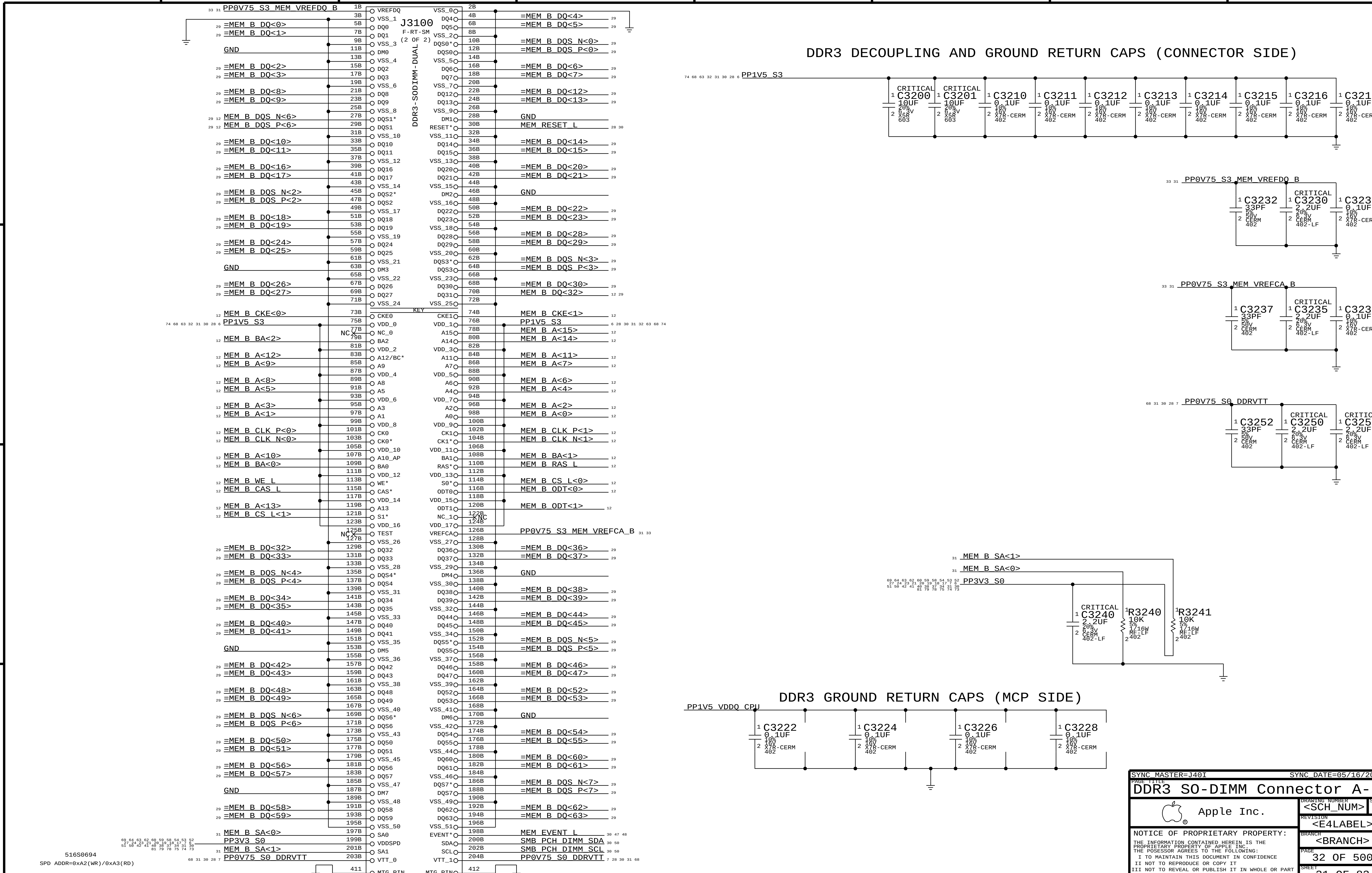
## DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)

## DDR3 GROUND RETURN CAPS (MCP SIDE)




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

# DDR3 SO-DIMM Connector A-2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

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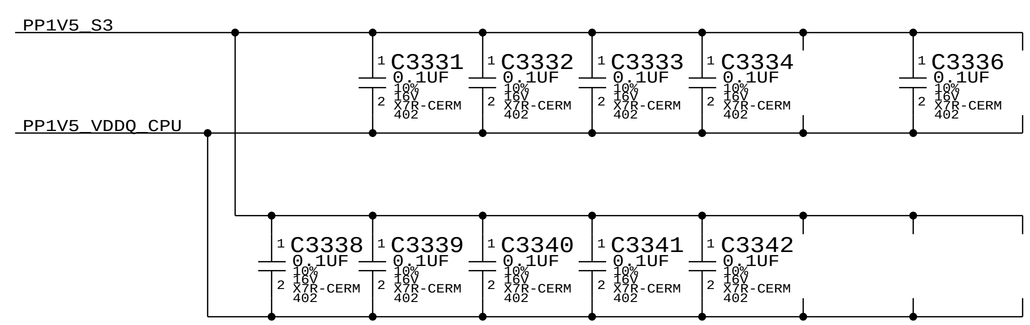


STICHING CAPS.

PLACE ACROSS PLANE SPLIT BETWEEN THESE RAIL
WHERE THE MEMORY ROUTES ARE.
EXTRAS CAN BE DELETED.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# DDR3 Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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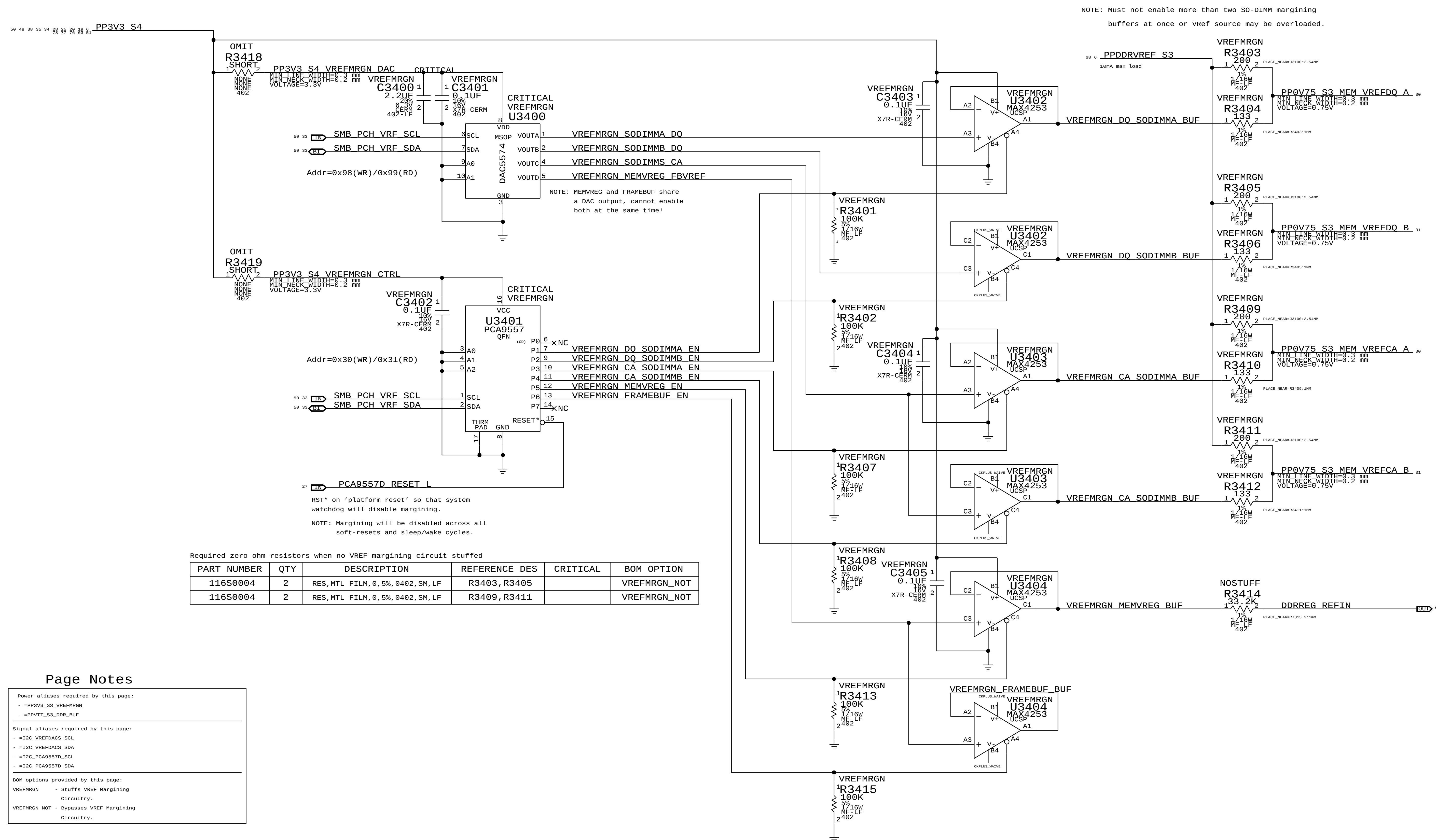


Required zero ohm resistors when no VREF margining circuit stuffed

| PART NUMBER | QTY | DESCRIPTION | REFERENCE DES | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 116S0004 | 2 | RES,MTL FILM,0,5%,0402,SM,LF | R3403,R3405 | | VREFMRGN_NOT |
| 116S0004 | 2 | RES,MTL FILM,0,5%,0402,SM,LF | R3409,R3411 | | VREFMRGN_NOT |

## Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- VREFMRGN - Stuffs VREF Margining Circuitry.
- VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

| | MEM A VREF DQ | MEM B VREF DQ | MEM A VREF CA | MEM B VREF CA | MEM VREG | GPU Frame Buffer (1.8V, 70% VRef) |
|---|---|---|---|---|---|---|
| DAC Channel: | A | B | C | C | D | D |
| PCA9557D Pin: | 1 | 2 | 3 | 4 | 5 | 6 |
| Nominal value | 0.75V (DAC: 0x3A) | | | | 1.5V (DAC: 0x3A) | 1.267V (DAC: 0x8B) |
| Margined target: | 0.300V - 1.200V (+/- 450mV) | | | | 1.998V - 1.002V (+/- 498mV) | 1.056V - 1.442V (+/- 180mV) |
| DAC range: | 0.000V - 1.501V (0x00 - 0x74) | | | | 0.000V - 1.501V (0x00 - 0x74) | 0.000V - 3.300V (0x00 - 0xFF) |
| VRef current: | +3.4mA - -3.4mA (- = sourced) | | | | +33uA - -33uA (- = sourced) | +6.0mA - -5.0mA (- = sourced) |
| DAC step size: | 7.69mV / step @ output | | | | 8.59mV / step @ output | 1.51mV / step @ output |

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FSB/DDR3/FRAMEBUF Vref Margining

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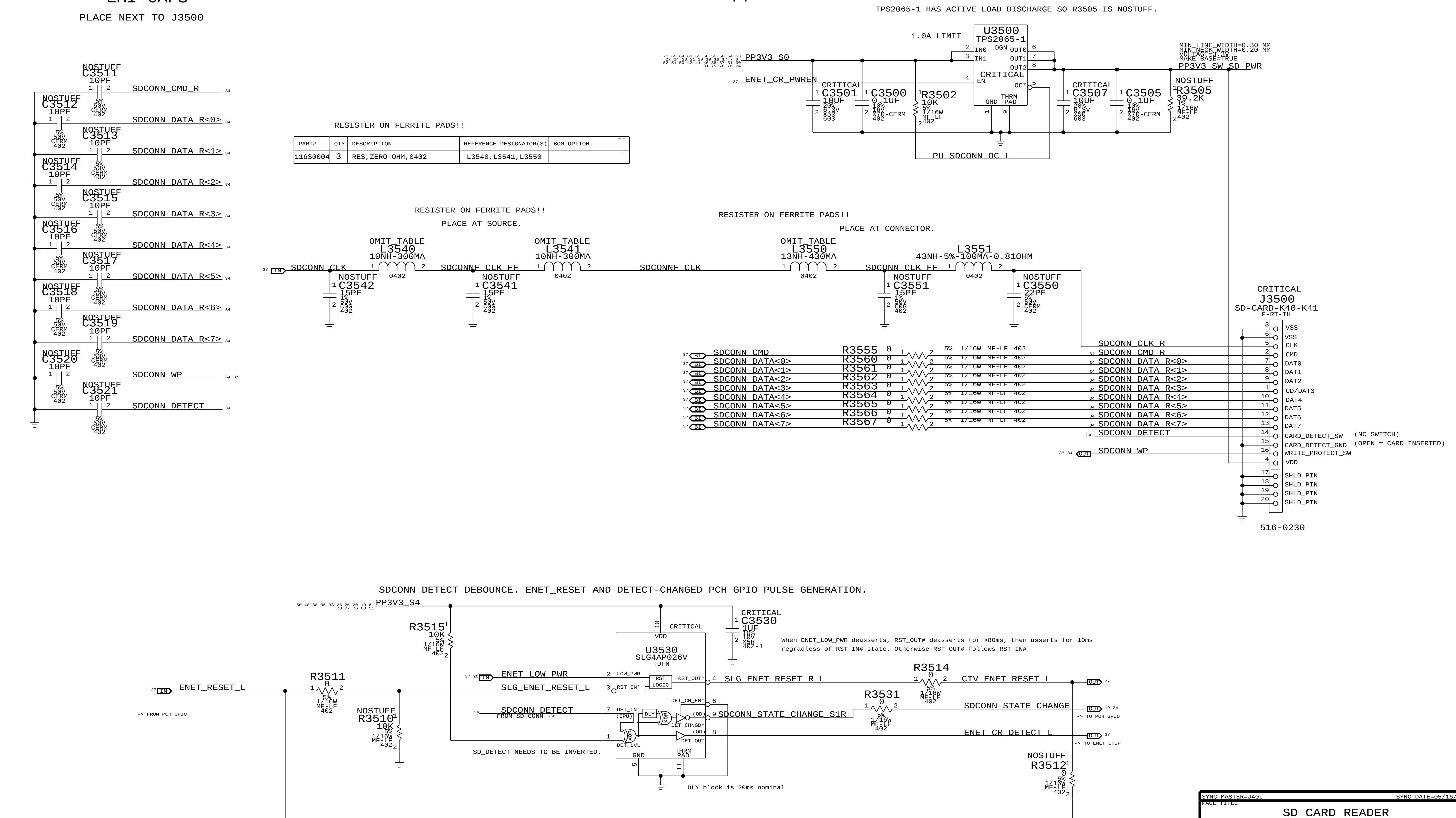
# Caesar IV Support

## EMI CAPS

PLACE NEXT TO J3500

RESISTER ON FERRITE PADS!!

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | BOM OPTION |
|---|---|---|---|---|
| 116S0004 | 3 | RES,ZERO OHM,0402 | L3540,L3541,L3550 | |

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SD CARD READER

Apple Inc.

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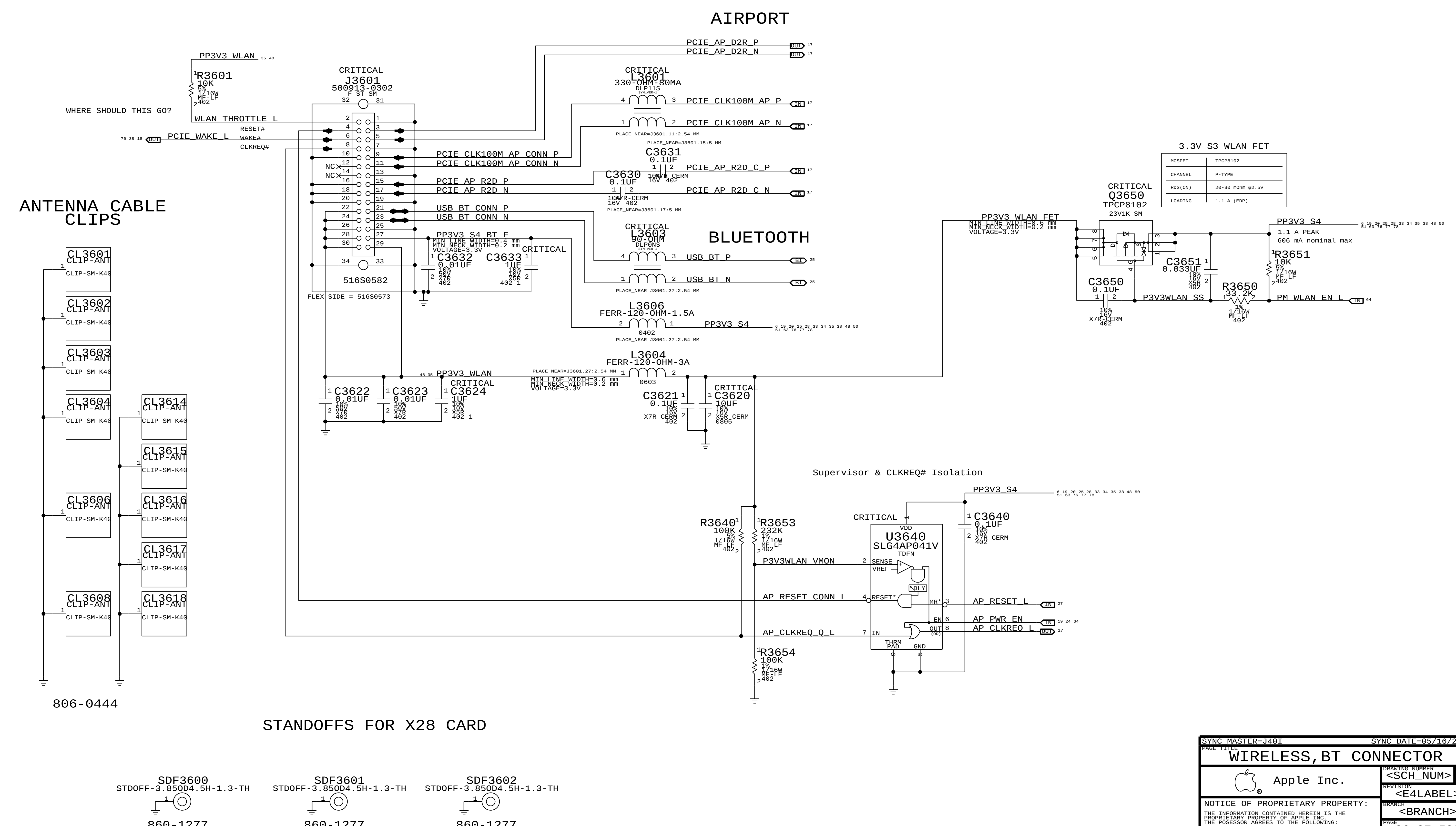


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PAGE TITLE
# WIRELESS,BT CONNECTOR

Apple Inc.

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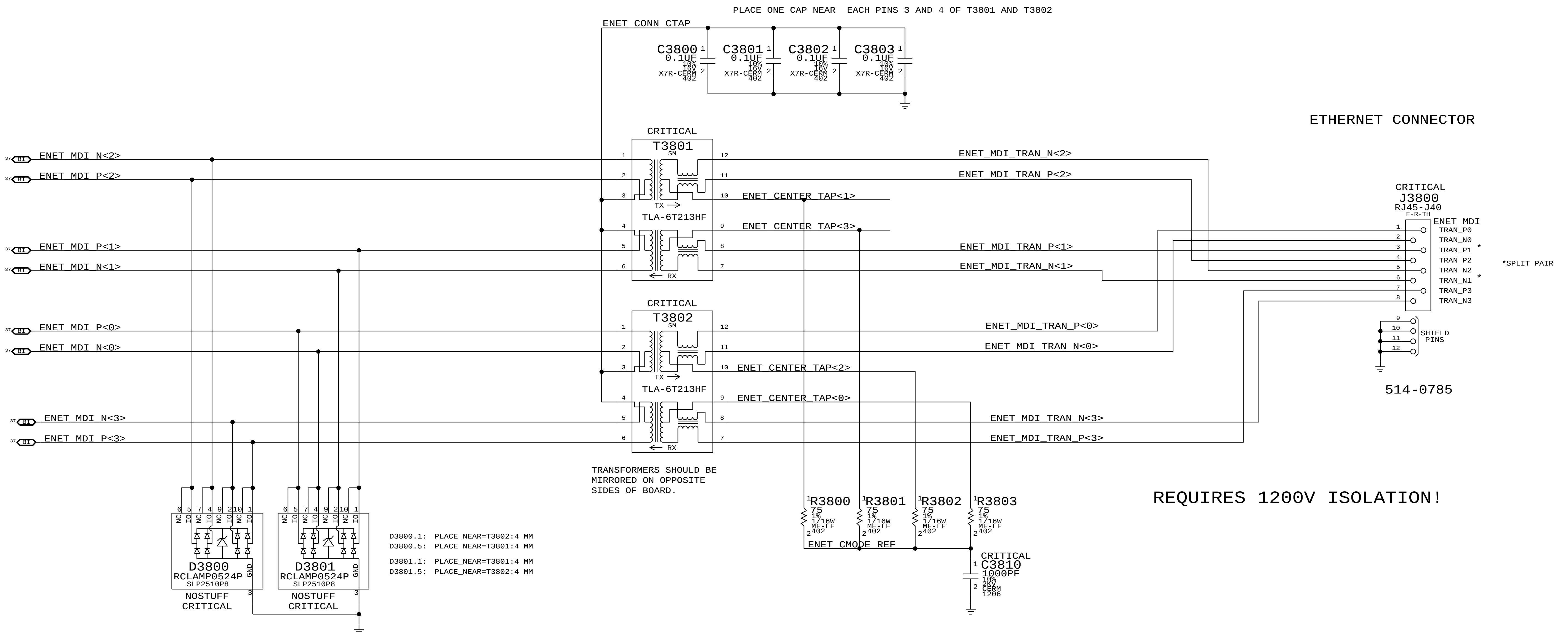


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# ETHERNET CONNECTOR

Apple Inc.

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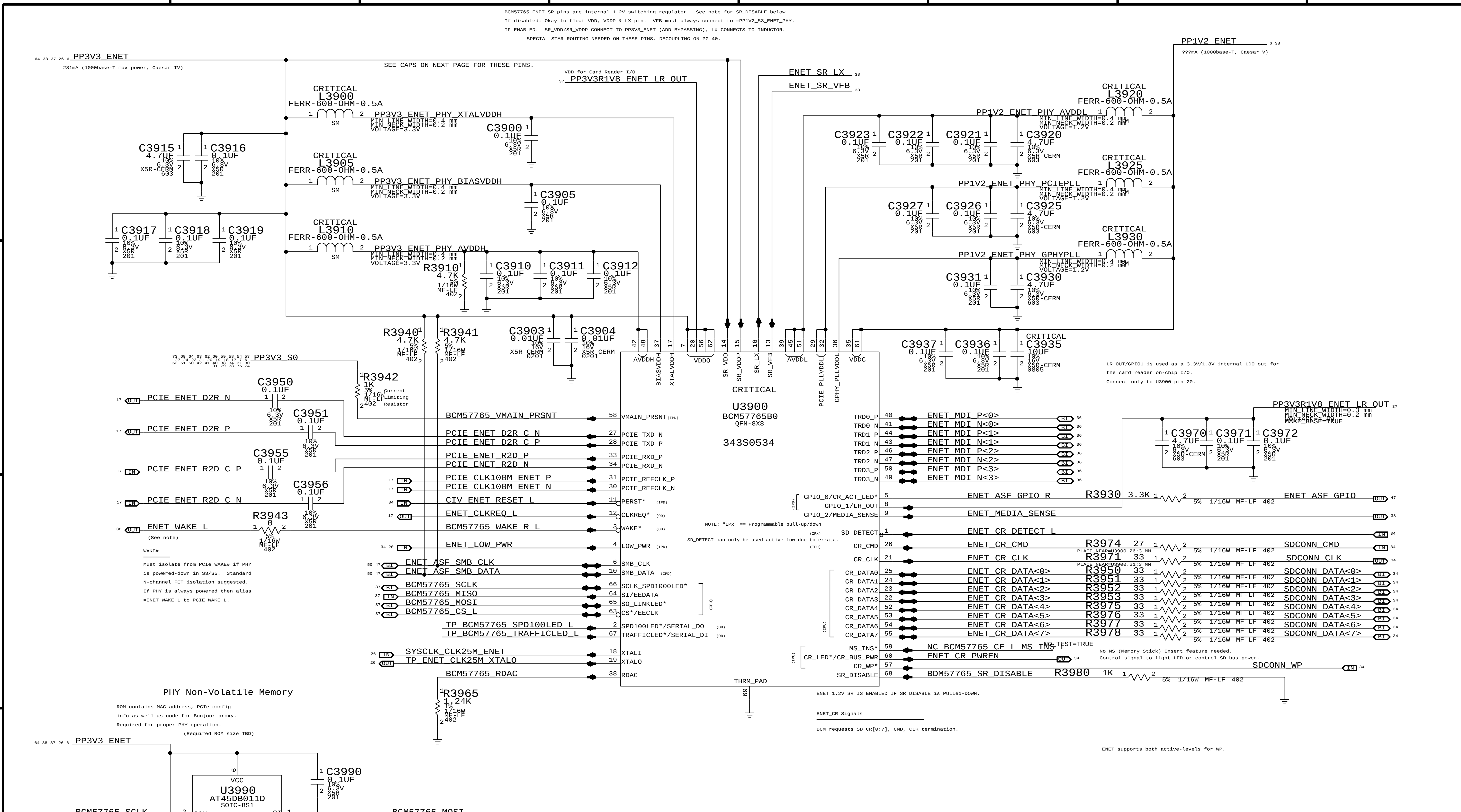


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ETHERNET PHY (CAESAR IV)

Apple Inc.

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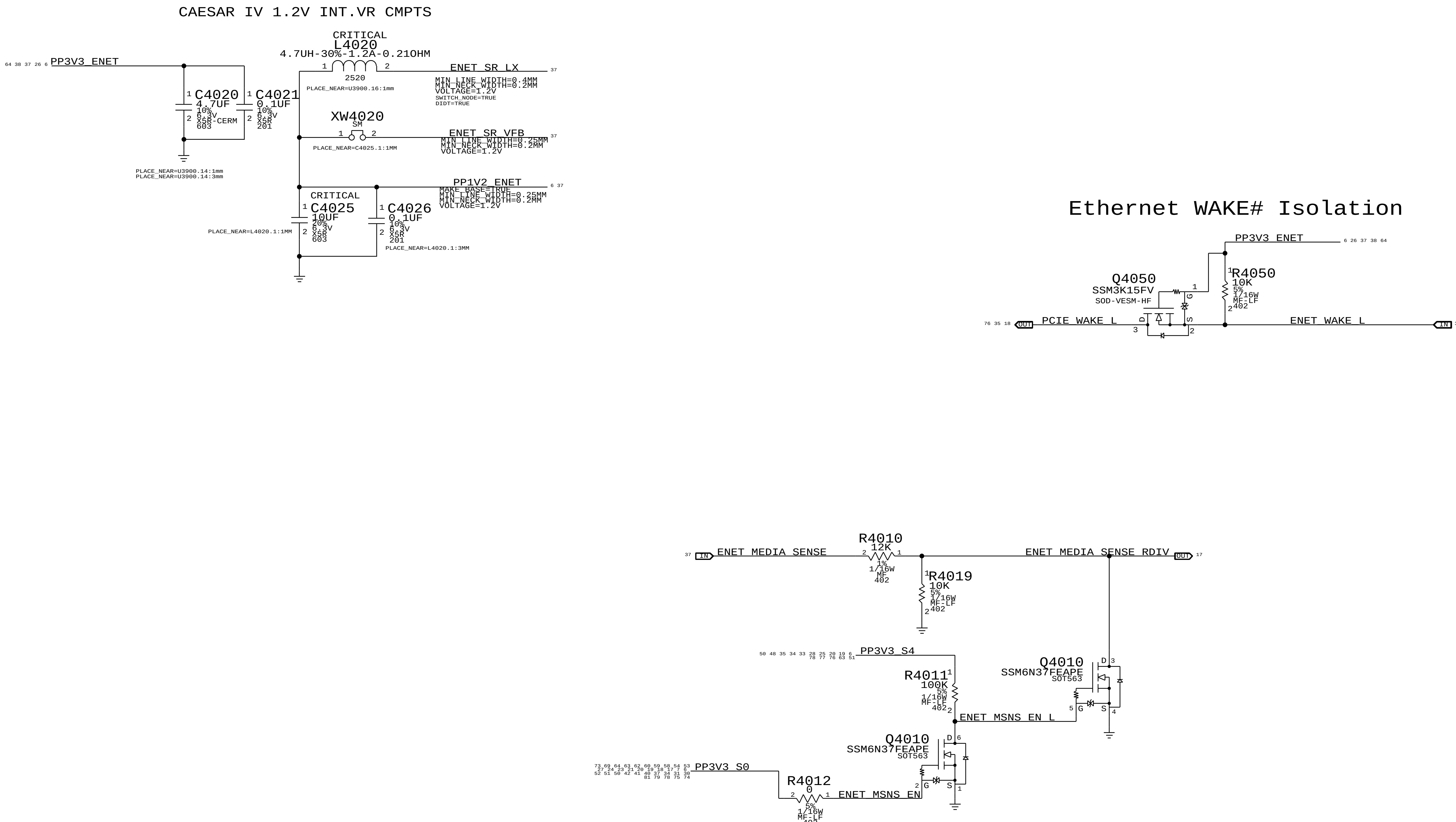


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# ETHERNET SUPPORT

Apple Inc.

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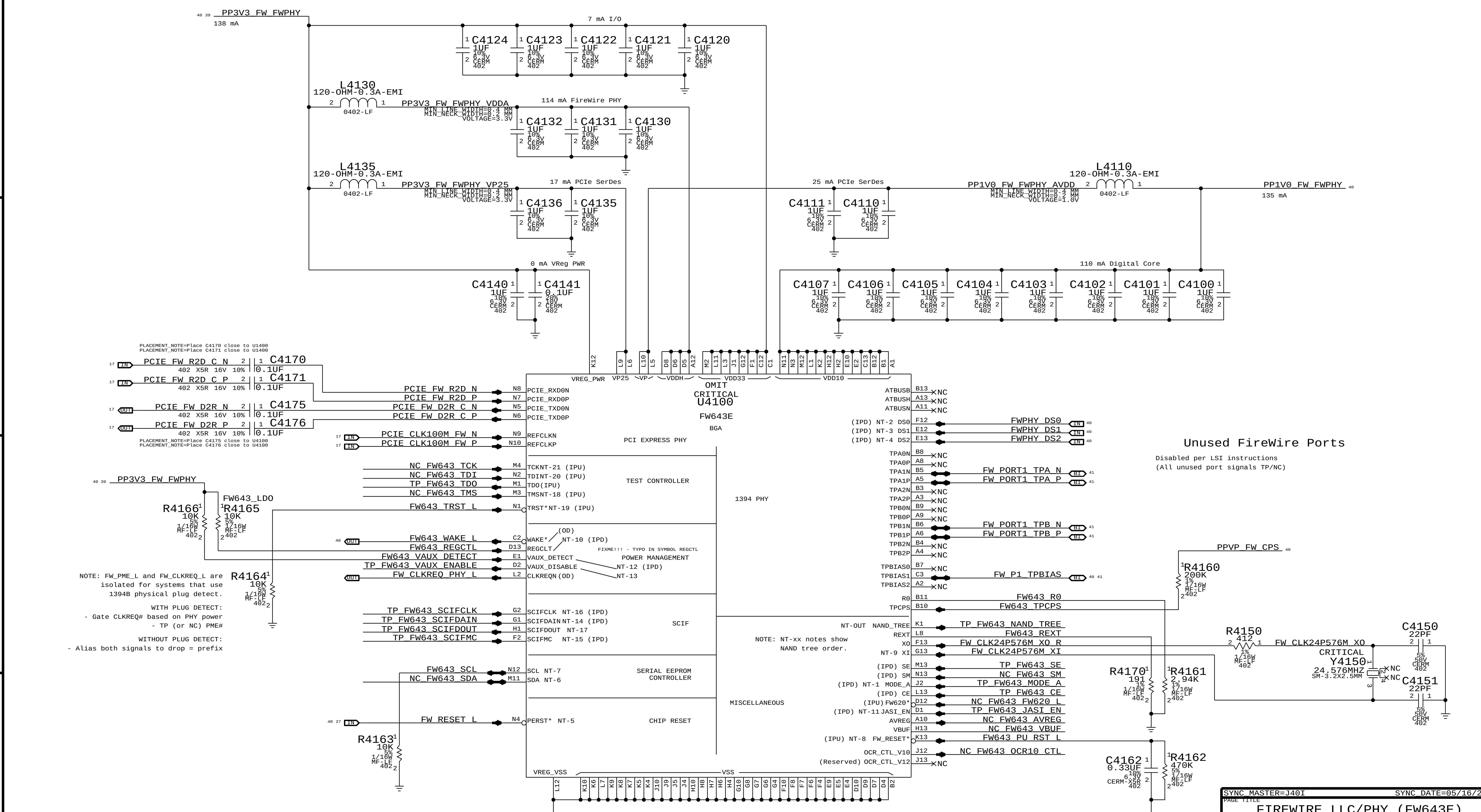


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PAGE TITLE: FIREWIRE LLC/PHY (FW643E)

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Apple Inc.

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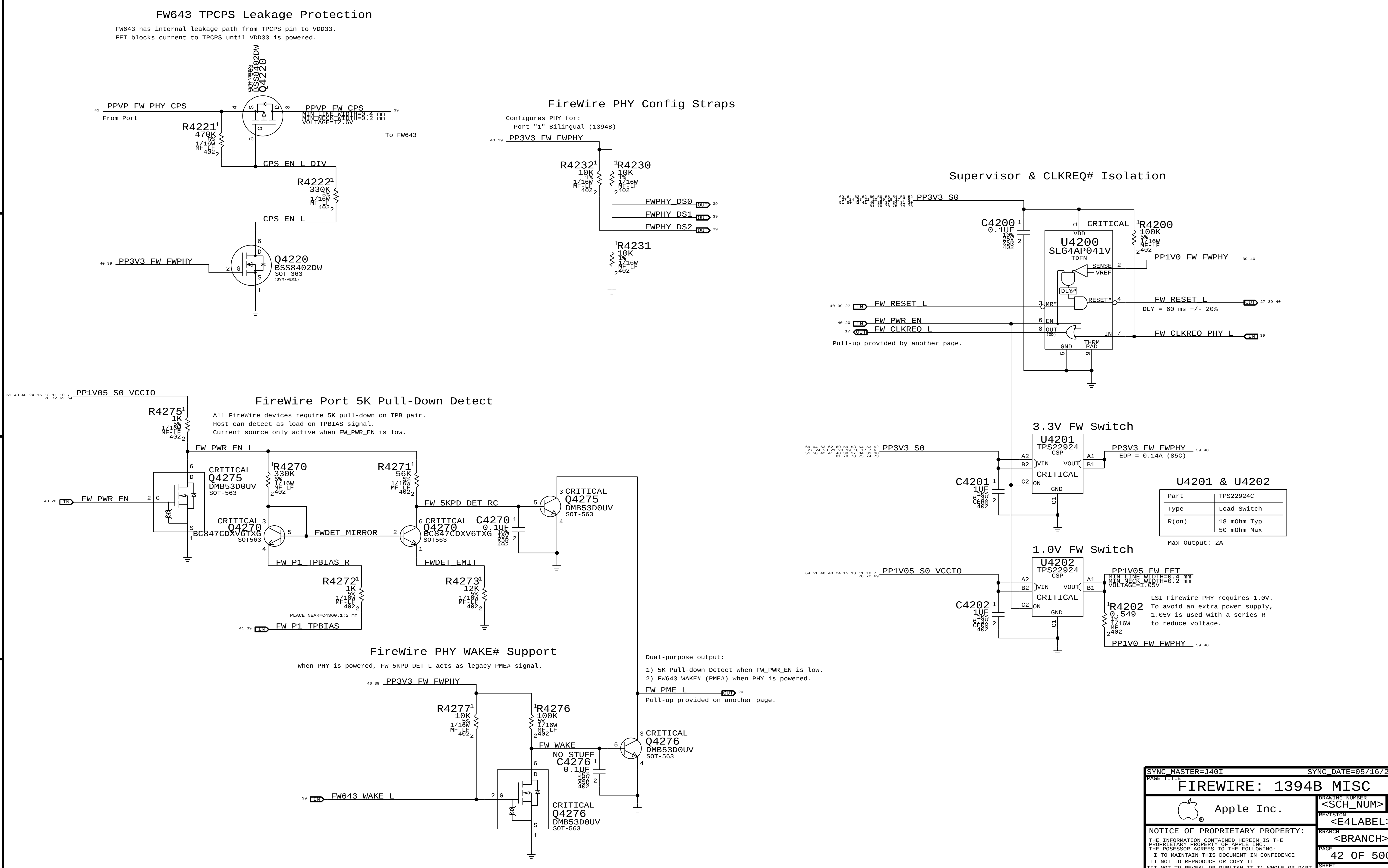


| Part | TPS22924C |
|---|---|
| Type | Load Switch |
| R(on) | 18 mOhm Typ<br>50 mOhm Max |



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# FIREWIRE: 1394B MISC

Apple Inc.

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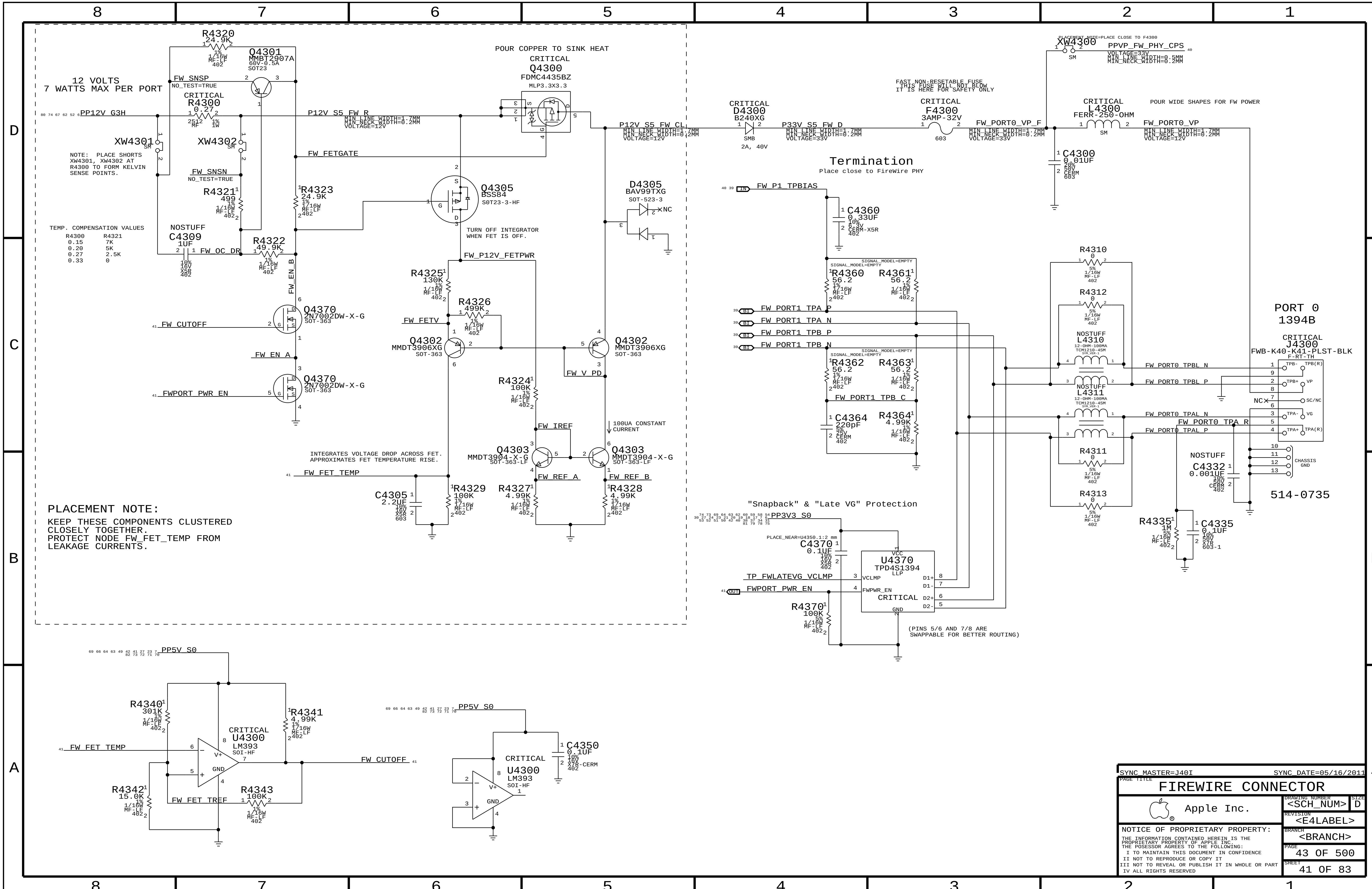


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FIREWIRE CONNECTOR

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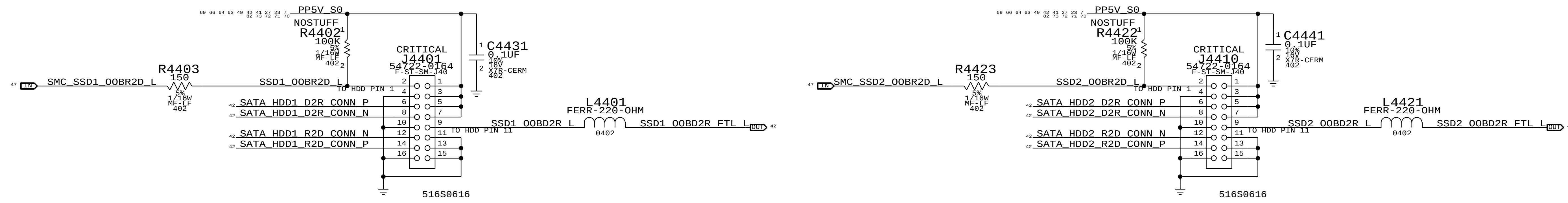
# SATA CONNECTOR (HDD1)

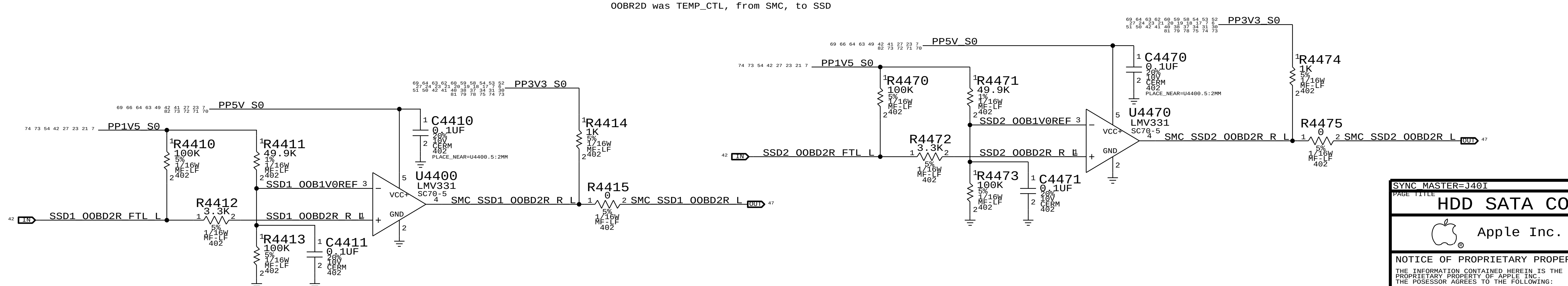
# SATA CONNECTOR (HDD2)





## SATA OOB Comparator

Notes:
OOBD2R was OOB_TEMP, from SSD, to SMC
OOBR2D was TEMP_CTL, from SMC, to SSD

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PAGE TITLE
## HDD SATA CONNECTORS

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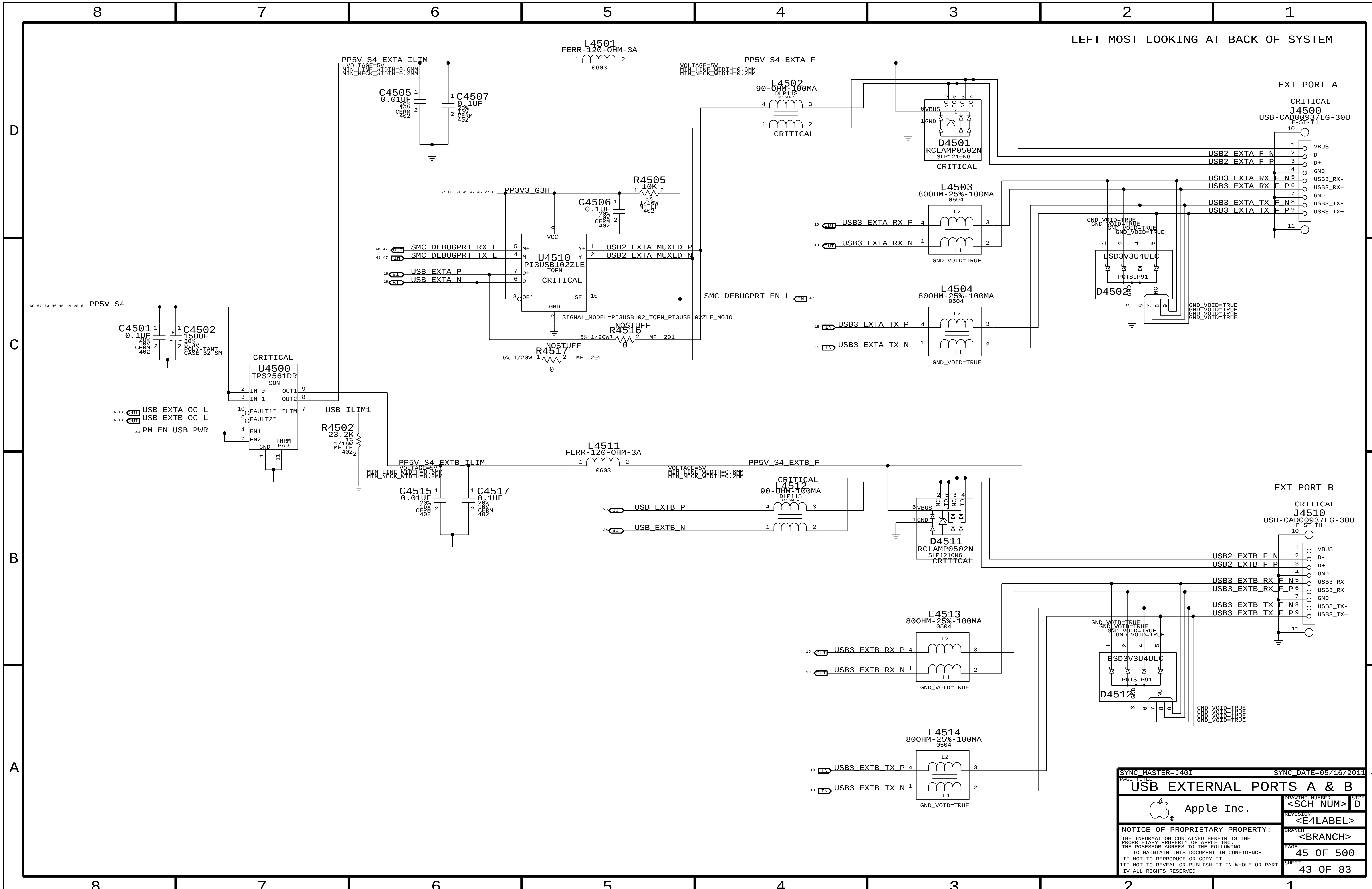


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# USB EXTERNAL PORTS A & B

Apple Inc.

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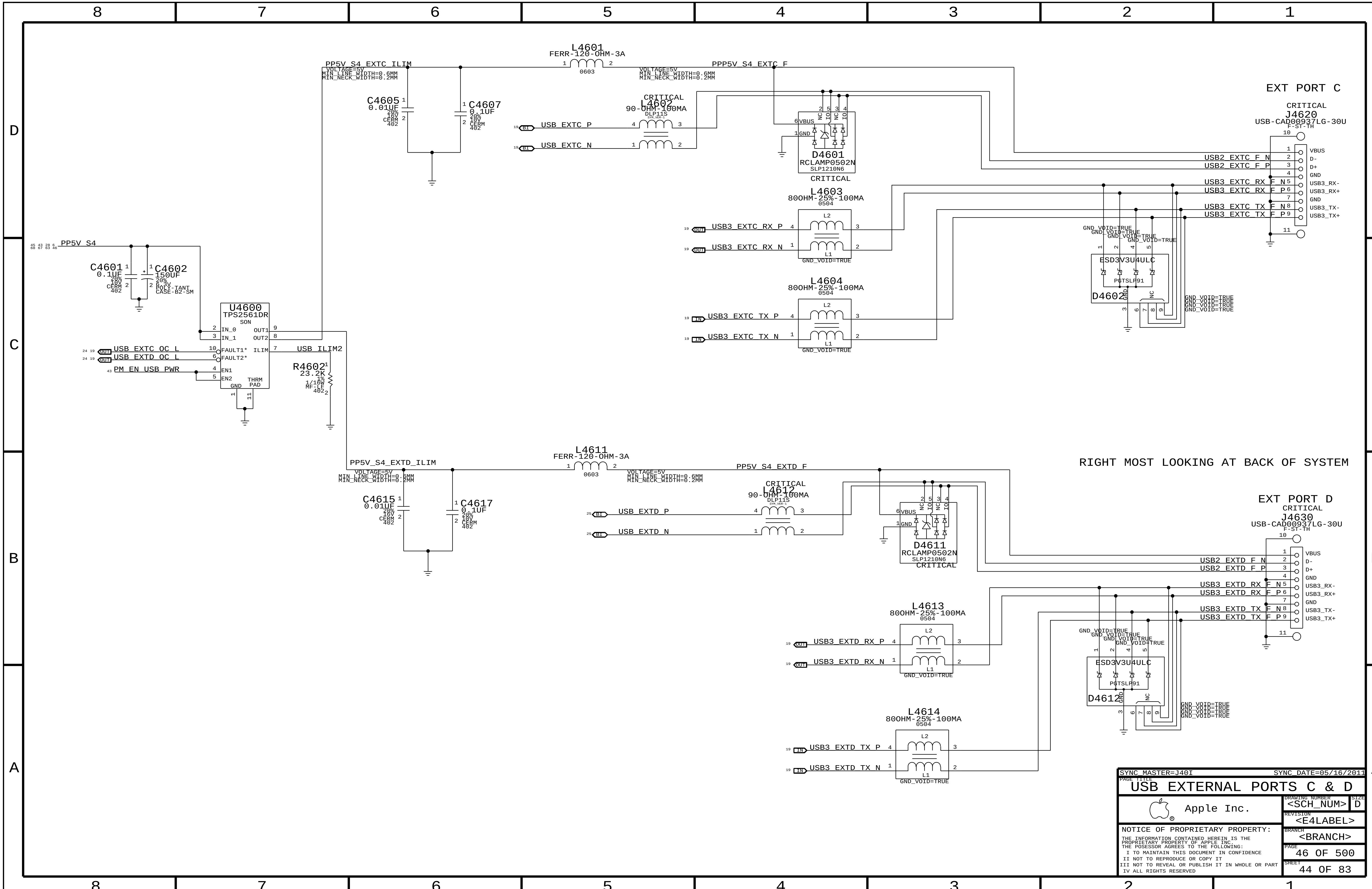

EXT PORT C
CRITICAL
J4620
USB-CAD00937LG-30U
F-ST-TH
VBUS
D-
D+
GND
USB3_RX-
USB3_RX+
GND
USB3_TX-
USB3_TX+
L4601
FERR-120-OHM-3A
0603
PP5V_S4_EXTC_ILIM
VOLTAGE=5V
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
PPP5V_S4_EXTC_F
C4605
0.01UF
C4607
0.1UF
CRITICAL
L4602
90-OHM-100MA
DLP11S
USB_EXTC_P
USB_EXTC_N
D4601
RCLAMP0502N
SLP1210N6
CRITICAL
L4603
80OHM-25%-100MA
0504
USB3_EXTC_RX_P
USB3_EXTC_RX_N
GND_VOID=TRUE
L4604
80OHM-25%-100MA
0504
USB3_EXTC_TX_P
USB3_EXTC_TX_N
USB2_EXTC_F_N
USB2_EXTC_F_P
USB3_EXTC_RX_F_N
USB3_EXTC_RX_F_P
USB3_EXTC_TX_F_N
USB3_EXTC_TX_F_P
ESD3V3U4ULC
PGTSLP91
D4602
PP5V_S4
C4601
0.1UF
C4602
150UF
POLY-TANT
CASE-B2-SM
U4600
TPS2561DR
SON
IN_0
IN_1
FAULT1*
FAULT2*
EN1
EN2
GND
THRM PAD
OUT1
OUT2
ILIM
USB_EXTC_OC_L
USB_EXTD_OC_L
PM_EN_USB_PWR
USB_ILIM2
R4602
23.2K
1%
1/16W
MF-LF
402
RIGHT MOST LOOKING AT BACK OF SYSTEM
L4611
FERR-120-OHM-3A
0603
PP5V_S4_EXTD_ILIM
VOLTAGE=5V
MIN_LINE_WIDTH=0.6MM
MIN_NECK_WIDTH=0.2MM
PP5V_S4_EXTD_F
C4615
0.01UF
C4617
0.1UF
CRITICAL
L4612
90-OHM-100MA
DLP11S
USB_EXTD_P
USB_EXTD_N
D4611
RCLAMP0502N
SLP1210N6
CRITICAL
EXT PORT D
CRITICAL
J4630
USB-CAD00937LG-30U
F-ST-TH
USB2_EXTD_F_N
USB2_EXTD_F_P
USB3_EXTD_RX_F_N
USB3_EXTD_RX_F_P
USB3_EXTD_TX_F_N
USB3_EXTD_TX_F_P
L4613
80OHM-25%-100MA
0504
USB3_EXTD_RX_P
USB3_EXTD_RX_N
ESD3V3U4ULC
PGTSLP91
D4612
L4614
80OHM-25%-100MA
0504
USB3_EXTD_TX_P
USB3_EXTD_TX_N
SYNC_MASTER=J40I
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USB EXTERNAL PORTS C & D
Apple Inc.
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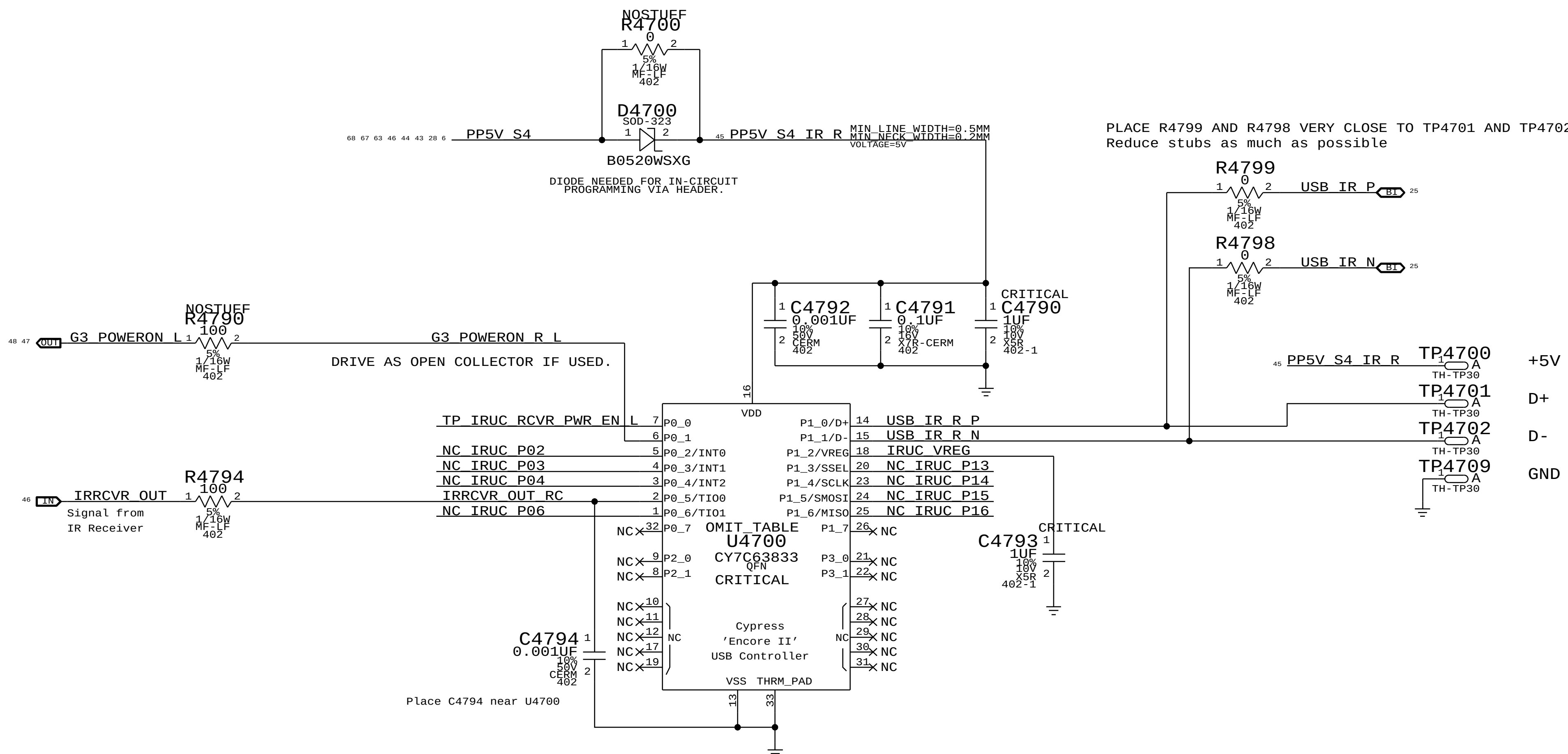
# IR Controller

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IR CONTROLLER

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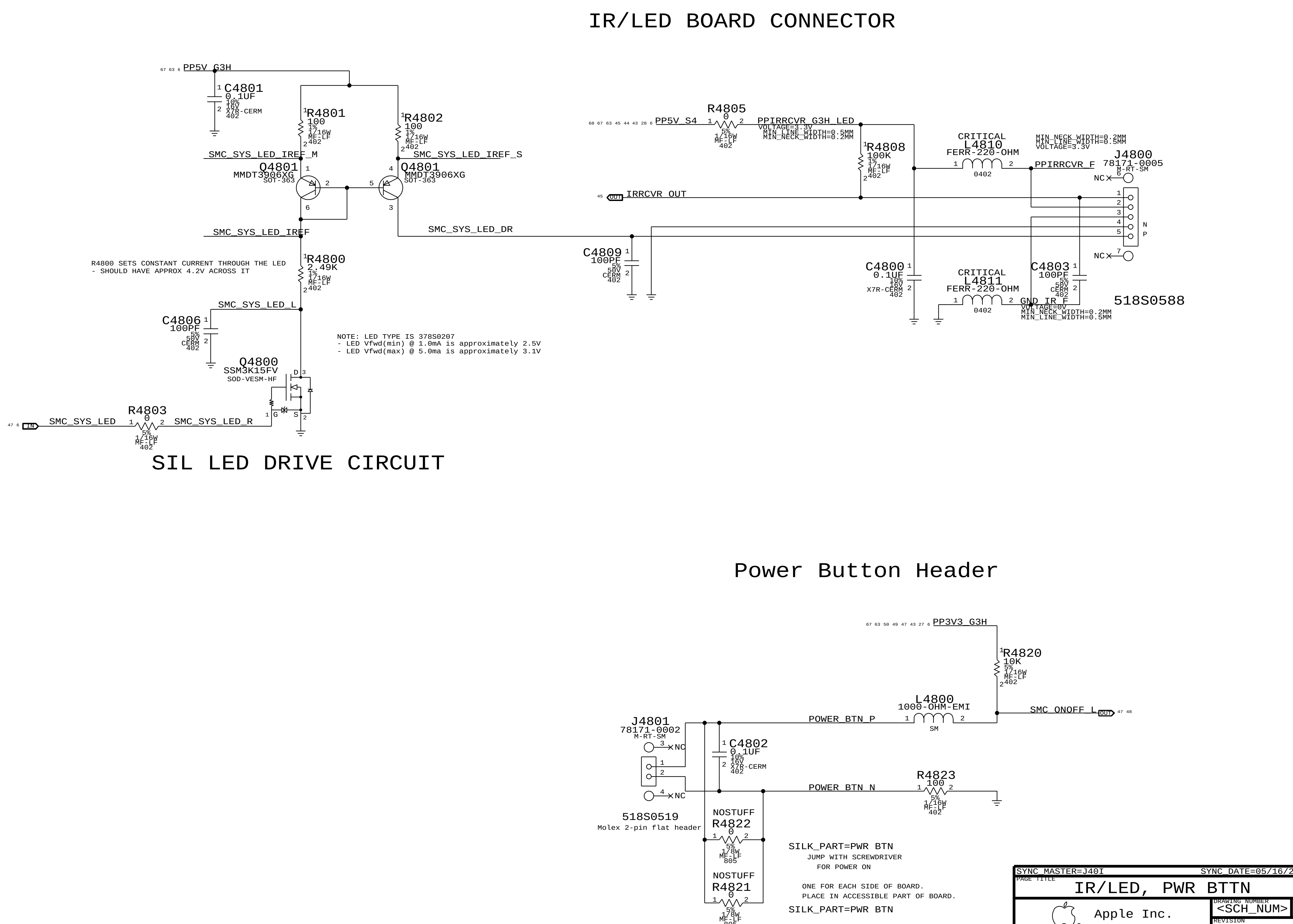


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# IR/LED, PWR BTTN

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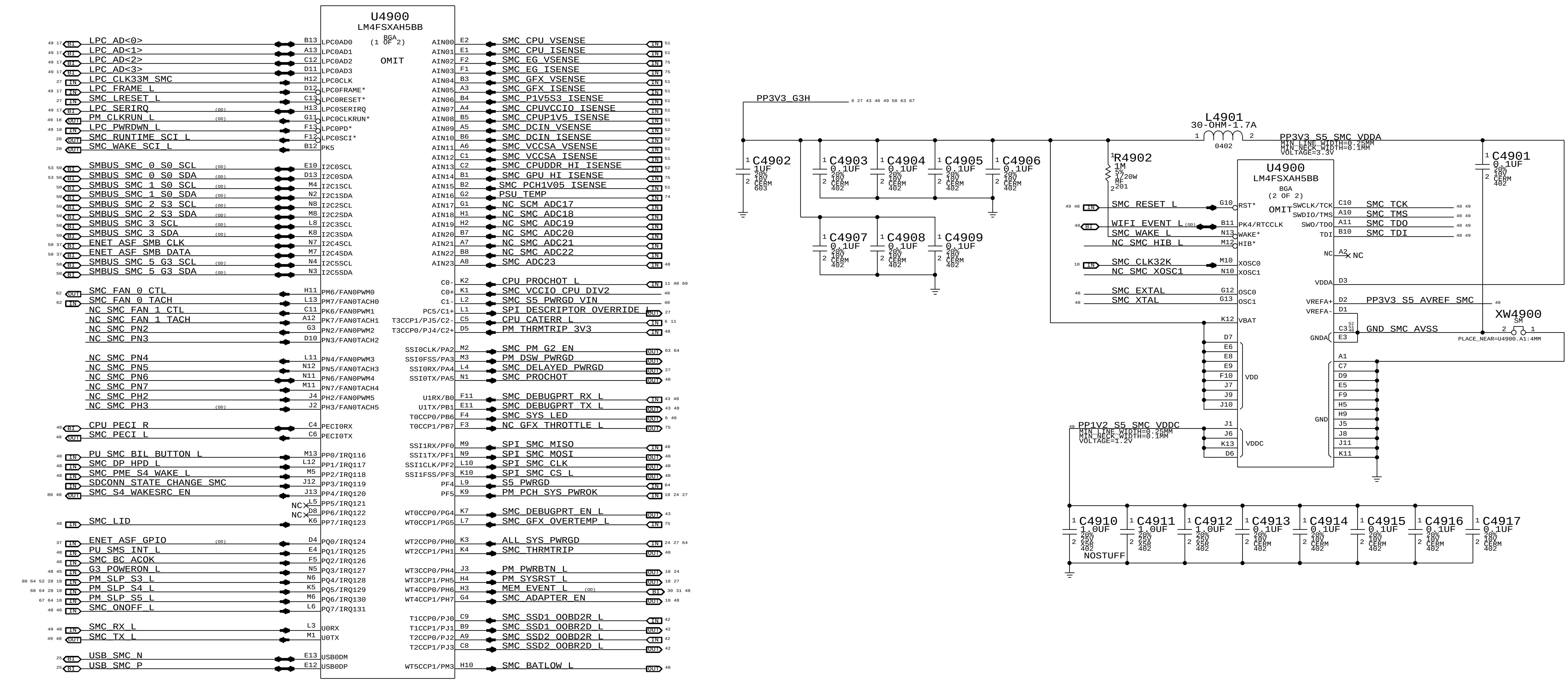


NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

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SMC

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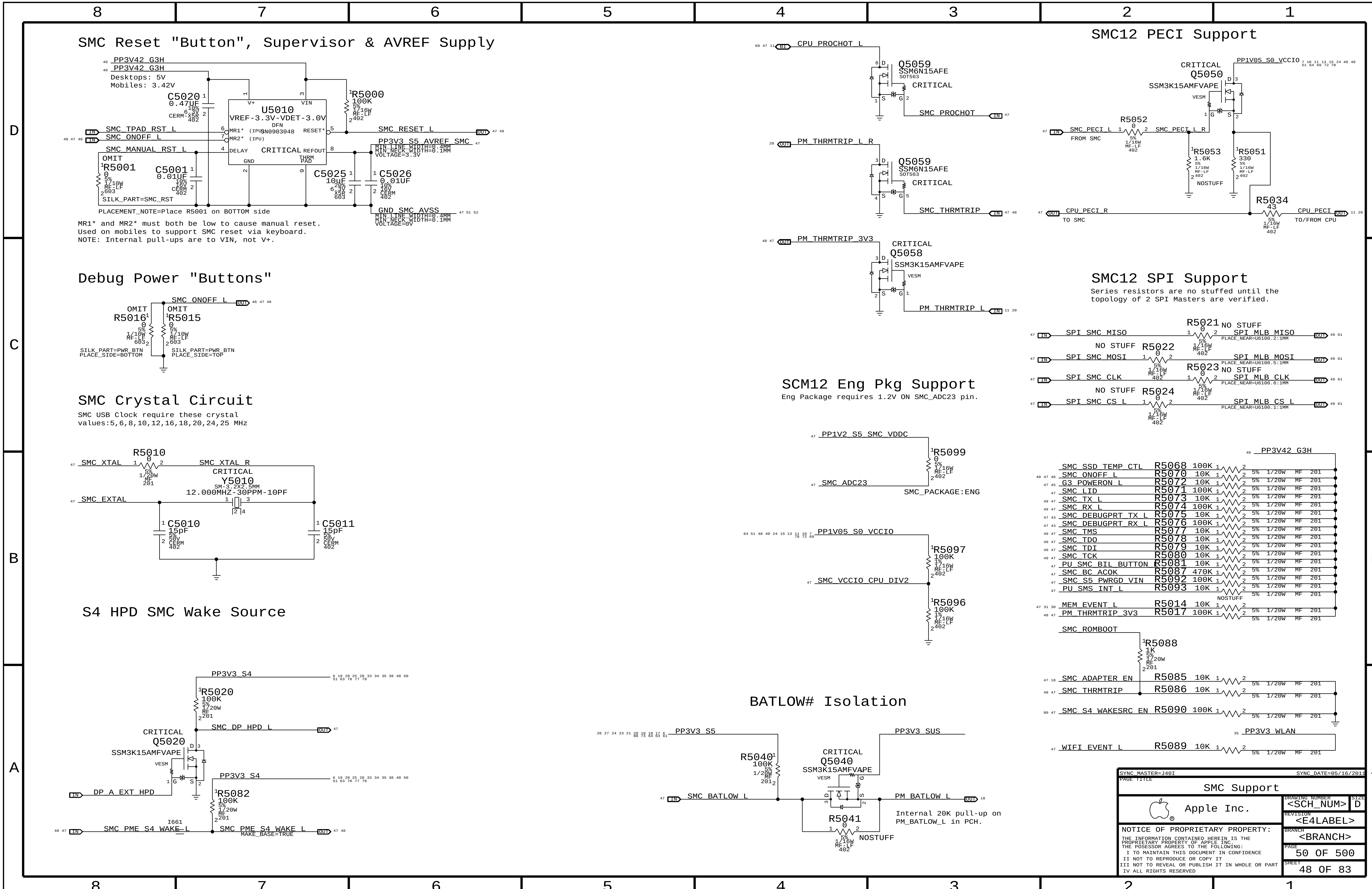

SMC Reset "Button", Supervisor & AVREF Supply
PP3V42_G3H
Desktops: 5V
Mobiles: 3.42V
C5020 0.47UF
U5010 VREF-3.3V-VDET-3.0V
SN0903048
R5000 100K
SMC_TPAD_RST_L
SMC_ONOFF_L
SMC_RESET_L
SMC_MANUAL_RST_L
PP3V3_S5_AVREF_SMC
MIN_LINE_WIDTH=0.4MM
MIN_NECK_WIDTH=0.1MM
VOLTAGE=3.3V
R5001
C5001 0.01UF
C5025 10uF
C5026 0.01UF
GND_SMC_AVSS
VOLTAGE=0V
SILK_PART=SMC_RST
PLACEMENT_NOTE=Place R5001 on BOTTOM side
MR1* and MR2* must both be low to cause manual reset.
Used on mobiles to support SMC reset via keyboard.
NOTE: Internal pull-ups are to VIN, not V+.
Debug Power "Buttons"
SMC_ONOFF_L
R5016
R5015
SILK_PART=PWR_BTN
PLACE_SIDE=BOTTOM
PLACE_SIDE=TOP
SMC Crystal Circuit
SMC USB Clock require these crystal
values:5,6,8,10,12,16,18,20,24,25 MHz
R5010
SMC_XTAL
SMC_XTAL_R
Y5010
12.000MHZ-30PPM-10PF
SMC_EXTAL
C5010 15pF
C5011 15pF
S4 HPD SMC Wake Source
PP3V3_S4
R5020 100K
SMC_DP_HPD_L
Q5020 SSM3K15AMFVAPE
DP_A_EXT_HPD
R5082 100K
SMC_PME_S4_WAKE_L
MAKE_BASE=TRUE
CPU_PROCHOT_L
Q5059 SSM6N15AFE
SMC_PROCHOT
PM_THRMTRIP_L_R
SMC_THRMTRIP
PM_THRMTRIP_3V3
Q5058 SSM3K15AMFVAPE
PM_THRMTRIP_L
SCM12 Eng Pkg Support
Eng Package requires 1.2V ON SMC_ADC23 pin.
PP1V2_S5_SMC_VDDC
R5099
SMC_ADC23
SMC_PACKAGE:ENG
PP1V05_S0_VCCIO
R5097 100K
SMC_VCCIO_CPU_DIV2
R5096 100K
BATLOW# Isolation
PP3V3_S5
PP3V3_SUS
R5040 100K
Q5040 SSM3K15AMFVAPE
SMC_BATLOW_L
PM_BATLOW_L
R5041 NOSTUFF
Internal 20K pull-up on
PM_BATLOW_L in PCH.
SMC12 PECI Support
Q5050 SSM3K15AMFVAPE
PP1V05_S0_VCCIO
R5052
SMC_PECI_L
FROM SMC
SMC_PECI_L_R
R5053 1.6K
R5051 330
NOSTUFF
R5034 43
CPU_PECI_R
TO SMC
CPU_PECI
TO/FROM CPU
SMC12 SPI Support
Series resistors are no stuffed until the
topology of 2 SPI Masters are verified.
SPI_SMC_MISO R5021 NO STUFF SPI_MLB_MISO
SPI_SMC_MOSI R5022 NO STUFF SPI_MLB_MOSI
SPI_SMC_CLK R5023 NO STUFF SPI_MLB_CLK
SPI_SMC_CS_L R5024 NO STUFF SPI_MLB_CS_L
PP3V42_G3H
SMC_SSD_TEMP_CTL R5068 100K
SMC_ONOFF_L R5070 10K
G3_POWERON_L R5072 10K
SMC_LID R5071 100K
SMC_TX_L R5073 10K
SMC_RX_L R5074 100K
SMC_DEBUGPRT_TX_L R5075 10K
SMC_DEBUGPRT_RX_L R5076 100K
SMC_TMS R5077 10K
SMC_TDO R5078 10K
SMC_TDI R5079 10K
SMC_TCK R5080 10K
PU_SMC_BIL_BUTTON_L R5081 10K
SMC_BC_ACOK R5087 470K
SMC_S5_PWRGD_VIN R5092 100K
PU_SMS_INT_L R5093 10K NOSTUFF
MEM_EVENT_L R5014 10K
PM_THRMTRIP_3V3 R5017 100K
SMC_ROMBOOT R5088 1K
SMC_ADAPTER_EN R5085 10K
SMC_THRMTRIP R5086 10K
SMC_S4_WAKESRC_EN R5090 100K
PP3V3_WLAN
WIFI_EVENT_L R5089 10K
SYNC_MASTER=J40I
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SMC Support
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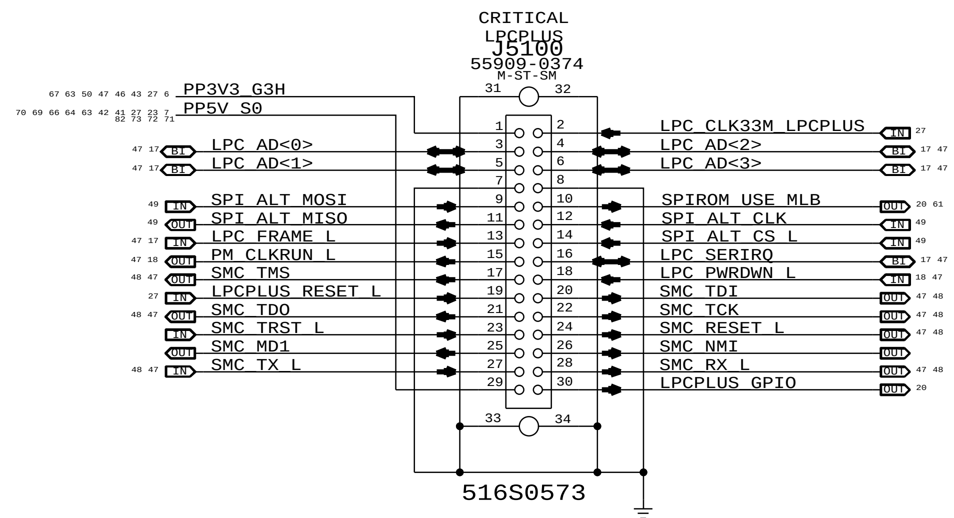
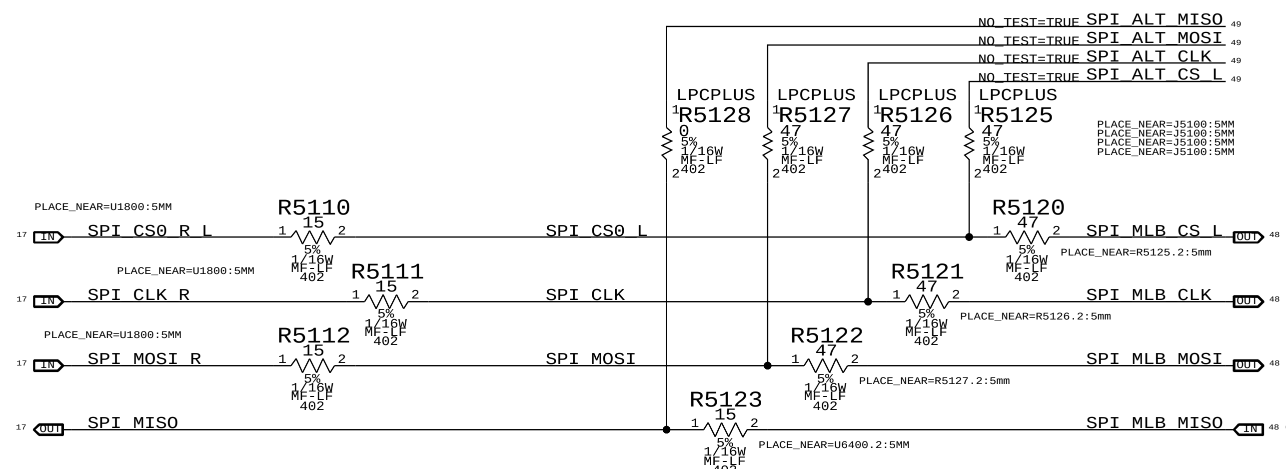
# MATT CARD
## LPC+SPI Connector




## SPI Bus Series Termination

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LPC+ Debug Connector

Apple Inc.

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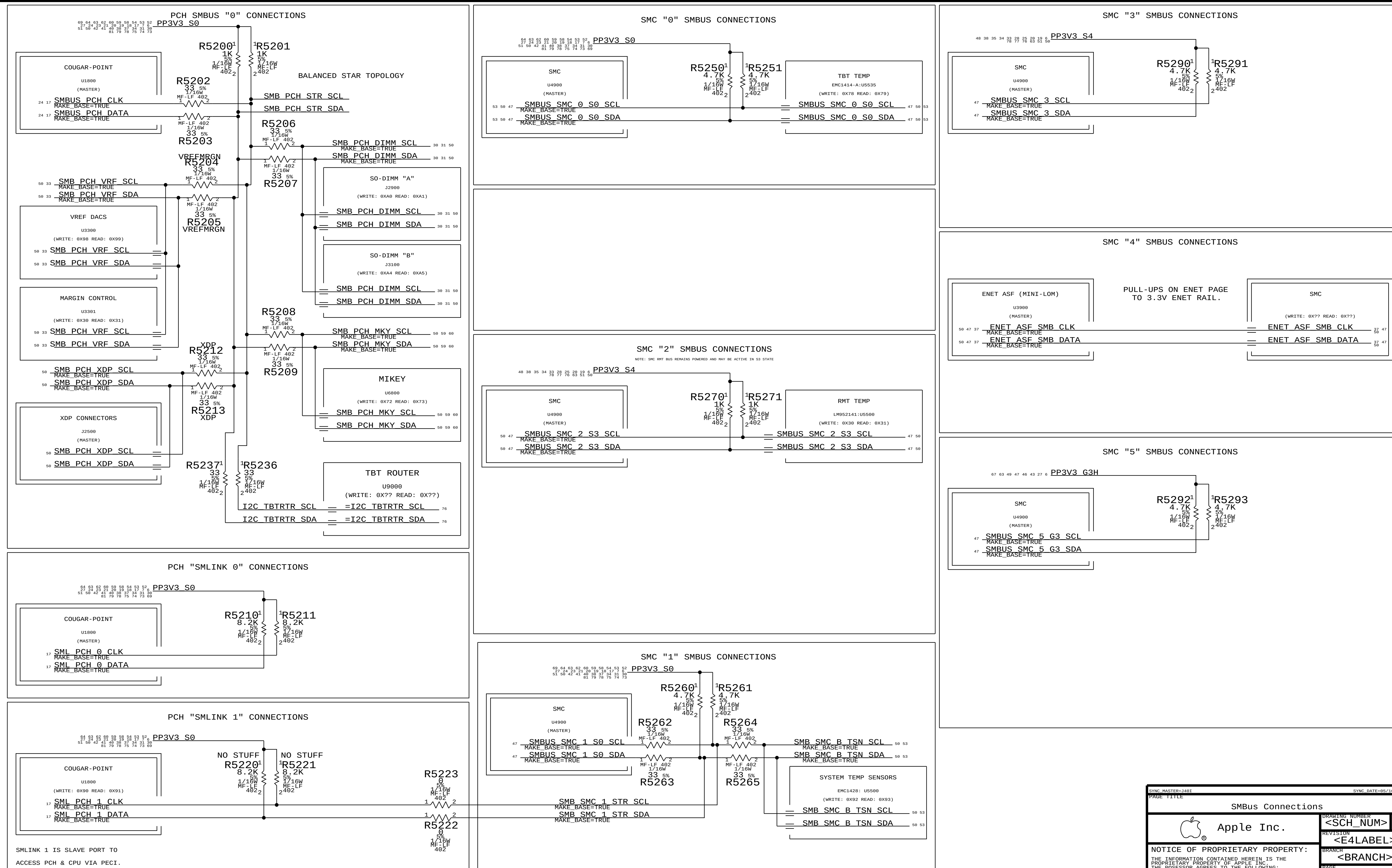


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SMBus Connections

Apple Inc.

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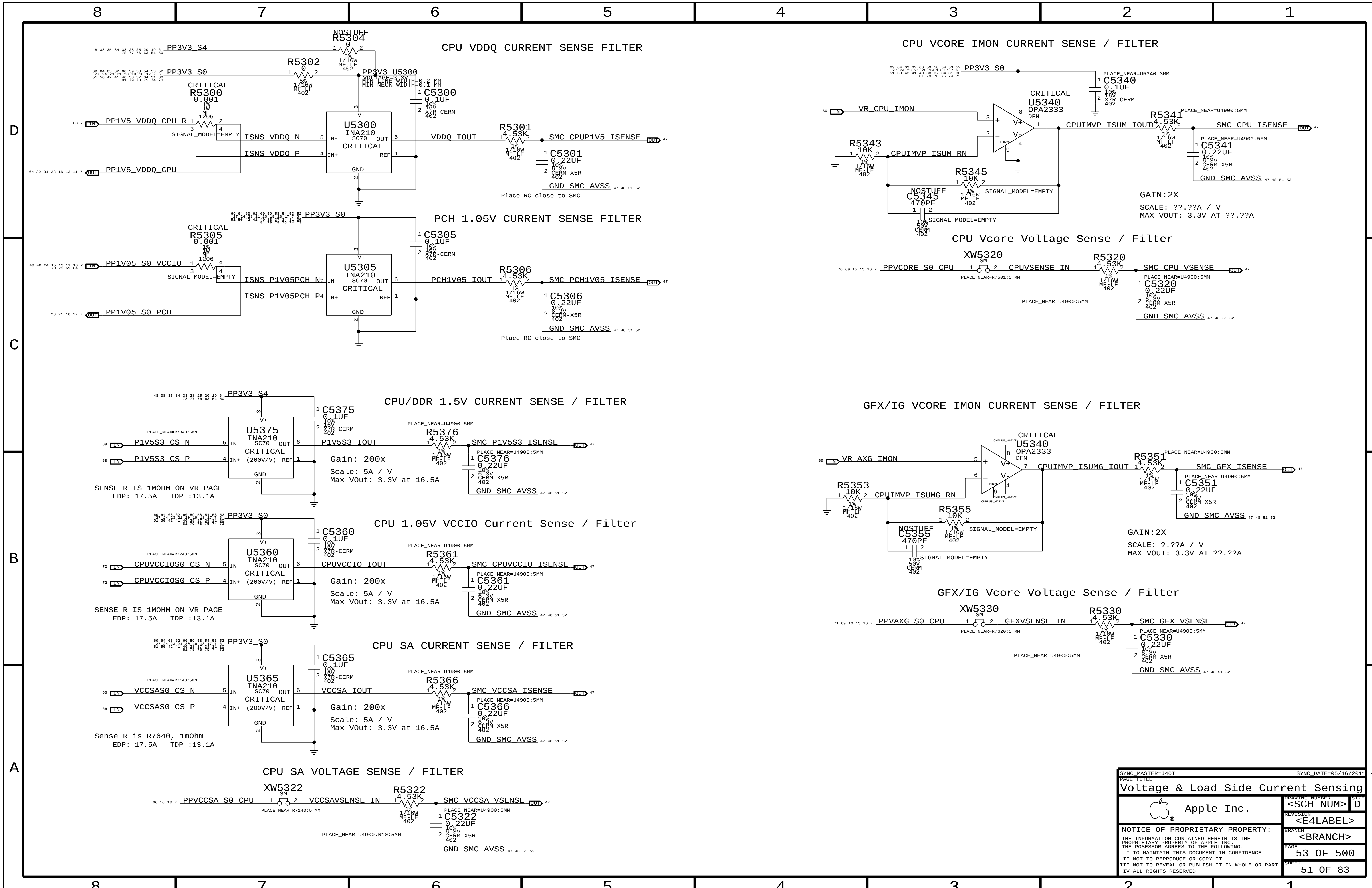


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Voltage & Load Side Current Sensing

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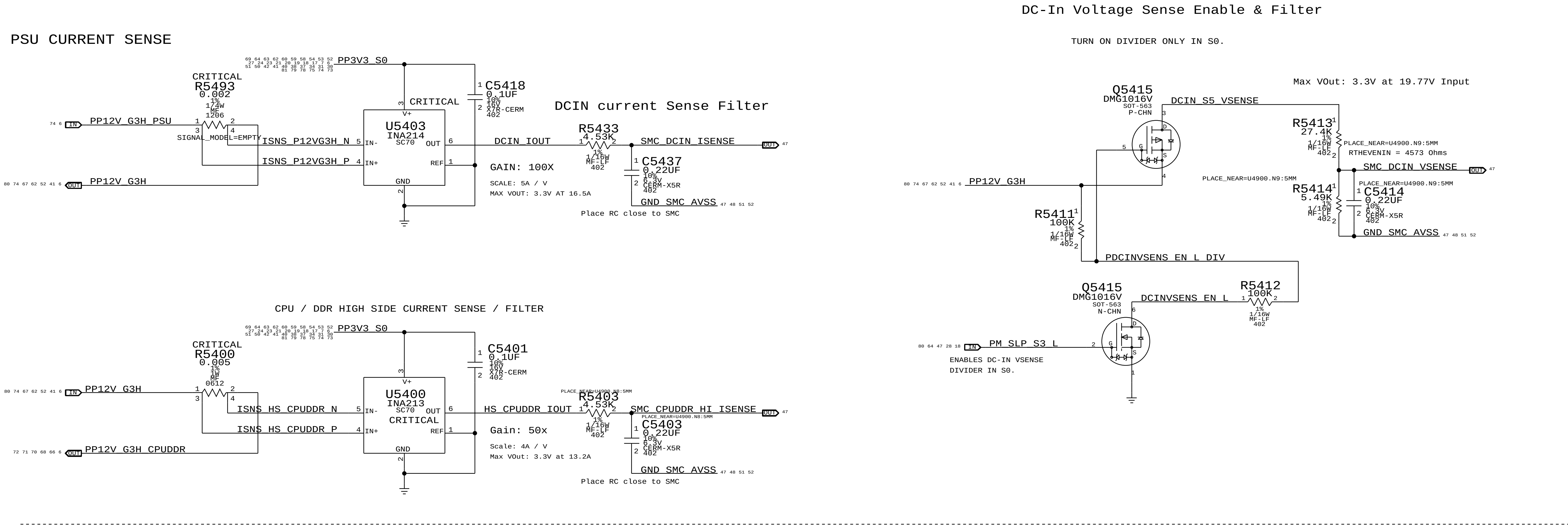
# J40I, J40S, AND J40

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High Side Current Sensing

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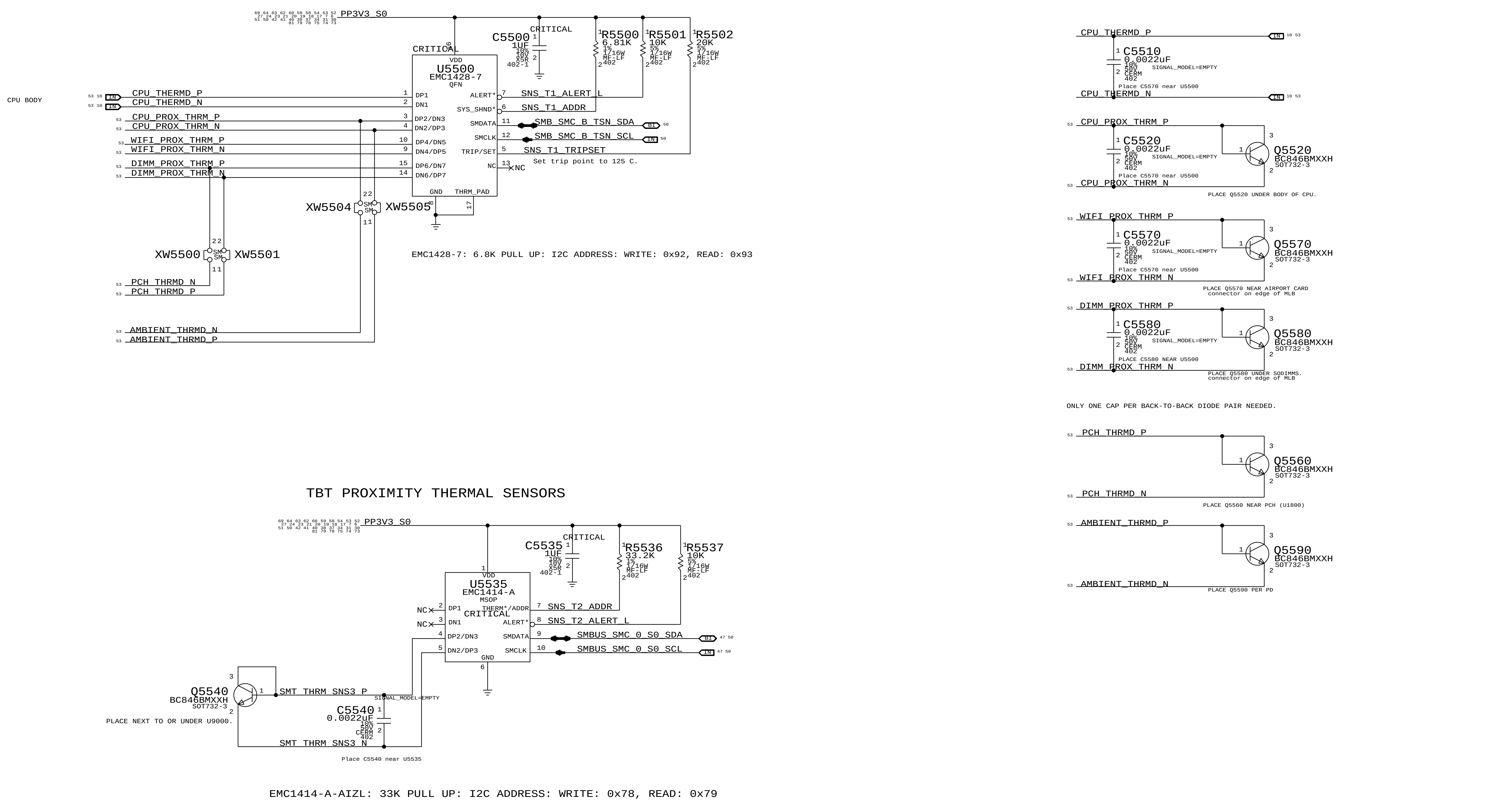


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# THERMAL SENSORS

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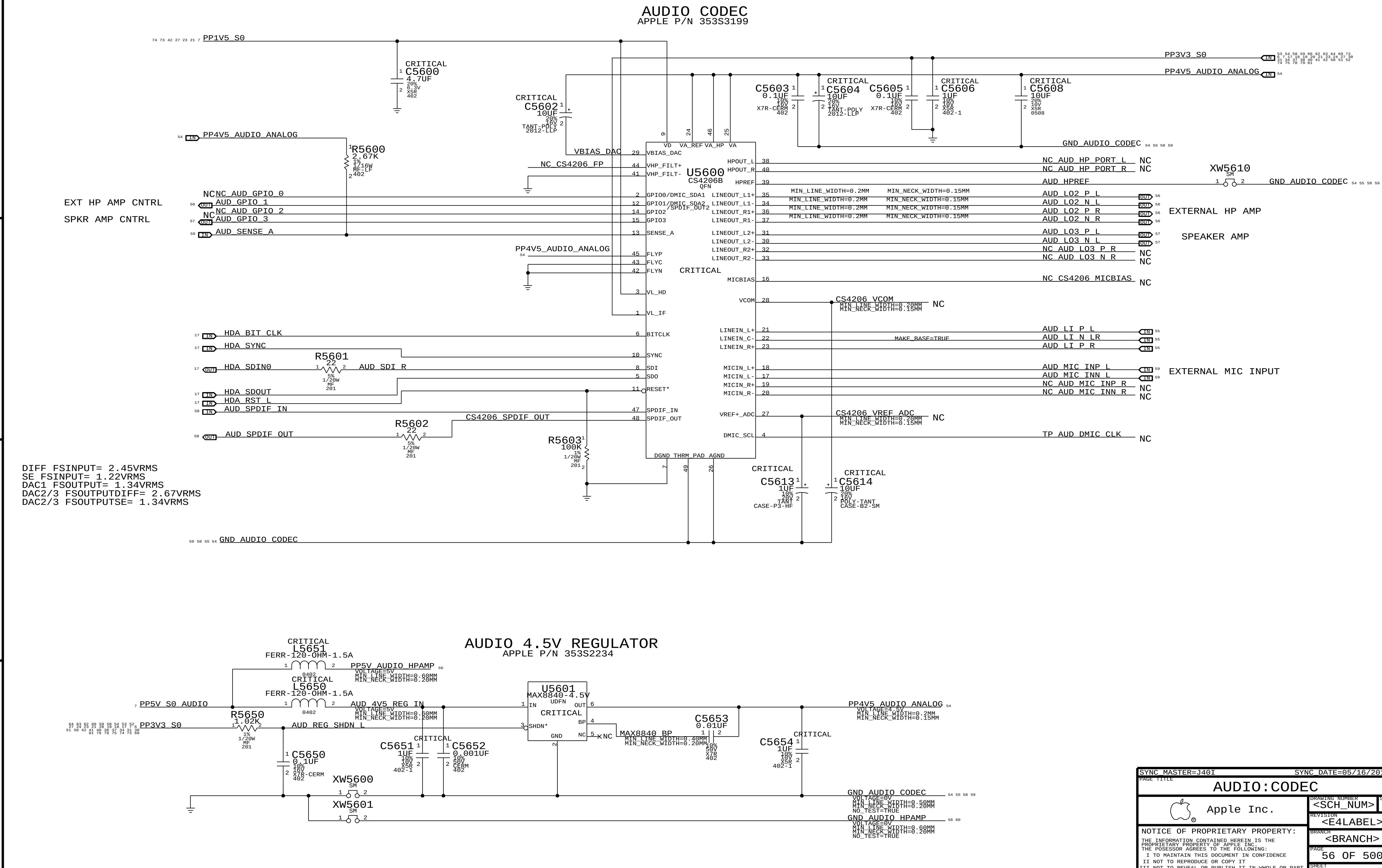
# AUDIO CODEC
APPLE P/N 353S3199




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: AUDIO:CODEC

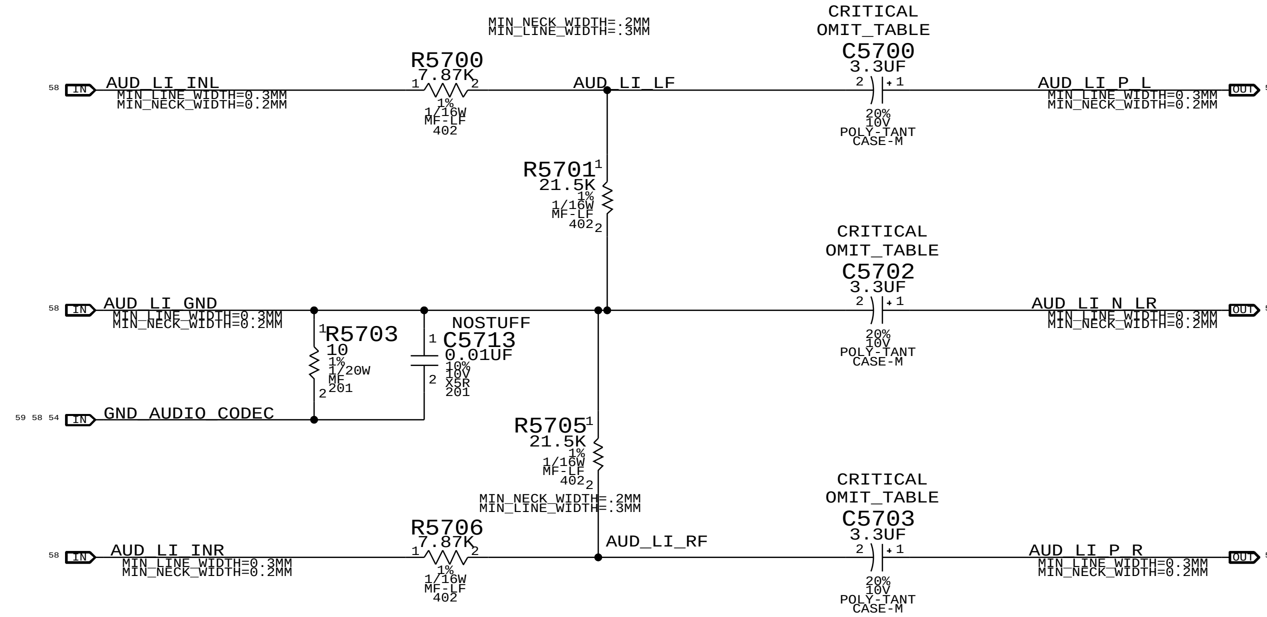
Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

CODEC Nom SE RIN = 20K OHMS
FC = 5 HZ Max
VIN = 2VRMS, CODEC VIN = 1.14 VRMS
NET RIN = 18K OHMS




| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 128S0309 | 3 | CAP,TANT,POLY,4.7UF,20%,10V,SMD | C5700,C5702,C5703 | CRITICAL | ? |

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# AUDIO:LINE-IN

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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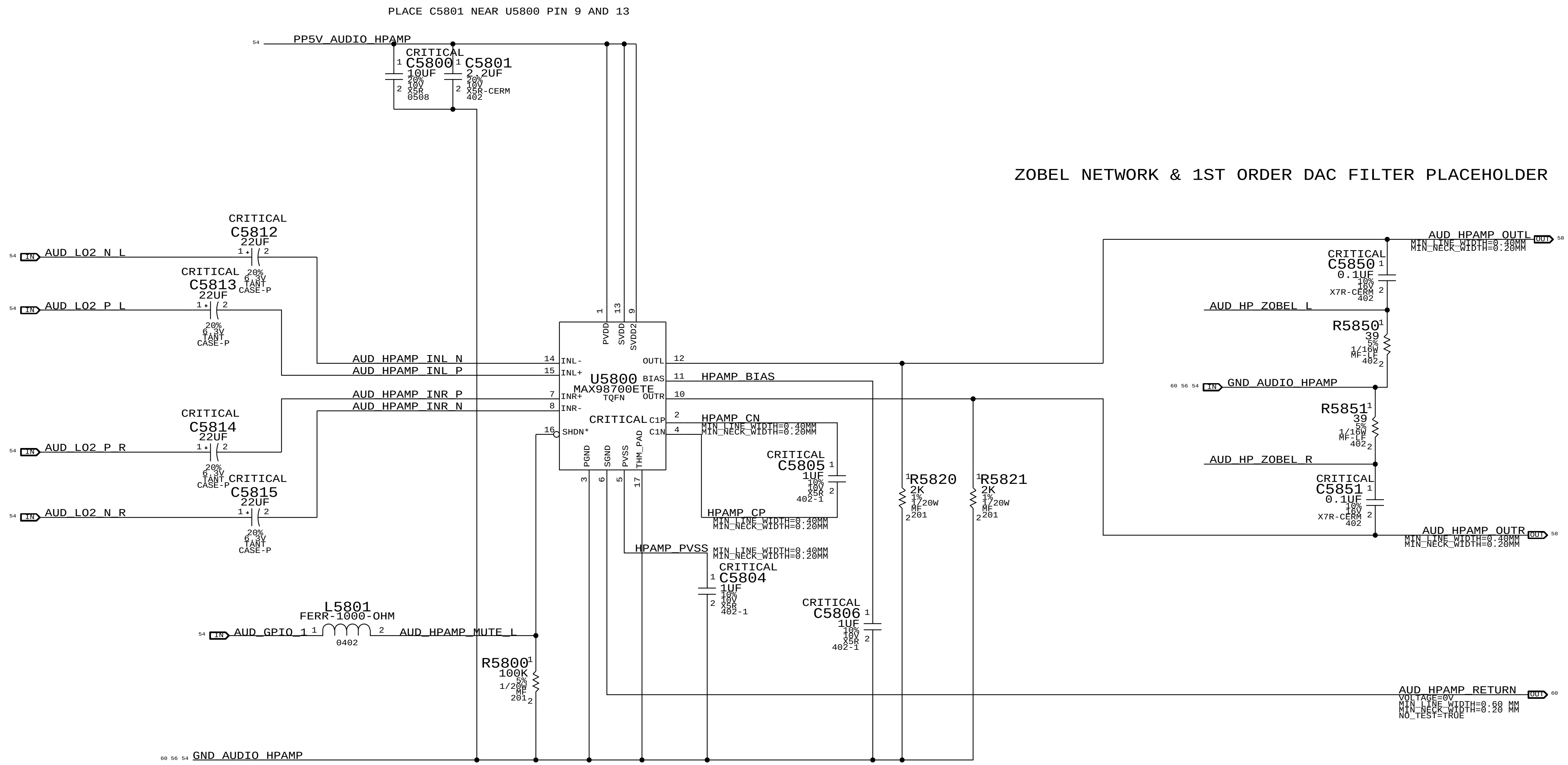
# HEADPHONE AMPLIFIER (MAX98700)

## APN:353S3244




DAC FILTER: @ 168KHZ

CODEC FILTER STOPBAND:26.256KHZ, ATTENUATION FROM STOPBAND TO 100KHZ IS 102DB

SYNC_MASTER=J40I SYNC_DATE=05/26/2011

PAGE TITLE

AUDIO:HEADPHONE AMP

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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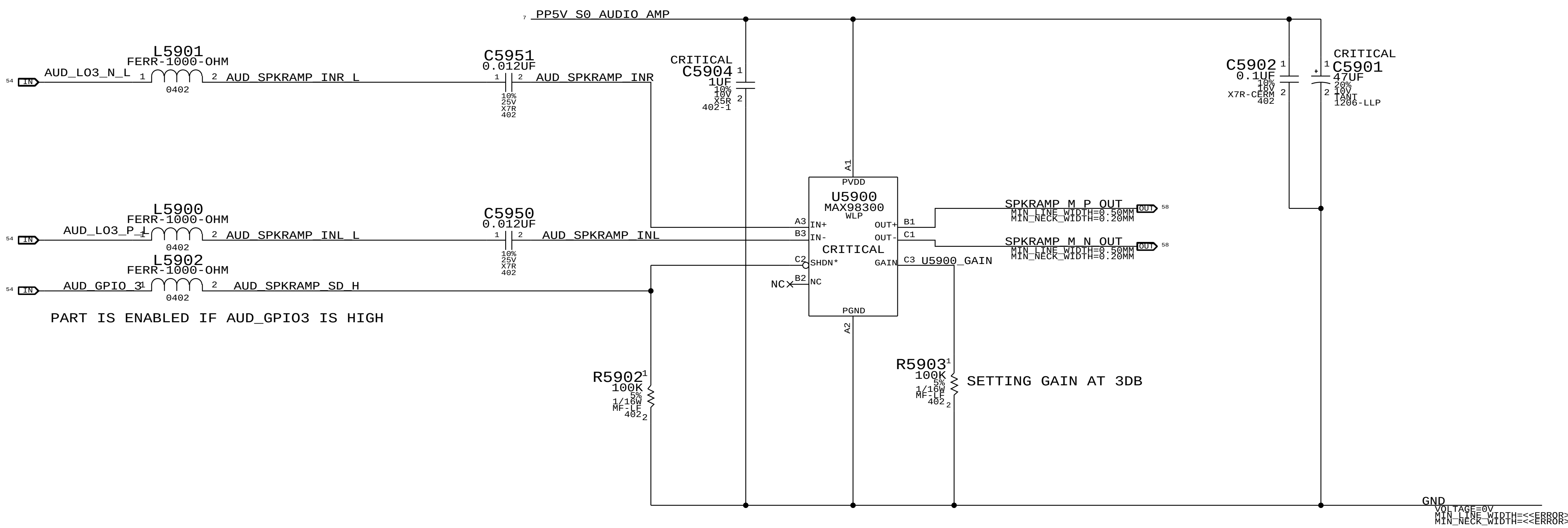
# MONO SPEAKER AMPLIFIER

APN 353S2888

GAIN = 3 DB NOM TO ALLOW A LITTLE MORE THAN 1% THD POWER = 2.1W AT 4OHMS.
FC = 470 HZ. RIN AT 3DB IS 28K.




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

AUDIO:SPEAKER AMP

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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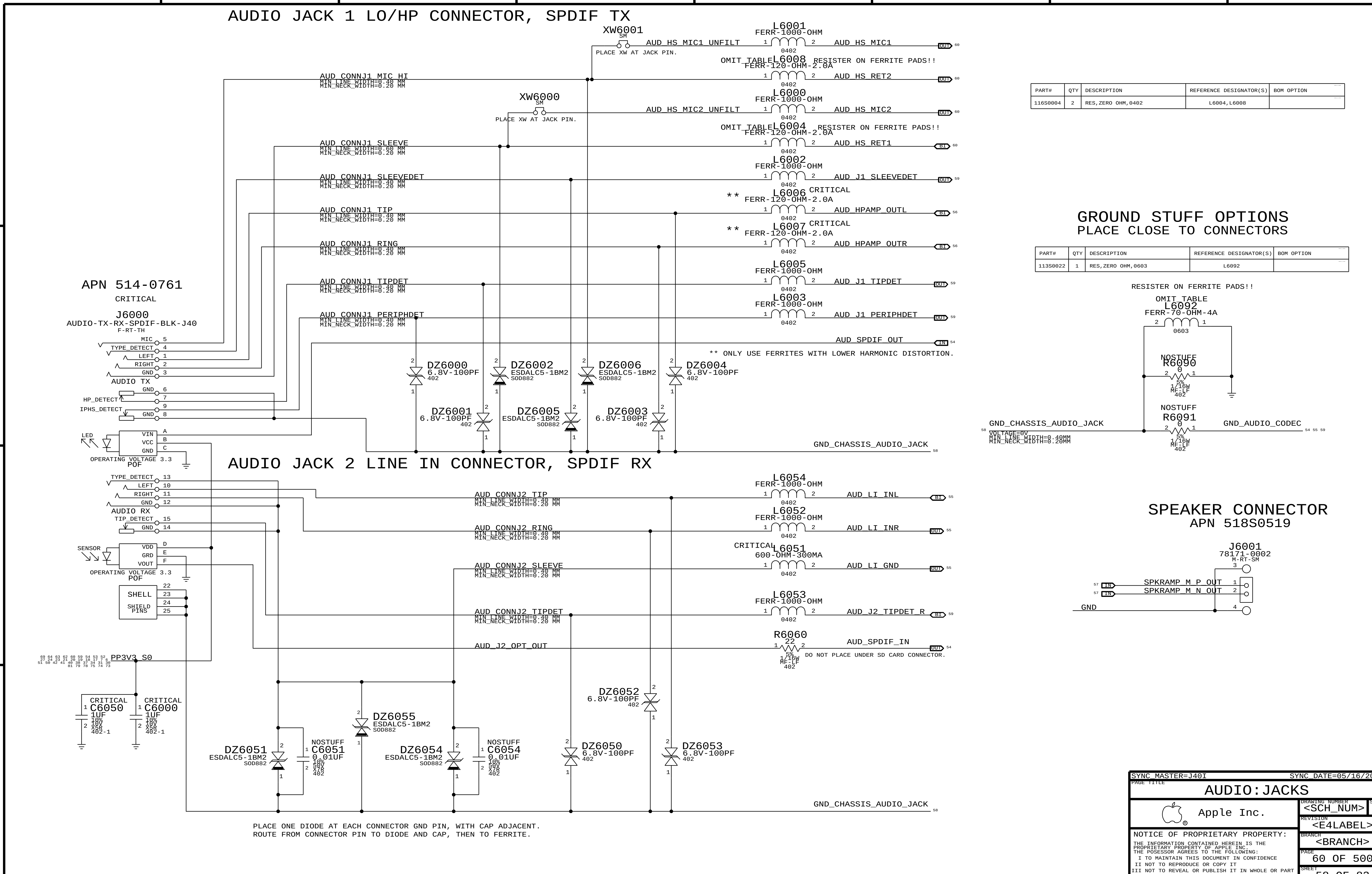


| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | BOM OPTION |
|---|---|---|---|---|
| 116S0004 | 2 | RES,ZERO OHM,0402 | L6004,L6008 | |

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | BOM OPTION |
|---|---|---|---|---|
| 113S0022 | 1 | RES,ZERO OHM,0603 | L6092 | |

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

AUDIO:JACKS

Apple Inc.

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REVISION <E4LABEL>

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# MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256

### CODEC OUTPUT SIGNAL PATHS

| FUNCTION | VOLUME/MUTE | CONVERTER | PIN COMPLEX | ~SHDN CONTROL | DET ASSIGNMENT |
|---|---|---|---|---|---|
| HP/LINE OUT | 0X03 (3) | 0X03 (03) | 0X0A (10) | GPIO_1 | 0X0A(D) |
| SPEAKER 1 | 0X04 (4) | 0X04 (04) | 0X0B (11) | GPIO_3 | N/A |
| SPDIF OUT | N/A | 0X08 (08) | 0X10 (16) | N/A | 0X0D(B) |

### COUGAR POINT OUTPUT SIGNAL PATHS

| FUNCTION | VOLUME/MUTE | CONVERTER | PIN COMPLEX | ~SHDN CONTROL | DET ASSIGNMENT |
|---|---|---|---|---|---|
| HDMI OUT | N/A | - | - | N/A | N/A |

### CODEC INPUT SIGNAL PATHS

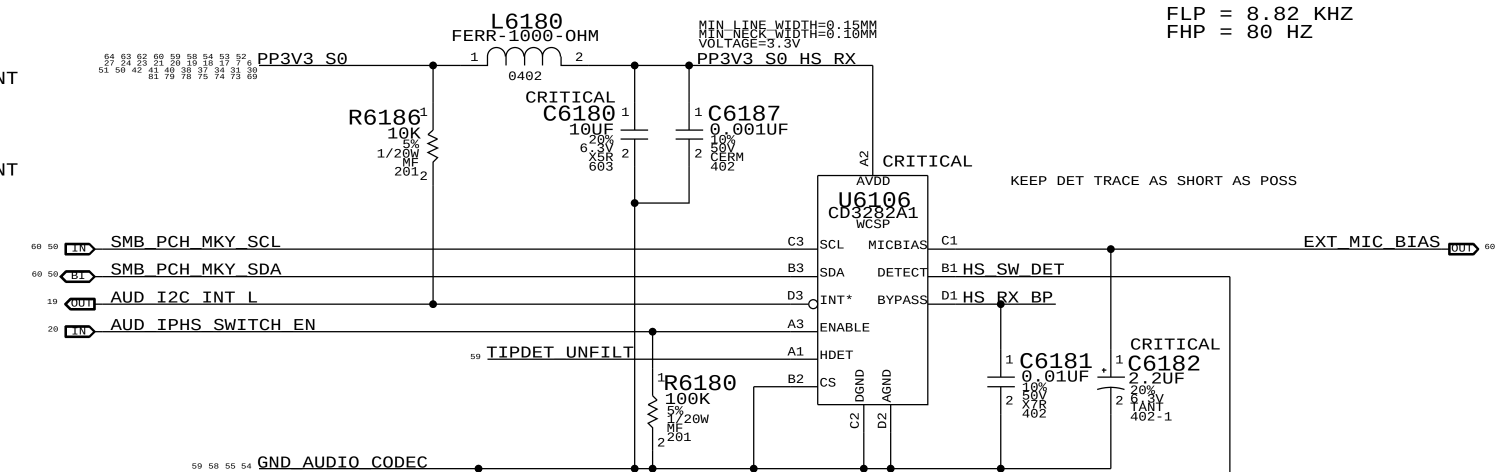
| FUNCTION | GAIN/MUTE | CONVERTER | PIN COMPLEX | VREF | DET ASSIGNMENT |
|---|---|---|---|---|---|
| LINE IN | 0X05 (5) | 0X05 (05) | 0X0C (12) | N/A | 0X0C(C) |
| EXT MIC IN | 0X06 (6) | 0X06 (06) | 0X0D (13,V22,B,LEFT) | MIKEY | MIKEY |
| SPDIF IN | N/A | 0X07 (07) | 0X0F (15) | N/A | N/A |

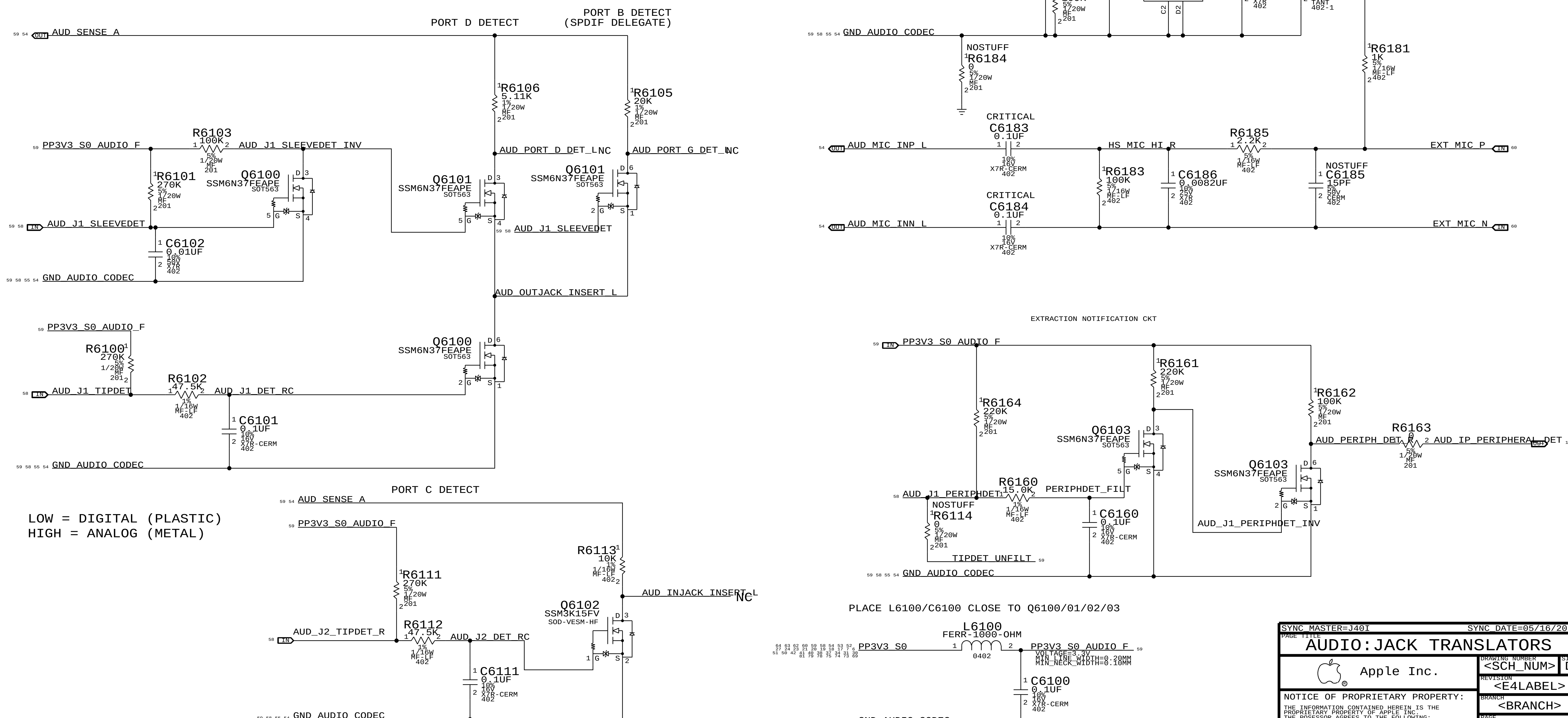
SYSTEM INT AND GPIO LINES

| FUNCTION | |
|---|---|
| MIKEY ENABLE | SATA4GP/GPIO16 |
| MIKEY INTERRUPT | PIRQG*/GPIO5 |
| PERIPHERAL DETECT | PIRQF*/GPIO3 |

WRITE: 0X72 READ: 0X73

FLP = 8.82 KHZ
FHP = 80 HZ




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: AUDIO:JACK TRANSLATORS

Apple Inc.

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REVISION <E4LABEL>

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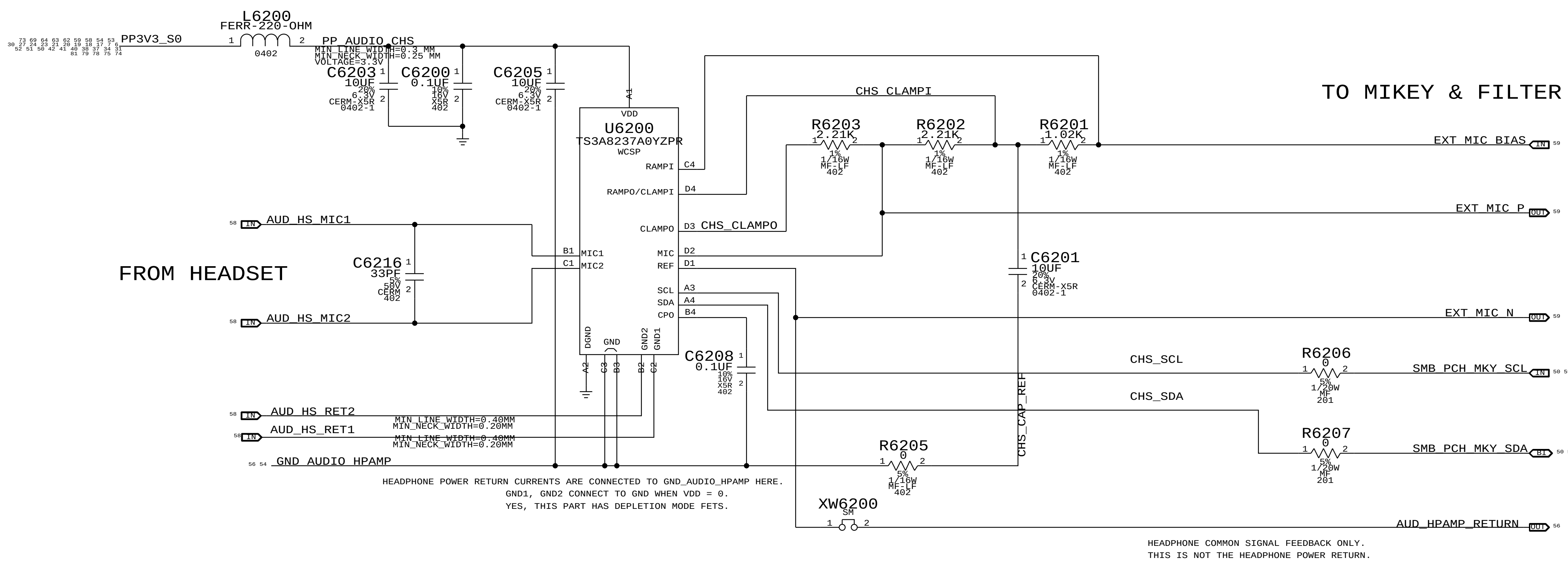
# EXTERNAL (HEADSET) MIC INPUT CIRCUITRY APN:353S3522




I2C ADDRESSES: CHS uses SMBus 0 connections

| | | | | | |
|---|---|---|---|---|---|
| CHS | U6200 | READ | 0111 | 0111 | 0X77 |
| CHS | U6200 | WRITE | 0111 | 0110 | 0X76 |

PAGE TITLE
AUDIO:DETECT/MIC BIAS

Apple Inc.

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REVISION <E4LABEL>
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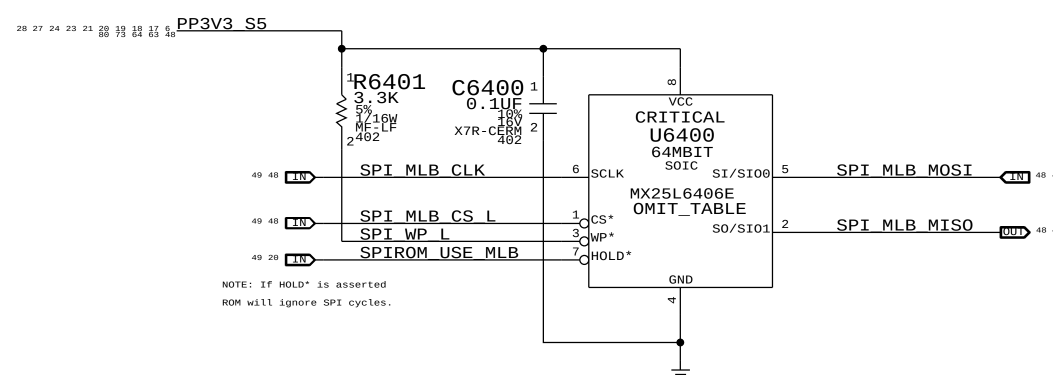


SYNC_MASTER=J40I | SYNC_DATE=05/16/2011

PAGE TITLE

# SPI BOOT ROM

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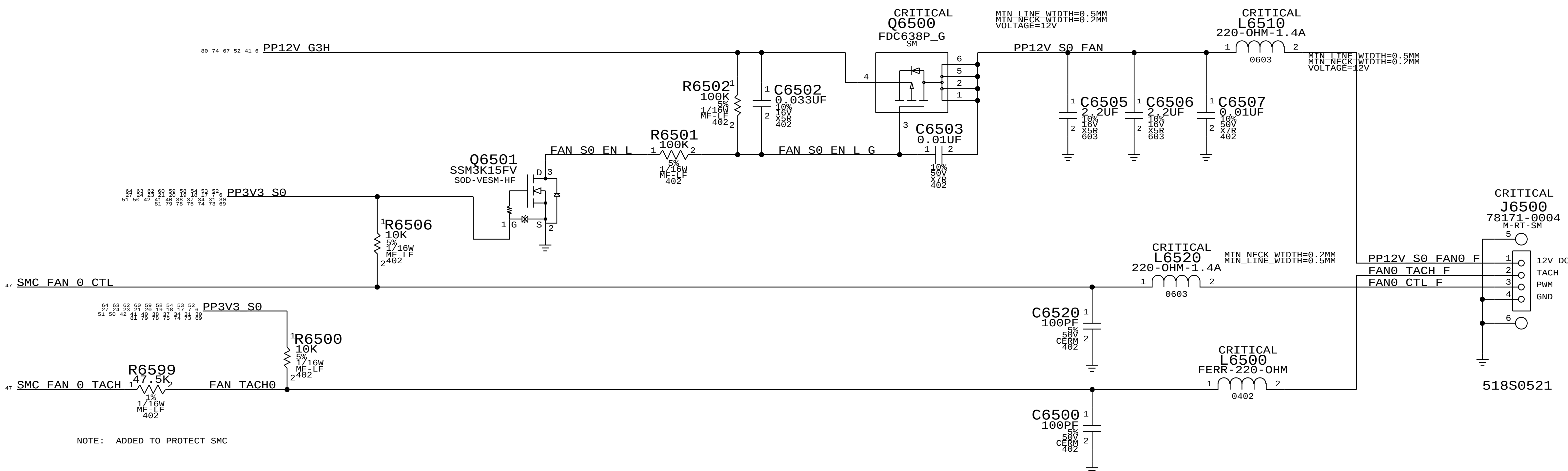
# FAN CONTROL CIRCUIT




NOTE: ADDED TO PROTECT SMC

| | |
|---|---|
| SYNC_MASTER=J40I | SYNC_DATE=05/16/2011 |
| PAGE TITLE: Fan Control Circuit | |
| Apple Inc. | DRAWING NUMBER: <SCH_NUM> SIZE: D |
| | REVISION: <E4LABEL> |
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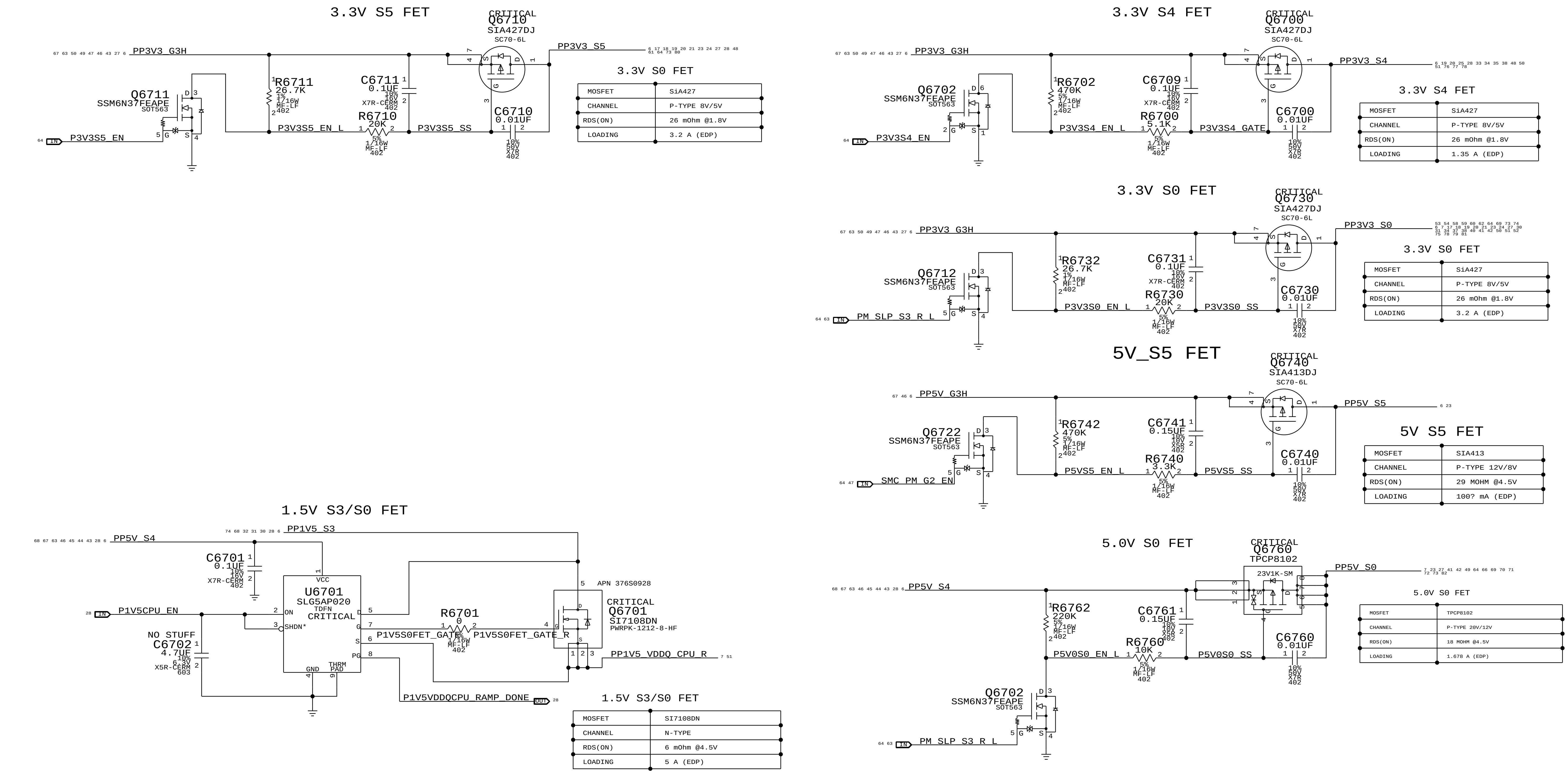
| | SHEET: 62 OF 83 |




3.3V S0 FET

| MOSFET | SiA427 |
|---|---|
| CHANNEL | P-TYPE 8V/5V |
| RDS(ON) | 26 mOhm @1.8V |
| LOADING | 3.2 A (EDP) |

3.3V S4 FET

| MOSFET | SiA427 |
|---|---|
| CHANNEL | P-TYPE 8V/5V |
| RDS(ON) | 26 mOhm @1.8V |
| LOADING | 1.35 A (EDP) |

3.3V S0 FET

| MOSFET | SiA427 |
|---|---|
| CHANNEL | P-TYPE 8V/5V |
| RDS(ON) | 26 mOhm @1.8V |
| LOADING | 3.2 A (EDP) |

5V S5 FET

| MOSFET | SIA413 |
|---|---|
| CHANNEL | P-TYPE 12V/8V |
| RDS(ON) | 29 MOHM @4.5V |
| LOADING | 100? mA (EDP) |

1.5V S3/S0 FET

| MOSFET | SI7108DN |
|---|---|
| CHANNEL | N-TYPE |
| RDS(ON) | 6 mOhm @4.5V |
| LOADING | 5 A (EDP) |

5.0V S0 FET

| MOSFET | TPCP8102 |
|---|---|
| CHANNEL | P-TYPE 20V/12V |
| RDS(ON) | 18 MOHM @4.5V |
| LOADING | 1.678 A (EDP) |

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# Power FETs

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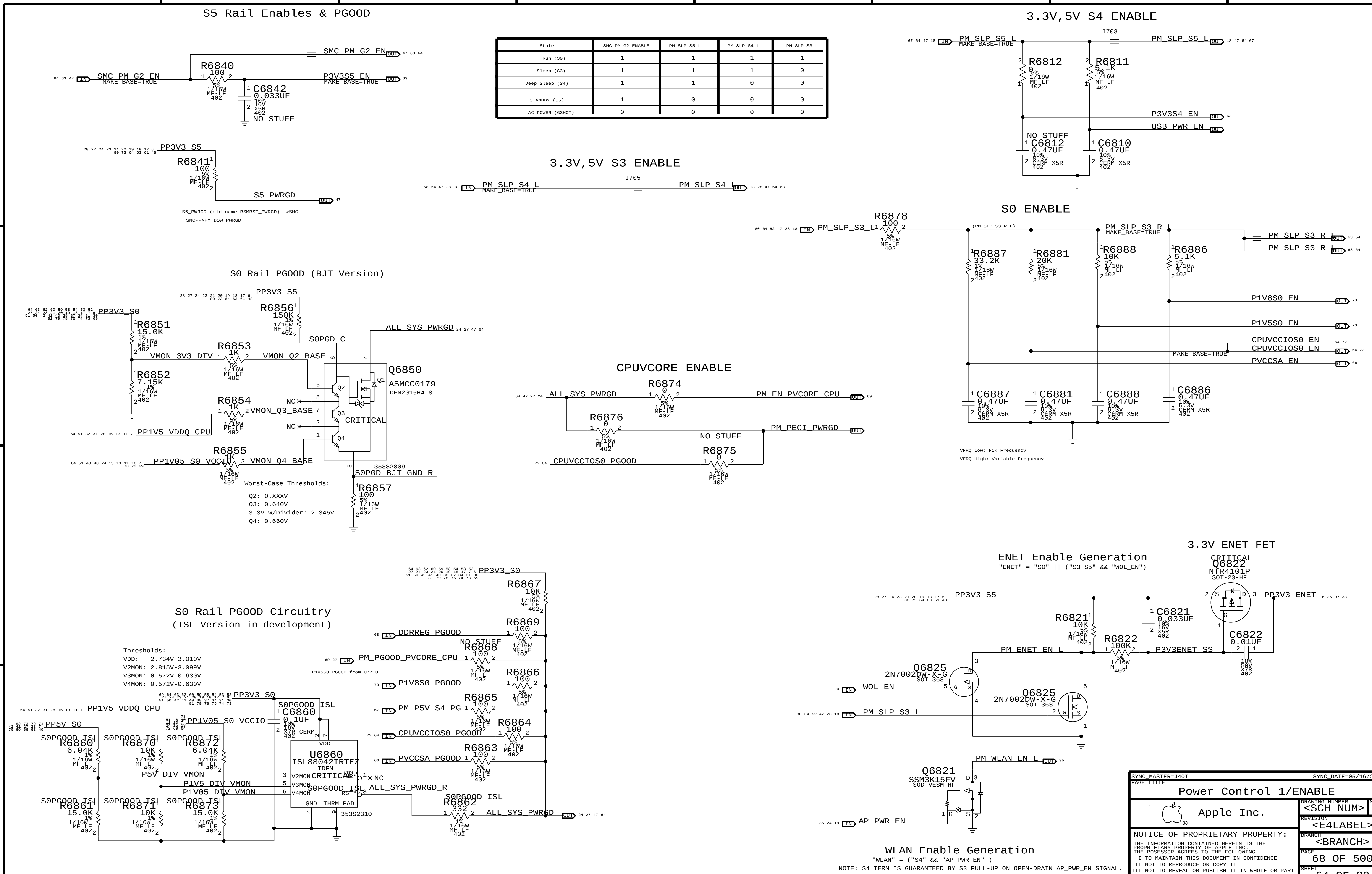


| State | SMC_PM_G2_ENABLE | PM_SLP_S5_L | PM_SLP_S4_L | PM_SLP_S3_L |
|---|---|---|---|---|
| Run (S0) | 1 | 1 | 1 | 1 |
| Sleep (S3) | 1 | 1 | 1 | 0 |
| Deep Sleep (S4) | 1 | 1 | 0 | 0 |
| STANDBY (S5) | 1 | 0 | 0 | 0 |
| AC POWER (G3HOT) | 0 | 0 | 0 | 0 |

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

Power Control 1/ENABLE

Apple Inc.

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ROOM FOR MORE?

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

POWER CONTROL CONT.

DRAWING NUMBER <SCH_NUM> SIZE D

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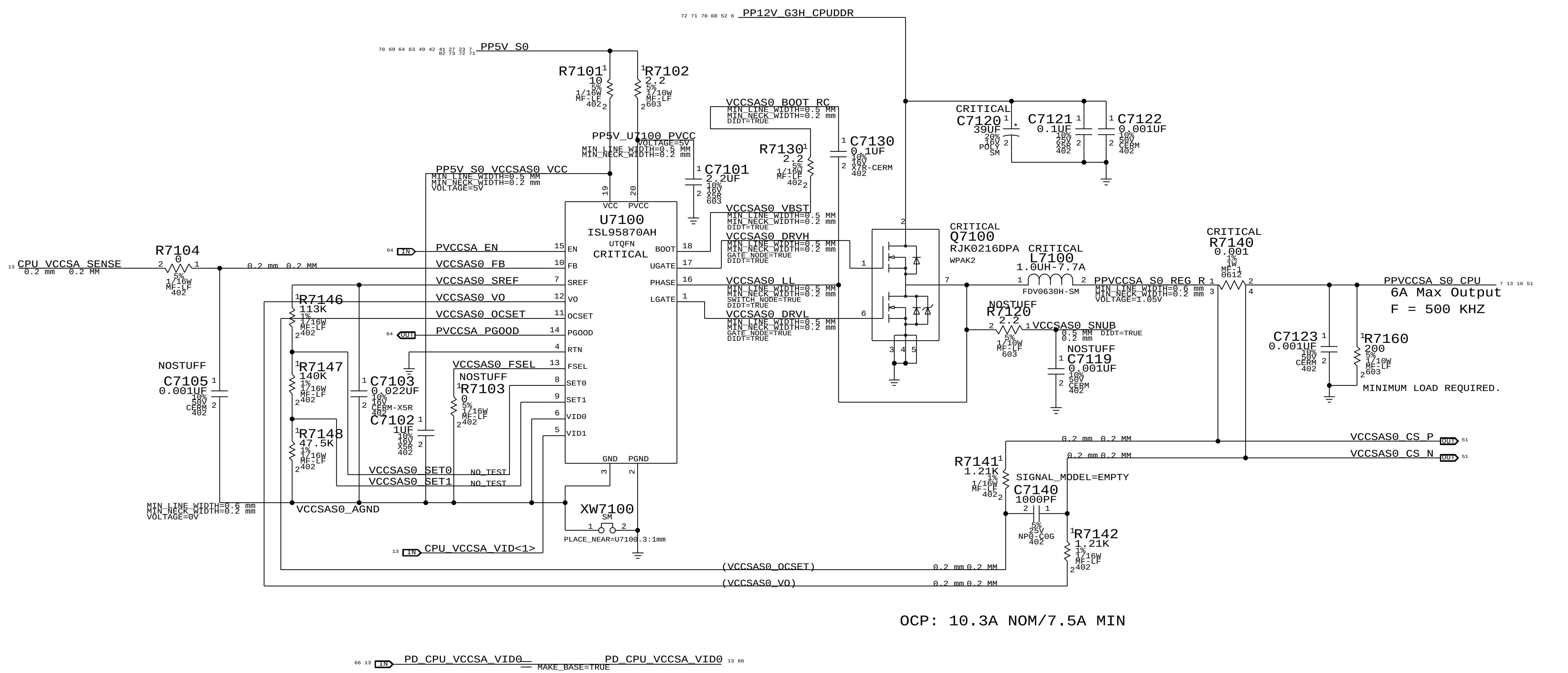
# System Agent Power Supply




OCP: 10.3A NOM/7.5A MIN

NOT USED FOR 2011 CPUS.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: VR - SYSTEM AGENT SUPPLY

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

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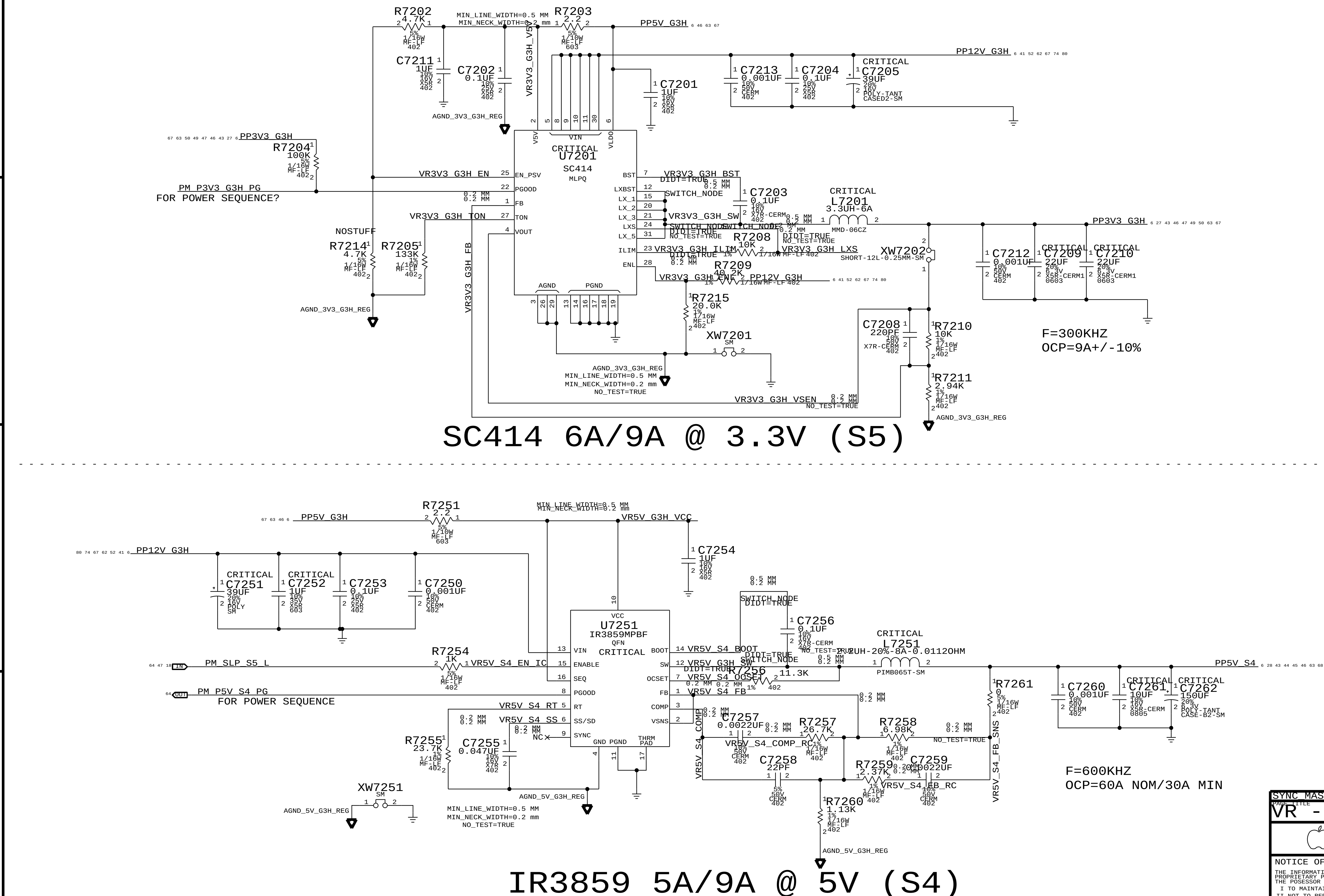


SYNC_MASTER=J401 SYNC_DATE=05/16/2011

PAGE TITLE: VR - 3.3V S5 AND 5V S4

Apple Inc.

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REVISION <E4LABEL>

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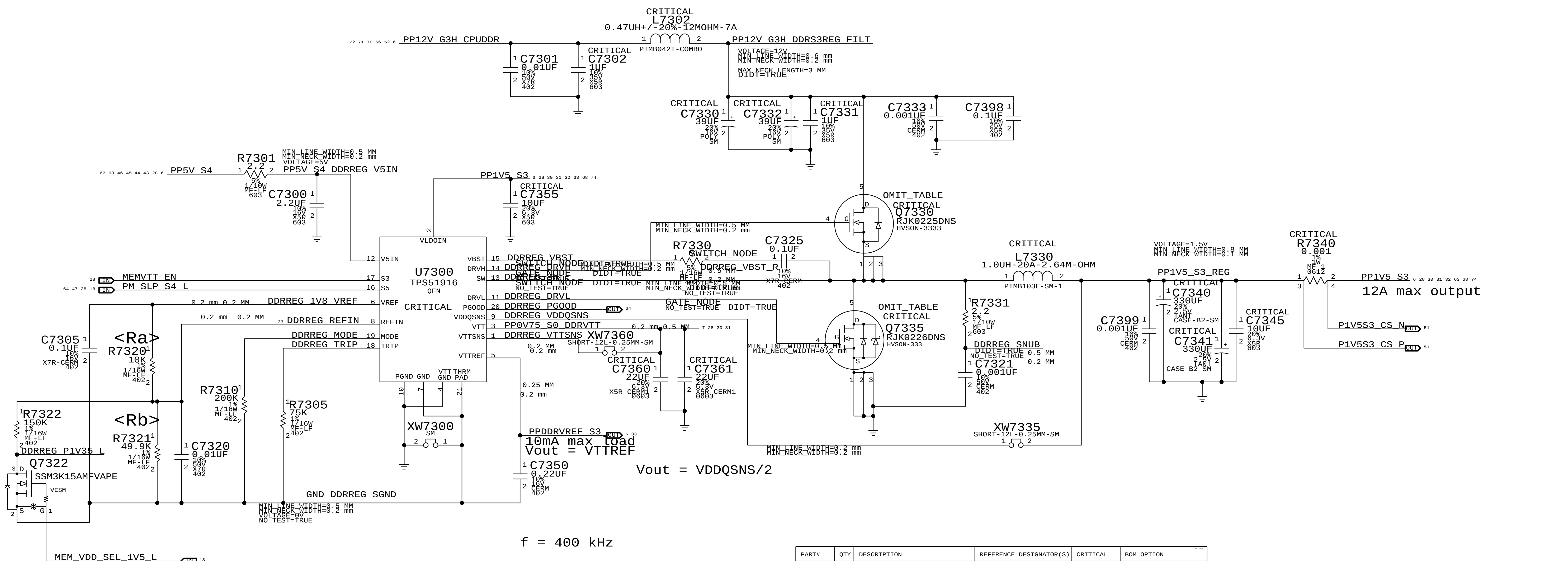


VDDQ = VREFIN = 1.8V * RB / (RA + RB)

OCP=27A NOM/19A MIN

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 376S0979 | 1 | TRA,FET,FAIRCHILD | Q7330 | CRITICAL | FET_SET_A:FAIRCHILD |
| 376S0874 | 1 | TRA,FET,FAIRCHILD | Q7335 | CRITICAL | FET_SET_A:FAIRCHILD |

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 376S0964 | 1 | TRA,FET,RENESAS | Q7330 | CRITICAL | FET_SET_A:RENESAS |
| 376S0965 | 1 | TRA,FET,RENESAS | Q7335 | CRITICAL | FET_SET_A:RENESAS |

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 085-3109 | 1 | COMP SET,FETS,FAIRCHILD | CSET1 | CRITICAL | FET_SET_A:USE_PHANTOMS |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 085-3110 | 085-3109 | FET_SET_A:USE_PHANTOMS | CSET1 | COMP SET,FETS,RENESAS |

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# DDR 1.5V VR

Apple Inc.

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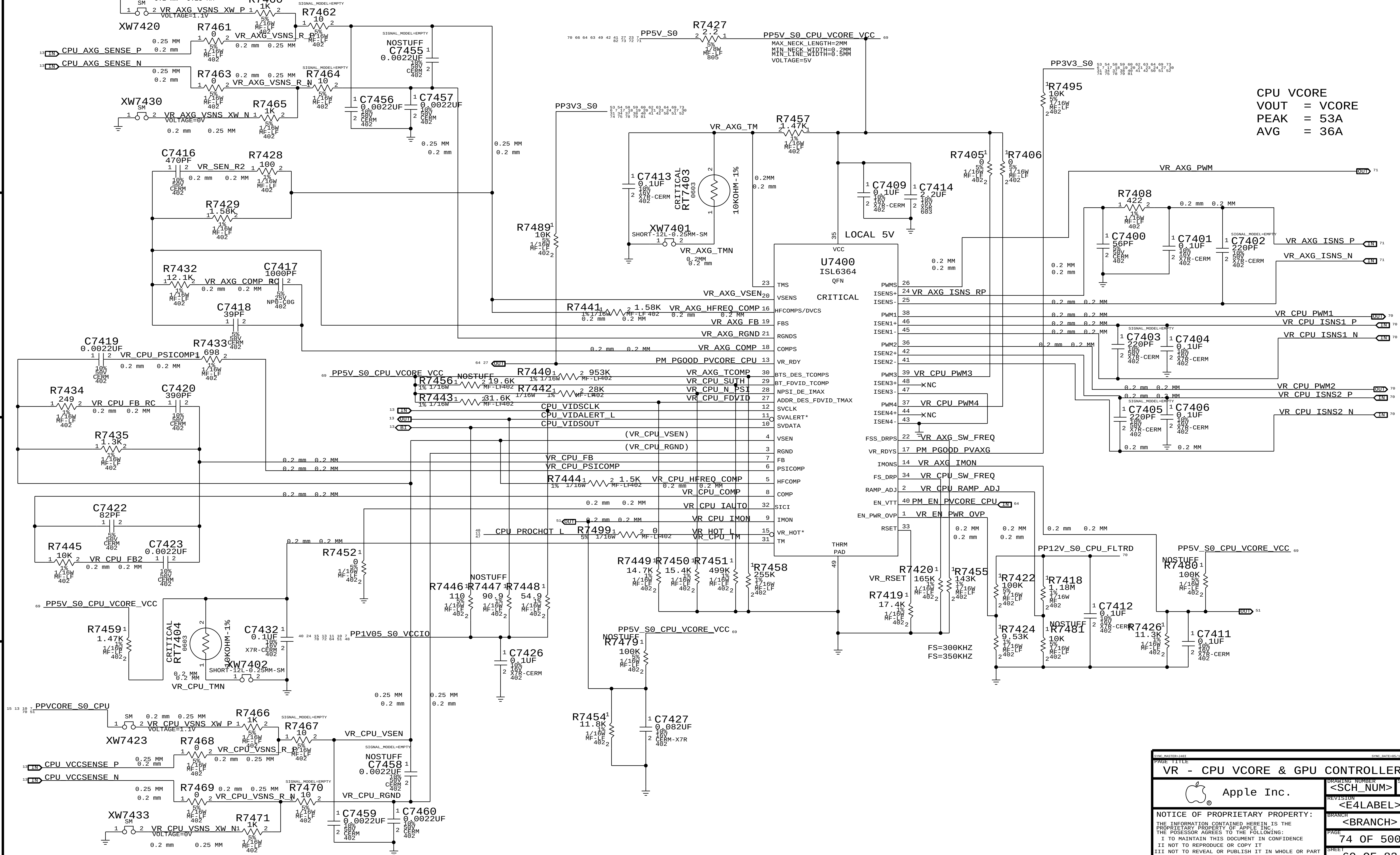
# CPU CORE/AXG REG 1.1V/53A   O/P= PPVCORE_S0_CPU_REG




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
VR - CPU VCORE & GPU CONTROLLER

Apple Inc.

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REVISION <E4LABEL>
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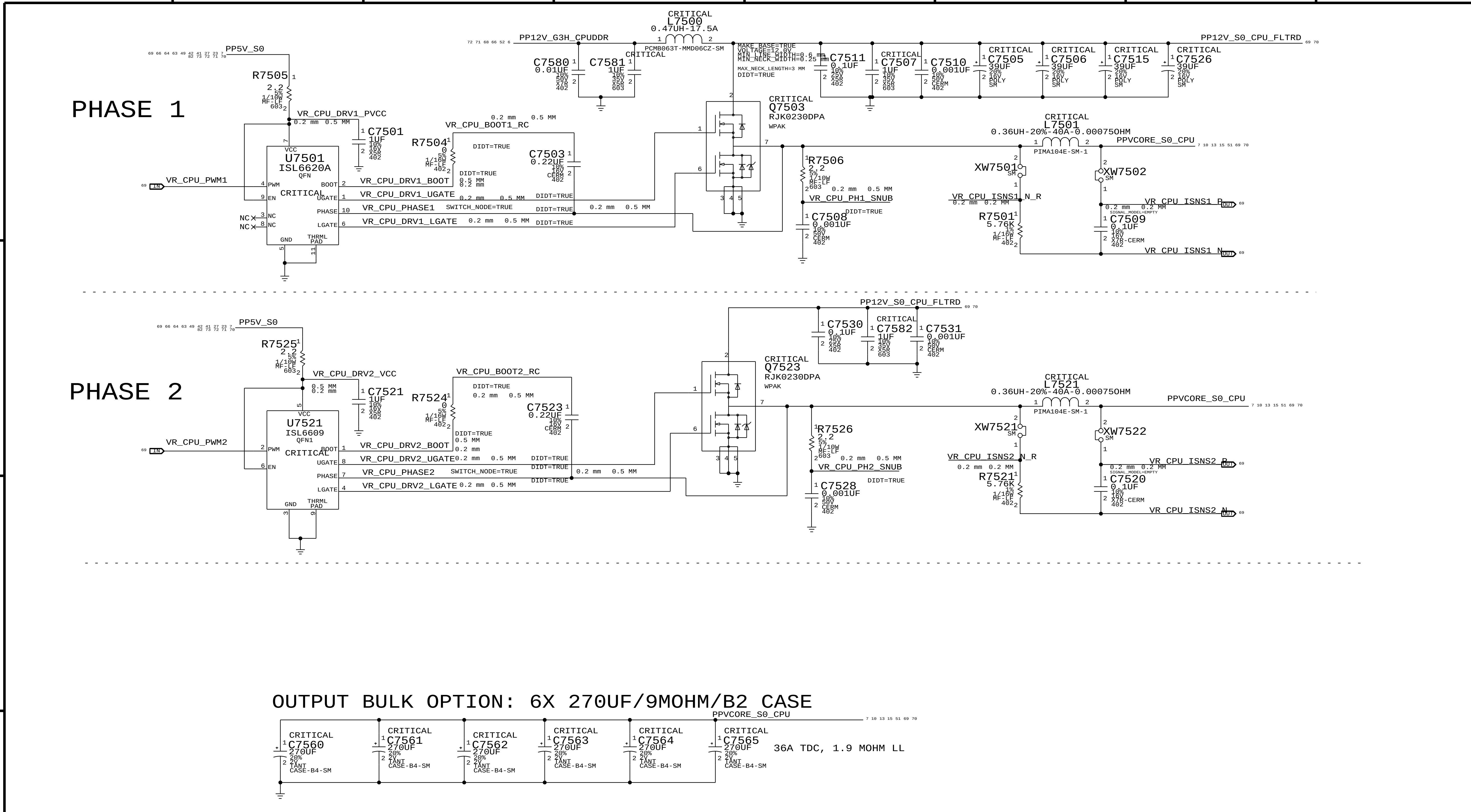


SYNC_MASTER=J401 SYNC_DATE=05/16/2011

PAGE TITLE
VR - CPU VCORE OUTPUT PHASES

Apple Inc.

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REVISION <E4LABEL>

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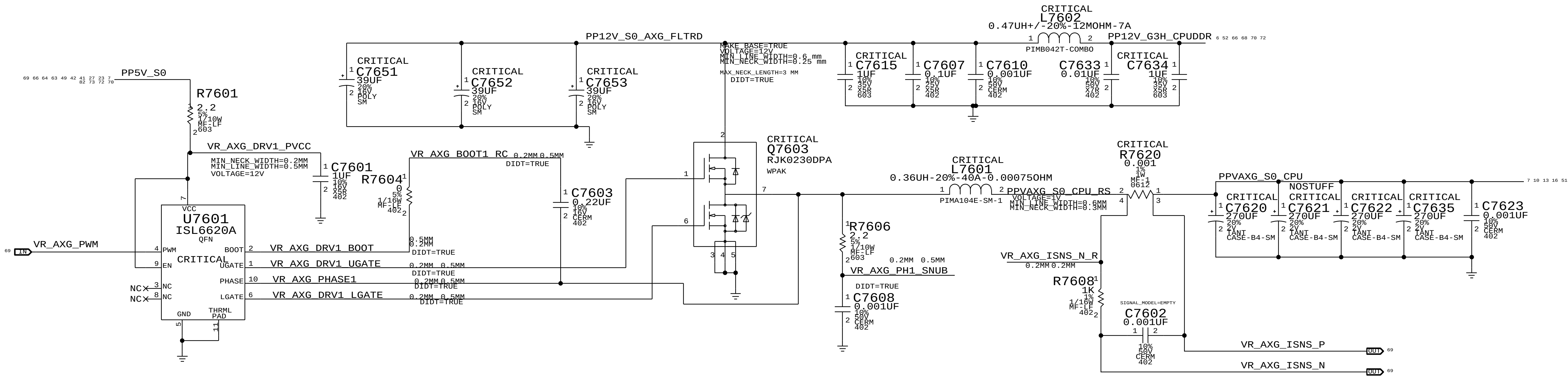
# AXG PHASE (22A AVG/33A PEAK, 3.9MOHM LL)




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

VR - GPU

Apple Inc.

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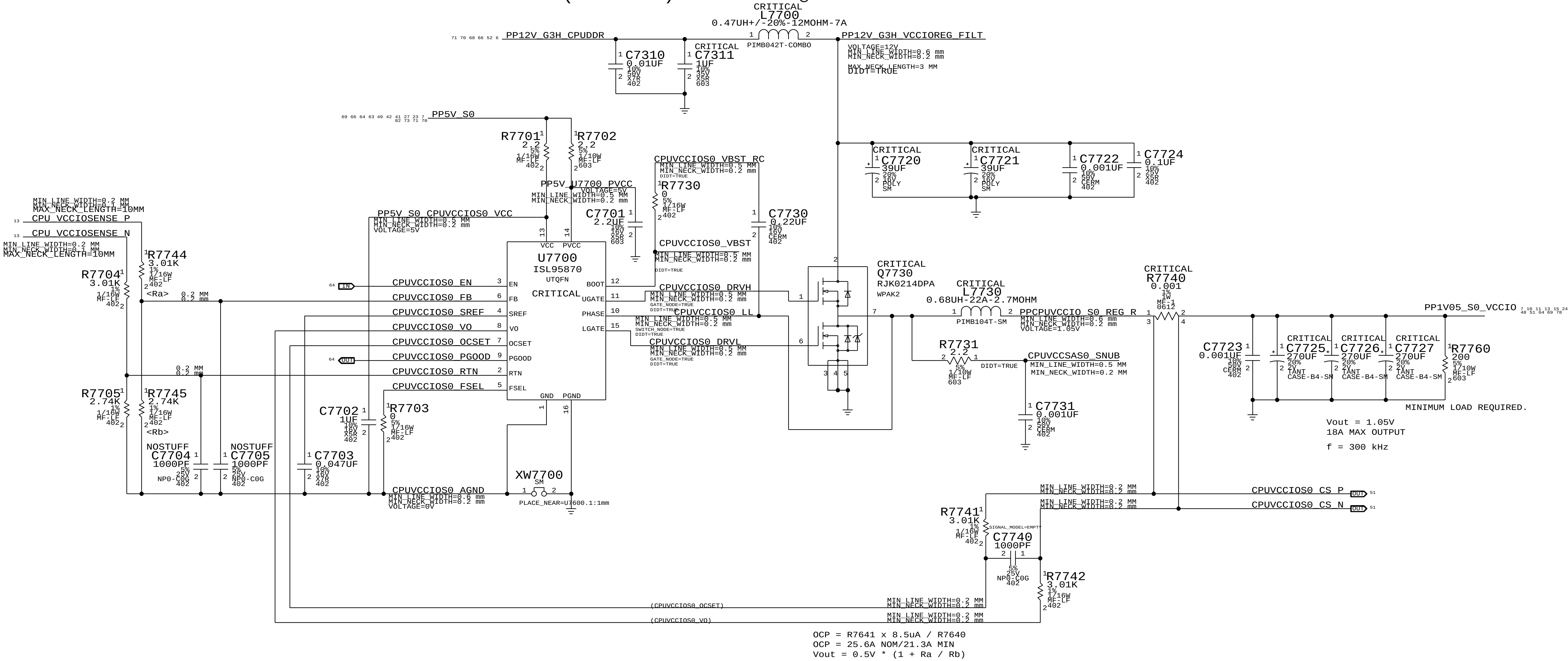
# CPU VCCIO (1.05V S0) REGULATOR @ 18A/22A




SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
VR - 1.05V S0 (VCCIO)

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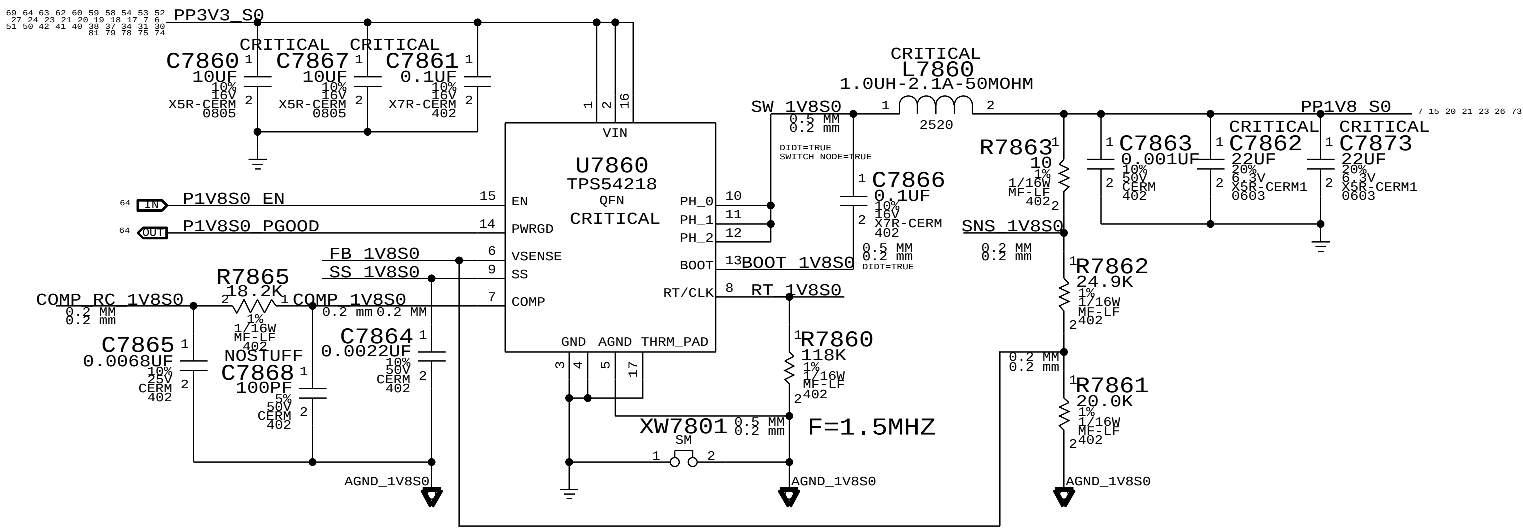
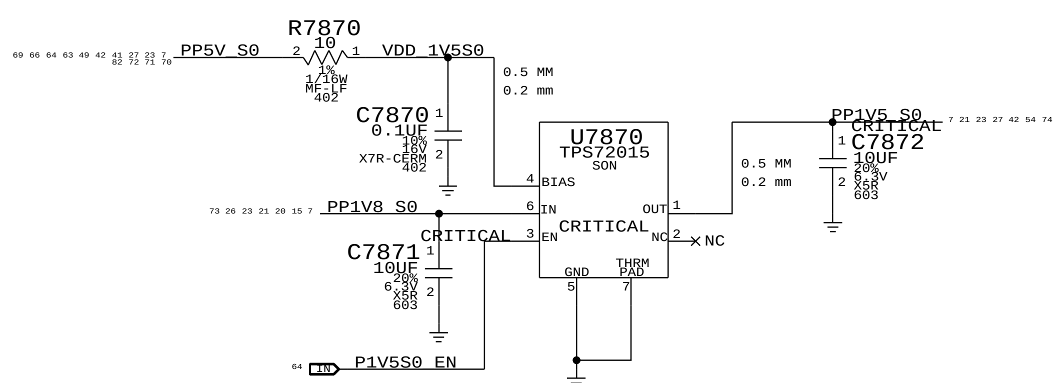
## 1.8V S0 SWITCHER @ 1.7A/2A

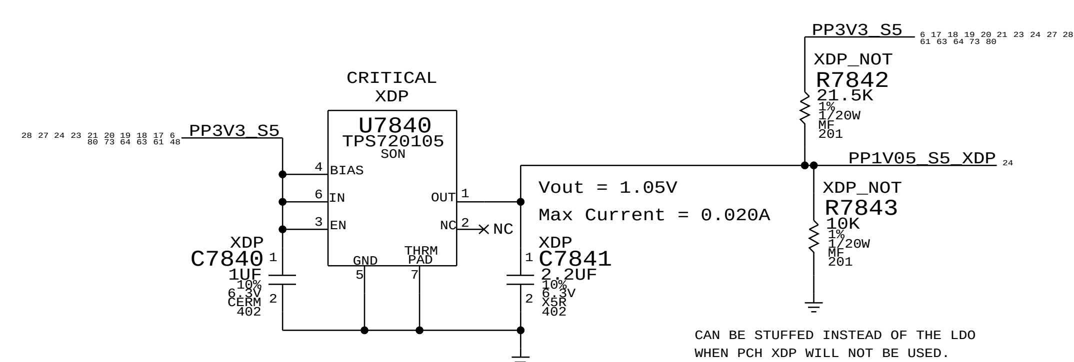



## 1.5V S0 @ 0.3A/0.5A




## 1.05V S5 XDP LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.




SYNC_MASTER=J401 SYNC_DATE=05/16/2011

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# VR - MISC

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SPRINKLE THESE CAPS SOMEWHAT EVENLY AROUND THE PLANES AND ROUTES FOR THESE RAILS.


PLACE CAPS AS CLOSE TO J7900 AS POSSIBLE.
BEST IF VIAS ARE AFTER THE CAPS.

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE
# PSU CONNECTOR, MISC CAPS

Apple Inc.

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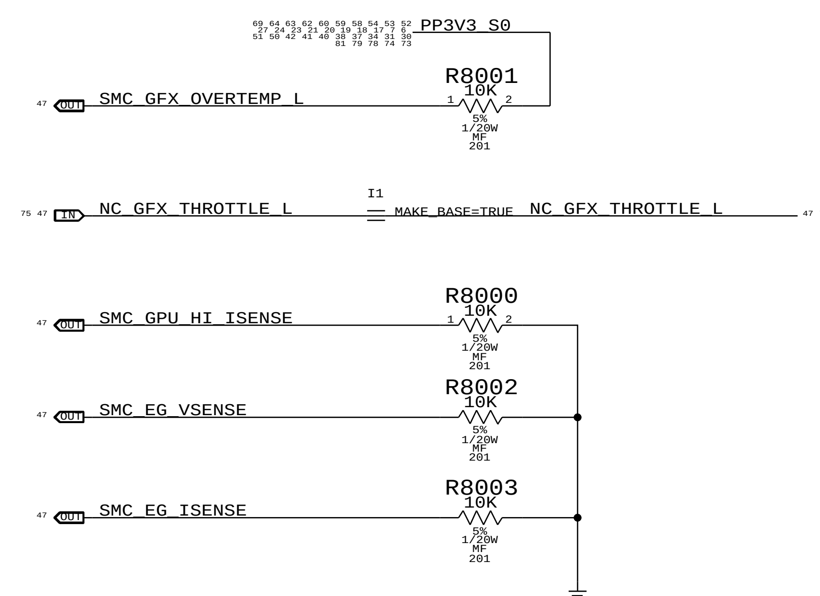


SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

# NO GPU GLUE

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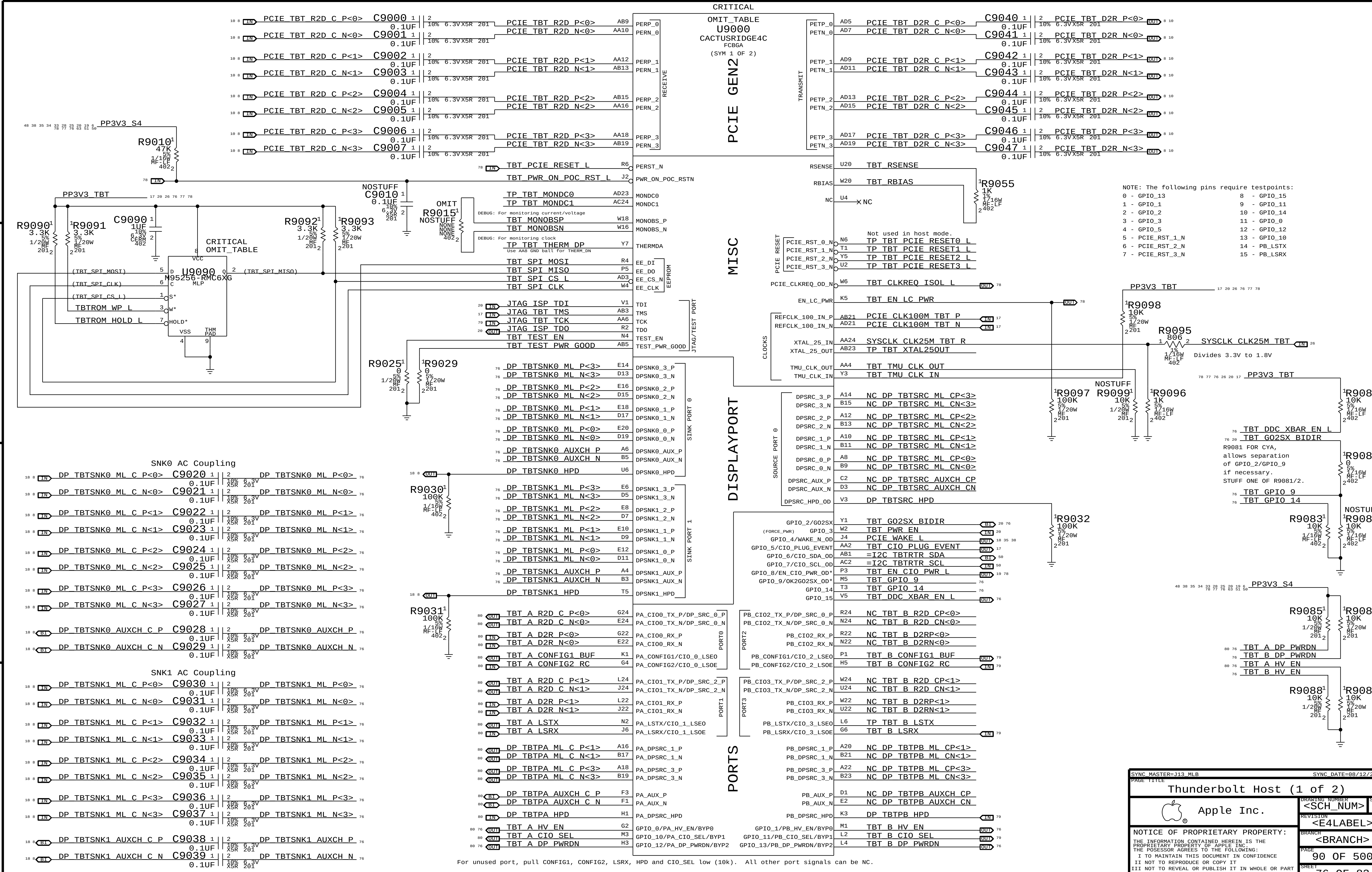


SYNC_MASTER=J13_MLB SYNC_DATE=08/12/2011

PAGE TITLE

Thunderbolt Host (1 of 2)

Apple Inc.

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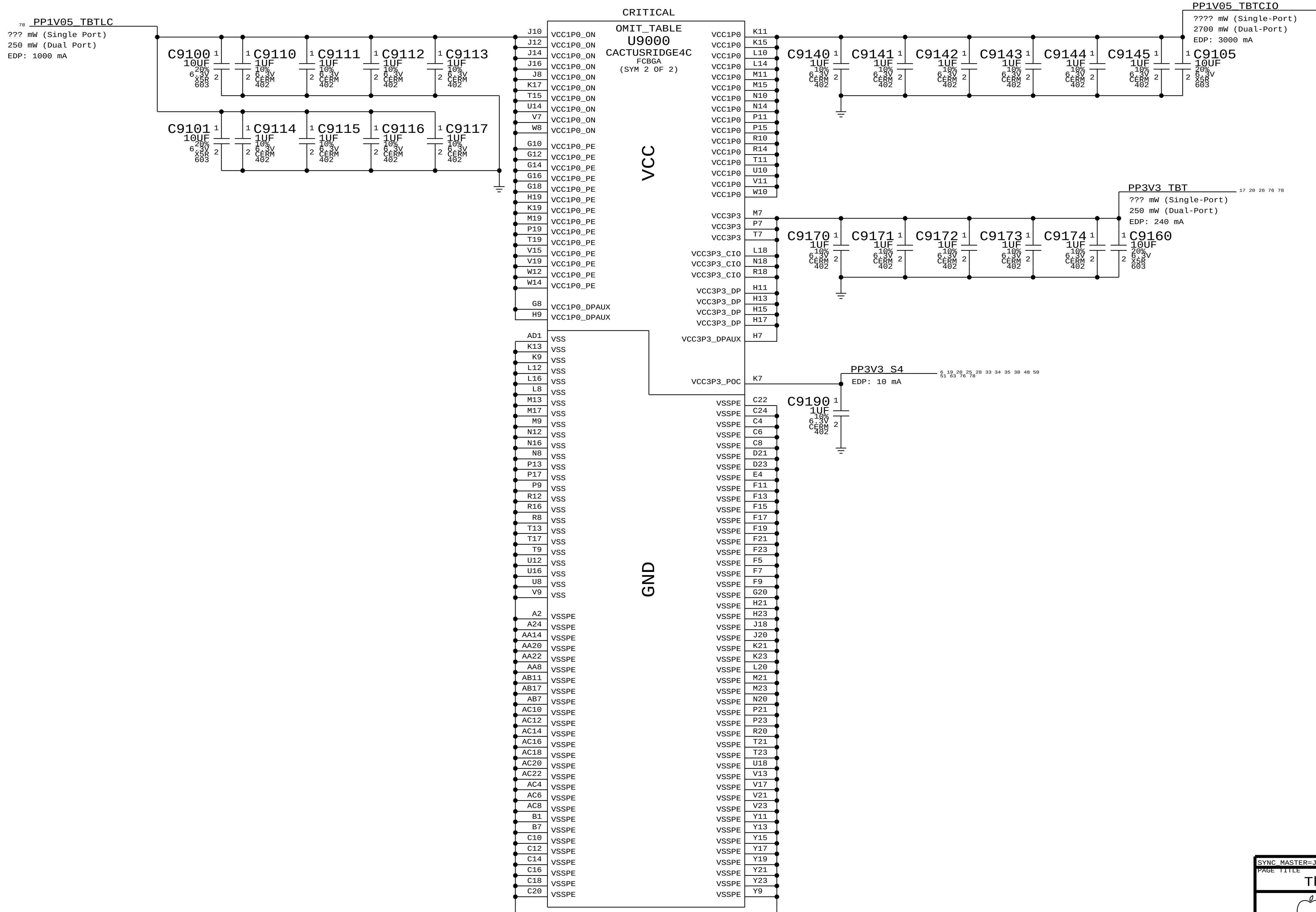


SYNC_MASTER=J13_MLB SYNC_DATE=08/12/2011

PAGE TITLE

Thunderbolt Host (2 of 2)

Apple Inc.

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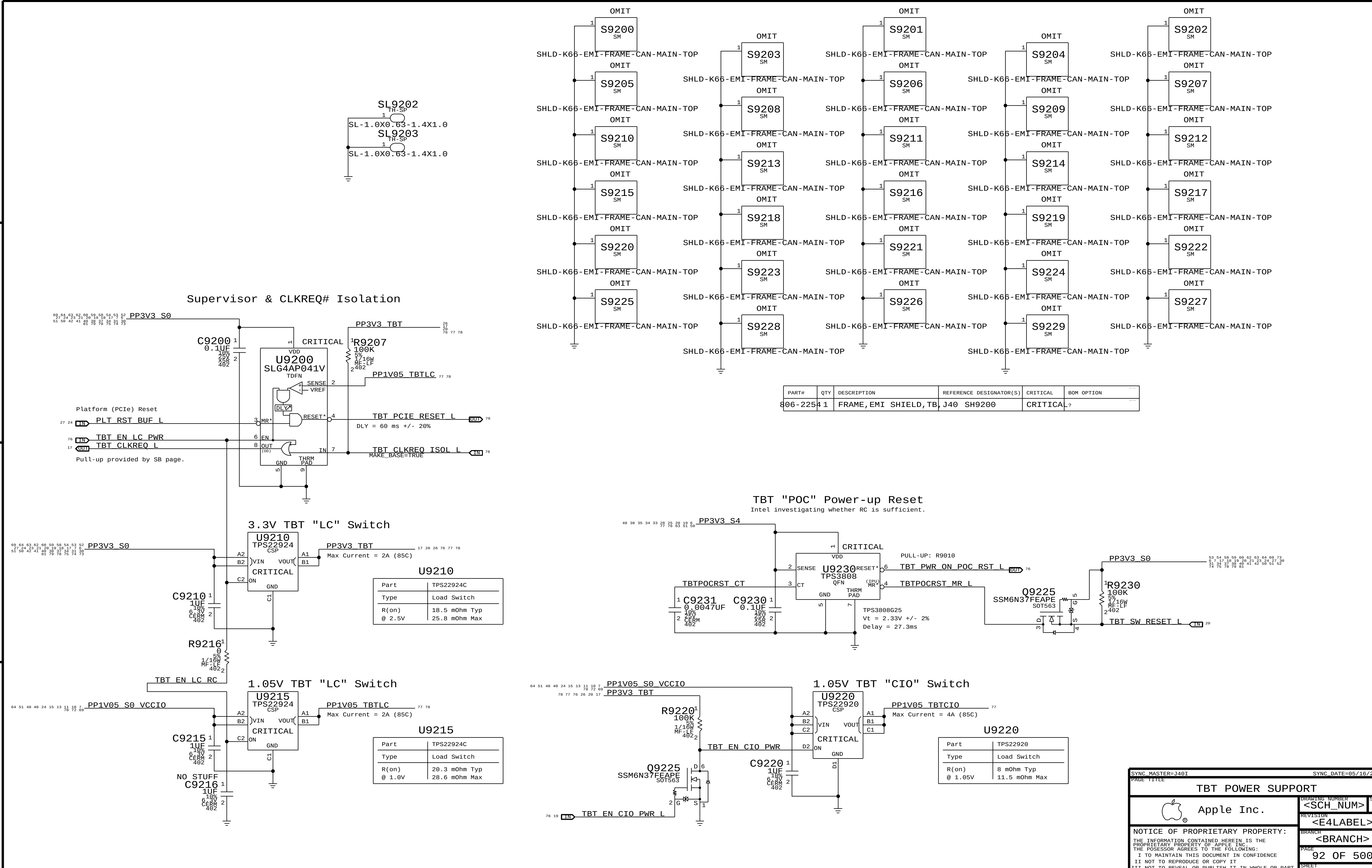


| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 806-2254 | 1 | FRAME,EMI SHIELD,TB | J40 SH9200 | CRITICAL | ? |

U9210

| Part | TPS22924C |
|---|---|
| Type | Load Switch |
| R(on) @ 2.5V | 18.5 mOhm Typ 25.8 mOhm Max |

U9215

| Part | TPS22924C |
|---|---|
| Type | Load Switch |
| R(on) @ 1.0V | 20.3 mOhm Typ 28.6 mOhm Max |

U9220

| Part | TPS22920 |
|---|---|
| Type | Load Switch |
| R(on) @ 1.05V | 8 mOhm Typ 11.5 mOhm Max |

SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE: TBT POWER SUPPORT

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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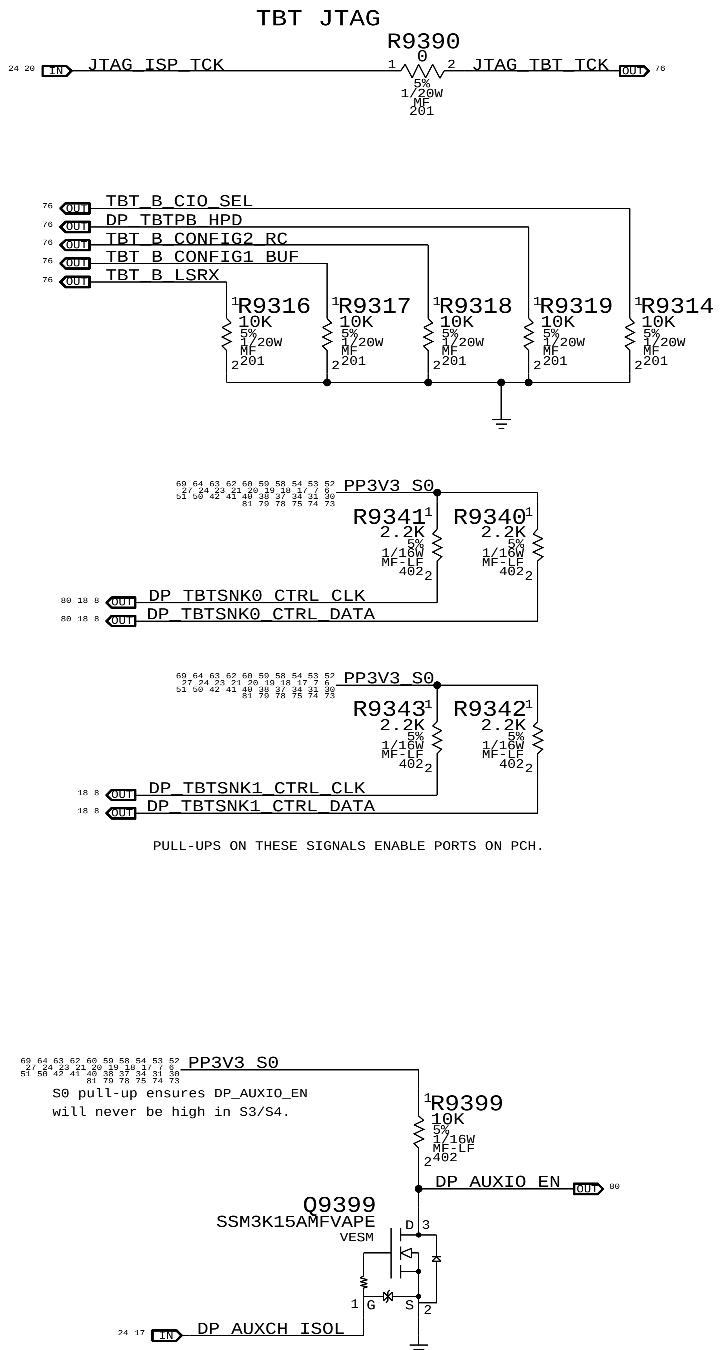


SYNC_MASTER=J40I SYNC_DATE=05/16/2011

PAGE TITLE

TBT MISC

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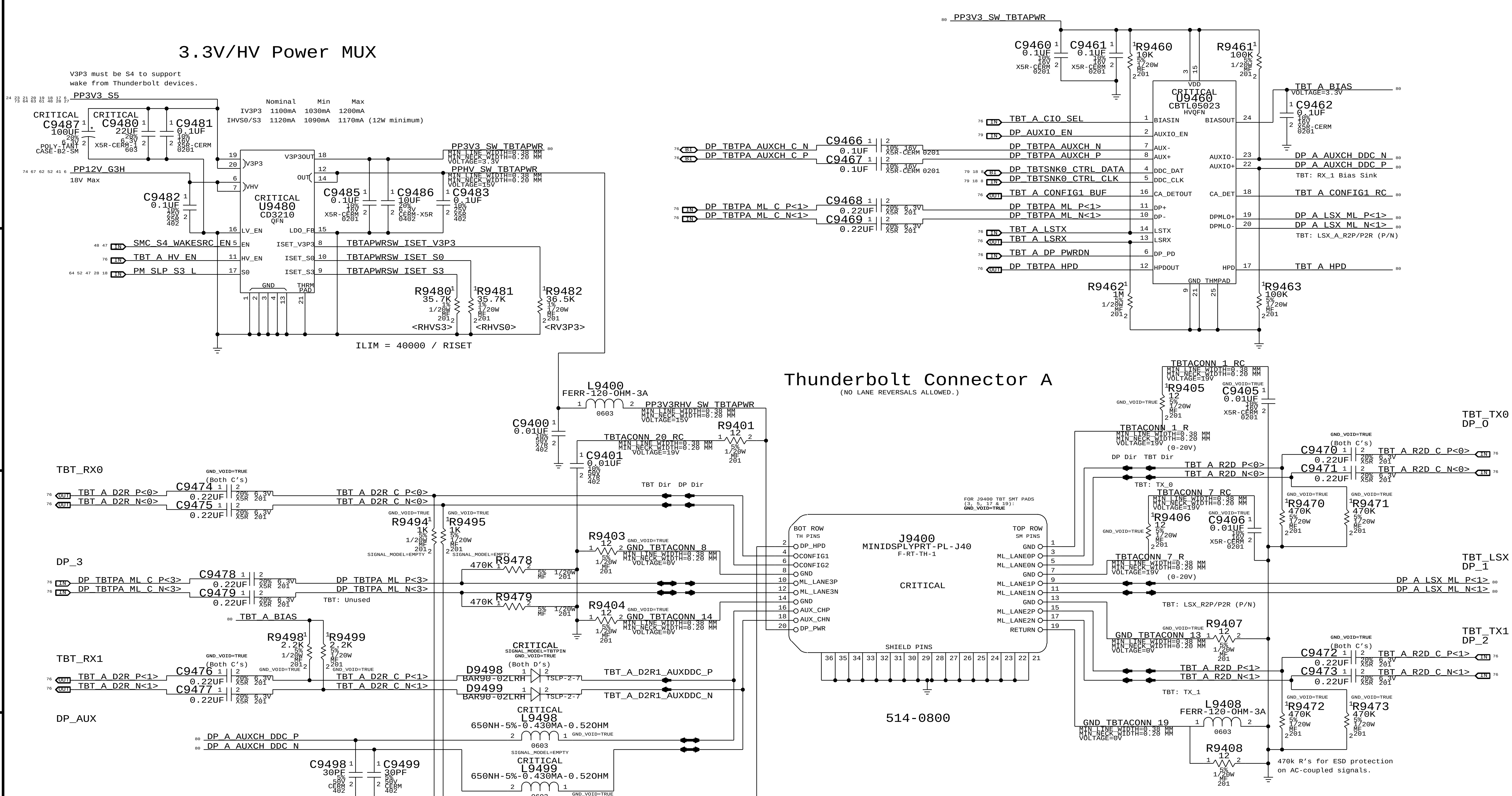


| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 870-2198 | 10 | SOLDER PREFORM 0201 | | CRITICAL | ? |

SF9400,SF9401,SF9402,SF9403,SF9404,SF9405,SF9406,SF9407,SF9408,SF9409

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 900-0064 | 870-2198 | ? | ? | ALTERNATE APN |

SYNC_MASTER=J5_KEPLER SYNC_DATE=07/20/2011

PAGE TITLE

# Thunderbolt Connector A

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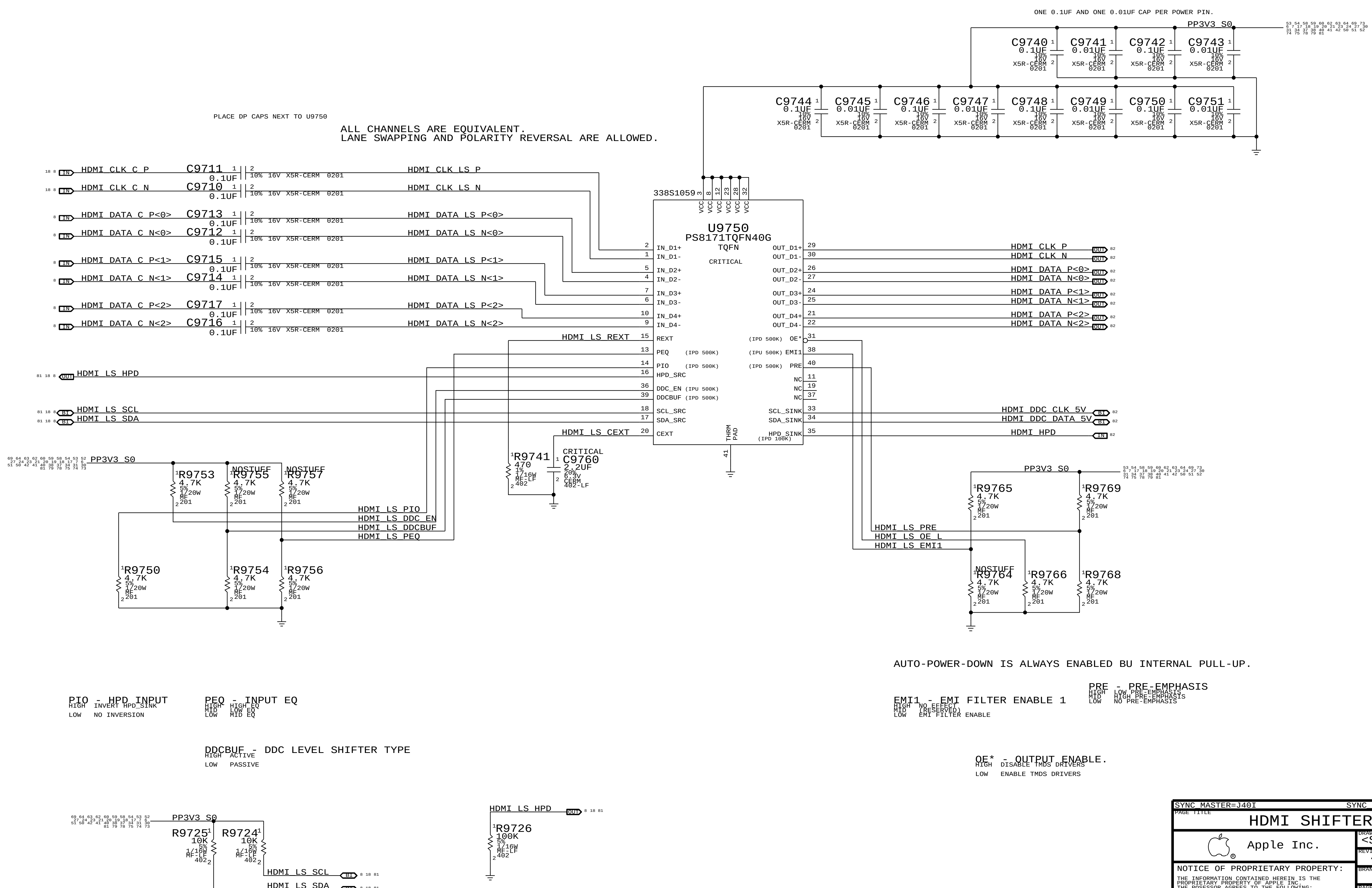


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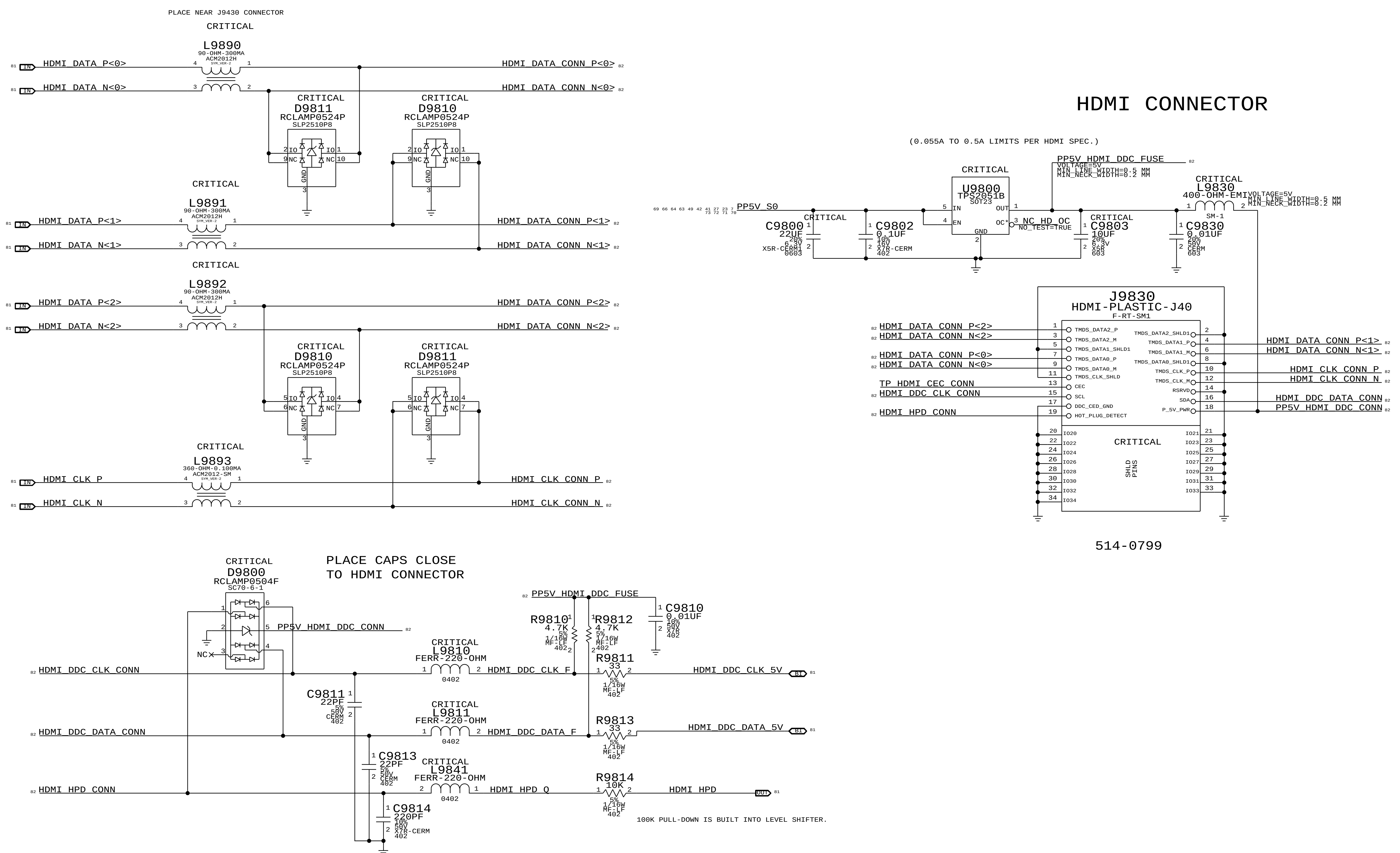


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SKIPS ANY TEMPORARY PAGES I MIGHT MAKE.

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